

Per-Element Trace Contributions for Deformed Diagonal K3 Surfaces

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For each of the 85 deformed diagonal K3 surfaces (Goto Table 10), we list all orbifold cohomology basis elements with their twisted Frobenius eigenvalues α_i .

Surface #1: $Q = (1, 1, 1, 1)$, $m = 4$

Equation: $x_0^4 + x_1^4 + x_2^4 + x_0x_3^3 = 0$

$p = 193$

Table 1: Trace contributions for $x_0^4 + x_1^4 + x_2^4 + x_0x_3^3 = 0$ at $p = 193$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p(\frac{1}{6}) \Gamma_p(\frac{1}{4}) \Gamma_p(\frac{1}{4}) \Gamma_p(\frac{1}{3})$	$-\Gamma_p(\frac{1}{6}) \Gamma_p(\frac{1}{4}) \Gamma_p(\frac{1}{4}) \Gamma_p(\frac{1}{3})$	$165+144\cdot 193+47\cdot 193^2+\dots$
$H^{1,1}$	$y_1^2 y_2^2 y_3^2 y_4^2$	p	p	193
	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{1}{4}) \Gamma_p(\frac{3}{4}) \Gamma_p(\frac{1}{2})$	$p \Gamma_p(\frac{1}{4}) \Gamma_p(\frac{3}{4}) \Gamma_p(\frac{1}{2})$	193
	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{1}{2})$	$p \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{1}{2})$	193
	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{1}{12}) \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{3}{4}) \Gamma_p(\frac{2}{3})$	$p \Gamma_p(\frac{1}{12}) \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{3}{4}) \Gamma_p(\frac{2}{3})$	-193
	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{3}{4}) \Gamma_p(\frac{1}{4}) \Gamma_p(\frac{1}{2})$	$p \Gamma_p(\frac{3}{4}) \Gamma_p(\frac{1}{4}) \Gamma_p(\frac{1}{2})$	193
	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{1}{12}) \Gamma_p(\frac{3}{4}) \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{2}{3})$	$p \Gamma_p(\frac{1}{12}) \Gamma_p(\frac{3}{4}) \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{2}{3})$	-193
	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{1}{6}) \Gamma_p(\frac{3}{4}) \Gamma_p(\frac{3}{2}) \Gamma_p(\frac{1}{3})$	$p \Gamma_p(\frac{1}{6}) \Gamma_p(\frac{3}{4}) \Gamma_p(\frac{3}{2}) \Gamma_p(\frac{1}{3})$	$55\cdot 193+2\cdot 193^2+\dots$
	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{1}{4}) \Gamma_p(\frac{3}{4}) \Gamma_p(\frac{2}{3})$	$p \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{1}{4}) \Gamma_p(\frac{3}{4}) \Gamma_p(\frac{2}{3})$	193
	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{2}{3})$	$p \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{2}{3})$	193
	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{5}{12}) \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{3}{4}) \Gamma_p(\frac{1}{3})$	$p \Gamma_p(\frac{5}{12}) \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{3}{4}) \Gamma_p(\frac{1}{3})$	-193
	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{3}{4}) \Gamma_p(\frac{1}{4}) \Gamma_p(\frac{2}{3})$	$p \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{3}{4}) \Gamma_p(\frac{1}{4}) \Gamma_p(\frac{2}{3})$	193
	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{5}{12}) \Gamma_p(\frac{3}{4}) \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{1}{3})$	$p \Gamma_p(\frac{5}{12}) \Gamma_p(\frac{3}{4}) \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{1}{3})$	-193
	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{7}{12}) \Gamma_p(\frac{1}{4}) \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{2}{3})$	$p \Gamma_p(\frac{7}{12}) \Gamma_p(\frac{1}{4}) \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{2}{3})$	-193
	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{4}) \Gamma_p(\frac{3}{4}) \Gamma_p(\frac{1}{3})$	$p \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{4}) \Gamma_p(\frac{3}{4}) \Gamma_p(\frac{1}{3})$	193
	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{1}{12}) \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{1}{4}) \Gamma_p(\frac{2}{3})$	$p \Gamma_p(\frac{1}{12}) \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{1}{4}) \Gamma_p(\frac{2}{3})$	-193
	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{1}{3})$	$p \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{1}{3})$	193
	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{3}{4}) \Gamma_p(\frac{1}{4}) \Gamma_p(\frac{1}{3})$	$p \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{3}{4}) \Gamma_p(\frac{1}{4}) \Gamma_p(\frac{1}{3})$	193
	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{5}{6}) \Gamma_p(\frac{1}{4}) \Gamma_p(\frac{1}{4}) \Gamma_p(\frac{1}{3})$	$p \Gamma_p(\frac{5}{6}) \Gamma_p(\frac{1}{4}) \Gamma_p(\frac{1}{4}) \Gamma_p(\frac{1}{3})$	$186\cdot 193+80\cdot 193^2+\dots$
	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{11}{12}) \Gamma_p(\frac{1}{4}) \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{1}{3})$	$p \Gamma_p(\frac{11}{12}) \Gamma_p(\frac{1}{4}) \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{1}{3})$	-193
	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{11}{12}) \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{1}{4}) \Gamma_p(\frac{1}{3})$	$p \Gamma_p(\frac{11}{12}) \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{1}{4}) \Gamma_p(\frac{1}{3})$	-193
$H^{2,0}$	$x_1^4 x_2^3 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$-p^2 \Gamma_p(\frac{5}{6}) \Gamma_p(\frac{3}{4}) \Gamma_p(\frac{3}{4}) \Gamma_p(\frac{2}{3})$	$-p^2 \Gamma_p(\frac{5}{6}) \Gamma_p(\frac{3}{4}) \Gamma_p(\frac{3}{4}) \Gamma_p(\frac{2}{3})$	$62\cdot 193^2+\dots$
$H^{2,2}$	$y_1^3 y_2^3 y_3^3 y_4^3$	p^2	p^2	37249

Surface #2: $Q = (1, 1, 1, 2)$, $m = 5$

Equation: $x_0^5 + x_1^5 + x_2^5 + x_0x_3^2 = 0$

$p = 251$

Table 2: Trace contributions for $x_0^5 + x_1^5 + x_2^5 + x_0x_3^2 = 0$ at $p = 251$.

[illegible]

Surface #3: $Q = (1, 1, 3, 1)$, $m = 6$

Equation: $x_0^6 + x_1^6 + x_2^2 + x_0 x_3^5 = 0$

$p = 1801$

Table 3: Trace contributions for $x_0^6 + x_1^6 + x_2^2 + x_0 x_3^5 = 0$ at $p = 1801$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{2}{15}\right) \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{5}\right)$	$-\Gamma_p\left(\frac{2}{15}\right) \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{5}\right)$	$699+1290 \cdot 1801+1086 \cdot 1801^2+\dots$
$H^{1,1}$	$y_1^3 y_2^3 y_3^1 y_4^3$	p	p	1801
	$x_1^1 x_2^3 x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{1}\right)$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{1}\right)$	1801
	$x_1^1 x_2^4 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{30}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{4}{5}\right)$	$p \Gamma_p\left(\frac{1}{30}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{4}{5}\right)$	$1115 \cdot 1801+14 \cdot 1801^2+\dots$
	$x_1^1 x_2^5 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{15}\right) \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{3}{5}\right)$	$p \Gamma_p\left(\frac{1}{15}\right) \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{3}{5}\right)$	$86 \cdot 1801+290 \cdot 1801^2+\dots$
	$x_1^1 x_2^3 x_3^4 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{5}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{5}\right)$	$p \Gamma_p\left(\frac{1}{5}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{5}\right)$	1801
	$x_1^2 x_2^3 x_3^4 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{3}{30}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{3}{5}\right)$	$p \Gamma_p\left(\frac{3}{30}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{3}{5}\right)$	$1234 \cdot 1801+159 \cdot 1801^2+\dots$
	$x_1^2 x_2^5 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{4}{15}\right) \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{5}\right)$	$p \Gamma_p\left(\frac{4}{15}\right) \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{5}\right)$	$964 \cdot 1801+554 \cdot 1801^2+\dots$
	$x_1^3 x_2^2 x_3^4 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{11}{30}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{4}{5}\right)$	$p \Gamma_p\left(\frac{11}{30}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{4}{5}\right)$	$336 \cdot 1801+705 \cdot 1801^2+\dots$
	$x_1^3 x_2^3 x_3^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{7}{5}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{3}{5}\right)$	$p \Gamma_p\left(\frac{7}{5}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{3}{5}\right)$	1801
	$x_1^3 x_2^4 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{13}{30}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{5}\right)$	$p \Gamma_p\left(\frac{13}{30}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{5}\right)$	$1461 \cdot 1801+1272 \cdot 1801^2+\dots$
	$x_1^1 x_2^5 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{8}{15}\right) \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{5}\right)$	$p \Gamma_p\left(\frac{8}{15}\right) \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{5}\right)$	$1782 \cdot 1801+1020 \cdot 1801^2+\dots$
	$x_1^1 x_2^1 x_3^4 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{17}{15}\right) \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{4}{5}\right)$	$p \Gamma_p\left(\frac{17}{15}\right) \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{4}{5}\right)$	$1327 \cdot 1801+1543 \cdot 1801^2+\dots$
	$x_1^4 x_2^2 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{17}{30}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{3}{5}\right)$	$p \Gamma_p\left(\frac{17}{30}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{3}{5}\right)$	$535 \cdot 1801+1293 \cdot 1801^2+\dots$
	$x_1^4 x_2^3 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{3}{5}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{5}\right)$	$p \Gamma_p\left(\frac{3}{5}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{5}\right)$	1801
	$x_1^4 x_2^4 x_3^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{19}{30}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{5}\right)$	$p \Gamma_p\left(\frac{19}{30}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{5}\right)$	$1667 \cdot 1801+500 \cdot 1801^2+\dots$
	$x_1^5 x_2^1 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{11}{15}\right) \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{3}{5}\right)$	$p \Gamma_p\left(\frac{11}{15}\right) \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{3}{5}\right)$	$312 \cdot 1801+693 \cdot 1801^2+\dots$
	$x_1^5 x_2^2 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{23}{30}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{5}\right)$	$p \Gamma_p\left(\frac{23}{30}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{5}\right)$	$1747 \cdot 1801+816 \cdot 1801^2+\dots$
	$x_1^5 x_2^3 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{4}{5}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{5}\right)$	$p \Gamma_p\left(\frac{4}{5}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{5}\right)$	1801
	$x_1^6 x_2^1 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{14}{15}\right) \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{5}\right)$	$p \Gamma_p\left(\frac{14}{15}\right) \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{5}\right)$	$1445 \cdot 1801+718 \cdot 1801^2+\dots$
	$x_1^6 x_2^2 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{29}{30}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{5}\right)$	$p \Gamma_p\left(\frac{29}{30}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{5}\right)$	$911 \cdot 1801+639 \cdot 1801^2+\dots$
$H^{2,0}$	$x_1^6 x_2^5 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$-p^2 \Gamma_p\left(\frac{13}{15}\right) \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{4}{5}\right)$	$-p^2 \Gamma_p\left(\frac{13}{15}\right) \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{4}{5}\right)$	$1296 \cdot 1801^2+\dots$
$H^{2,2}$	$y_1^5 y_2^5 y_3^1 y_4^5$	p^2	p^2	3243601

Surface #4: $Q = (3, 1, 1, 1)$, $m = 6$

Equation: $x_0^2 + x_1^6 + x_2^6 + x_0 x_3^3 = 0$

$p = 433$

Table 4: Trace contributions for $x_0^2 + x_1^6 + x_2^6 + x_0 x_3^3 = 0$ at $p = 433$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p(\frac{1}{3}) \Gamma_p(\frac{1}{6}) \Gamma_p(\frac{1}{6}) \Gamma_p(\frac{1}{3})$	$-\Gamma_p(\frac{1}{3}) \Gamma_p(\frac{1}{6}) \Gamma_p(\frac{1}{6}) \Gamma_p(\frac{1}{3})$	$4+431\cdot 433+107\cdot 433^2+\dots$
$H^{1,1}$	$y_1^1 y_2^3 y_3^3 y_4^2$	p	p	433
	$x_1^1 x_2^3 x_3^3 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{1}{6}) \Gamma_p(\frac{5}{6}) \Gamma_p(\frac{1}{3})$	$p \Gamma_p(\frac{1}{6}) \Gamma_p(\frac{5}{6}) \Gamma_p(\frac{1}{3})$	433
	$x_1^1 x_2^4 x_3^3 x_4^3 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{3})$	$p \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{3})$	433
	$x_1^1 x_2^5 x_3^3 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{1}{6}) \Gamma_p(\frac{5}{6}) \Gamma_p(\frac{2}{3})$	$p \Gamma_p(\frac{1}{6}) \Gamma_p(\frac{5}{6}) \Gamma_p(\frac{2}{3})$	433
	$x_1^1 x_2^3 x_3^3 x_4^3 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{1}{3})$	$p \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{1}{3})$	433
	$x_1^1 x_2^3 x_3^3 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{1}{6}) \Gamma_p(\frac{5}{6}) \Gamma_p(\frac{2}{3})$	$p \Gamma_p(\frac{1}{6}) \Gamma_p(\frac{5}{6}) \Gamma_p(\frac{2}{3})$	433
	$x_1^1 x_2^3 x_3^5 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{3})$	$p \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{3})$	433
	$x_1^1 x_2^4 x_3^3 x_4^3 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{3})$	$p \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{3})$	433
	$x_1^1 x_2^4 x_3^3 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{1}{6}) \Gamma_p(\frac{5}{6}) \Gamma_p(\frac{2}{3})$	$p \Gamma_p(\frac{1}{6}) \Gamma_p(\frac{5}{6}) \Gamma_p(\frac{2}{3})$	433
	$x_1^1 x_2^4 x_3^5 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{3})$	$p \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{3})$	433
	$x_1^1 x_2^5 x_3^3 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{1}{6}) \Gamma_p(\frac{5}{6}) \Gamma_p(\frac{2}{3})$	$p \Gamma_p(\frac{1}{6}) \Gamma_p(\frac{5}{6}) \Gamma_p(\frac{2}{3})$	433
	$x_1^1 x_2^5 x_3^5 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{3})$	$p \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{3})$	433
	$x_1^1 x_2^5 x_3^3 x_4^3 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{1}{6}) \Gamma_p(\frac{5}{6}) \Gamma_p(\frac{2}{3})$	$p \Gamma_p(\frac{1}{6}) \Gamma_p(\frac{5}{6}) \Gamma_p(\frac{2}{3})$	433
	$x_1^1 x_2^5 x_3^5 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{3})$	$p \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{3})$	433
	$x_1^1 x_2^5 x_3^3 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{1}{6}) \Gamma_p(\frac{5}{6}) \Gamma_p(\frac{2}{3})$	$p \Gamma_p(\frac{1}{6}) \Gamma_p(\frac{5}{6}) \Gamma_p(\frac{2}{3})$	433
	$x_1^1 x_2^5 x_3^5 x_4^3 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{3})$	$p \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{3})$	433
	$x_1^1 x_2^5 x_3^3 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{1}{6}) \Gamma_p(\frac{5}{6}) \Gamma_p(\frac{2}{3})$	$p \Gamma_p(\frac{1}{6}) \Gamma_p(\frac{5}{6}) \Gamma_p(\frac{2}{3})$	433
	$x_1^1 x_2^5 x_3^5 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{3})$	$p \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{3})$	433
	$x_1^1 x_2^5 x_3^3 x_4^3 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{1}{6}) \Gamma_p(\frac{5}{6}) \Gamma_p(\frac{2}{3})$	$p \Gamma_p(\frac{1}{6}) \Gamma_p(\frac{5}{6}) \Gamma_p(\frac{2}{3})$	433
	$x_1^1 x_2^5 x_3^5 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{3})$	$p \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{3})$	433
	$x_1^1 x_2^5 x_3^3 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{1}{6}) \Gamma_p(\frac{5}{6}) \Gamma_p(\frac{2}{3})$	$p \Gamma_p(\frac{1}{6}) \Gamma_p(\frac{5}{6}) \Gamma_p(\frac{2}{3})$	433
	$x_1^1 x_2^5 x_3^5 x_4^3 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{3})$	$p \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{3})$	433
$H^{2,0}$	$x_1^2 x_2^5 x_3^5 x_4^2 e_1 e_2 e_3 e_4$	$-p^2 \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{5}{6}) \Gamma_p(\frac{5}{6}) \Gamma_p(\frac{2}{3})$	$-p^2 \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{5}{6}) \Gamma_p(\frac{5}{6}) \Gamma_p(\frac{2}{3})$	$325\cdot 433^2+\dots$
$H^{2,2}$	$y_1^1 y_2^5 y_3^5 y_4^3$	p^2	p^2	187489

Surface #5: $Q = (1, 2, 2, 1)$, $m = 6$

Equation: $x_0^6 + x_1^3 + x_2^3 + x_0 x_3^5 = 0$

$p = 271$

Table 5: Trace contributions for $x_0^6 + x_1^3 + x_2^3 + x_0 x_3^5 = 0$ at $p = 271$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{2}{15}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{5}\right)$	$-\Gamma_p\left(\frac{2}{15}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{5}\right)$	$252+261\cdot 271+118\cdot 271^2+\dots$
$H^{1,1}$	$y_1^2 y_2^2 y_3^2 y_4^2$	p	p	271
	$y_1^4 y_2^4 y_3^4 y_4^4$	p	p	271
	$x_1^1 x_2^2 x_3^1 y_4^3 e_2 e_3$	$-p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	$-p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	271
	$x_2^2 x_3^1 y_1^3 y_4^3 e_2 e_3$	$-p \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	$-p \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	271
	$x_1^1 x_2^1 x_3^5 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	$p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	271
	$x_1^1 x_2^2 x_3^4 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	$p \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	271
	$x_1^1 x_2^2 x_3^4 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{15}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{3}{5}\right)$	$p \Gamma_p\left(\frac{1}{15}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{3}{5}\right)$	$142\cdot 271+161\cdot 271^2+\dots$
	$x_1^2 x_2^1 x_3^4 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{5}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{4}{5}\right)$	$p \Gamma_p\left(\frac{1}{5}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{4}{5}\right)$	271
	$x_2^2 x_2^1 x_3^4 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{5}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{4}{5}\right)$	$p \Gamma_p\left(\frac{1}{5}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{4}{5}\right)$	271
	$x_2^2 x_2^2 x_3^4 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{4}{15}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{2}{5}\right)$	$p \Gamma_p\left(\frac{4}{15}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{2}{5}\right)$	$141\cdot 271+224\cdot 271^2+\dots$
	$x_3^1 x_2^2 x_3^4 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{2}{5}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{3}{5}\right)$	$p \Gamma_p\left(\frac{2}{5}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{3}{5}\right)$	271
	$x_3^1 x_2^2 x_3^4 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{2}{5}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{3}{5}\right)$	$p \Gamma_p\left(\frac{2}{5}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{3}{5}\right)$	271
	$x_3^2 x_2^2 x_3^4 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{7}{15}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{5}\right)$	$p \Gamma_p\left(\frac{7}{15}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{5}\right)$	$26\cdot 271+194\cdot 271^2+\dots$
	$x_4^1 x_2^1 x_3^4 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{8}{15}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{4}{5}\right)$	$p \Gamma_p\left(\frac{8}{15}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{4}{5}\right)$	$73\cdot 271+70\cdot 271^2+\dots$
	$x_4^1 x_2^1 x_3^4 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{2}{5}\right)$	$p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{2}{5}\right)$	271
	$x_4^1 x_2^2 x_3^4 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{5}\right)$	$p \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{5}\right)$	271
	$x_5^1 x_2^1 x_3^4 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{11}{15}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{3}{5}\right)$	$p \Gamma_p\left(\frac{11}{15}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{3}{5}\right)$	$148\cdot 271+216\cdot 271^2+\dots$
	$x_5^1 x_2^1 x_3^4 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{4}{5}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{5}\right)$	$p \Gamma_p\left(\frac{4}{5}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{5}\right)$	271
	$x_1^1 x_2^2 x_3^4 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{5}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{5}\right)$	$p \Gamma_p\left(\frac{1}{5}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{5}\right)$	271
	$x_1^6 x_2^1 x_3^4 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{14}{15}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{5}\right)$	$p \Gamma_p\left(\frac{14}{15}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{5}\right)$	$21\cdot 271+41\cdot 271^2+\dots$
$H^{2,0}$	$x_1^6 x_2^2 x_3^4 x_4^1 e_1 e_2 e_3 e_4$	$-p^2 \Gamma_p\left(\frac{13}{15}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{4}{5}\right)$	$-p^2 \Gamma_p\left(\frac{13}{15}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{4}{5}\right)$	$57\cdot 271^2+\dots$
$H^{2,2}$	$y_1^5 y_2^2 y_3^2 y_4^5$	p^2	p^2	73441

Surface #6: $Q = (2, 1, 1, 2)$, $m = 6$

Equation: $x_0^3 + x_1^6 + x_2^6 + x_0x_3^2 = 0$

$p = 433$

Table 6: Trace contributions for $x_0^3 + x_1^6 + x_2^6 + x_0x_3^2 = 0$ at $p = 433$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{2}\right)$	$-\Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{2}\right)$	$4+431\cdot 433+107\cdot 433^2+\dots$
$H^{1,1}$	$y_1^1 y_2^4 y_3^4 y_4^1$	p	p	433
	$y_1^2 y_2^2 y_3^2 y_4^2$	p	p	433
	$y_1^1 x_2^2 y_3^3 y_4^1 e_1 e_4$	$-p \Gamma_p\left(\frac{1}{1}\right)$	$-p \Gamma_p\left(\frac{1}{1}\right)$	433
	$x_1^1 x_2^1 x_3^5 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{1}\right)$	$p \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{1}\right)$	433
	$x_1^1 x_2^2 x_3^4 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{1}\right)$	$p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{1}\right)$	433
	$x_1^1 x_2^3 x_3^3 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{1}\right)$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{1}\right)$	433
	$x_1^1 x_2^3 x_3^3 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{2}\right)$	433
	$x_1^1 x_2^4 x_3^2 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{1}\right)$	$p \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{1}\right)$	433
	$x_1^1 x_2^4 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	433
	$x_1^1 x_2^5 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{1}\right)$	$p \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{1}\right)$	433
	$x_1^1 x_2^5 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	433
	$x_1^2 x_2^1 y_3^3 y_4^1 e_1 e_4$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	433
	$x_1^2 x_2^2 x_3^5 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{2}\right)$	433
	$x_1^2 x_2^2 x_3^4 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	433
	$x_1^2 x_2^3 x_3^3 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	433
	$x_1^2 x_2^4 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	433
	$x_1^2 x_2^5 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{2}\right)$	433
	$x_1^3 x_2^1 x_3^3 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	433
	$x_1^3 x_2^2 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	433
	$x_1^3 x_2^3 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{2}\right)$	433
$H^{2,0}$	$x_1^3 x_2^5 x_3^5 x_4^1 e_1 e_2 e_3 e_4$	$-p^2 \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{2}\right)$	$-p^2 \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{2}\right)$	$325\cdot 433^2+\dots$
$H^{2,2}$	$y_1^2 y_2^5 y_3^5 y_4^2$	p^2	p^2	187489

Surface #7: $Q = (2, 1, 2, 1)$, $m = 6$

Equation: $x_0^3 + x_1^6 + x_2^3 + x_0 x_3^4 = 0$

$p = 433$

Table 7: Trace contributions for $x_0^3 + x_1^6 + x_2^3 + x_0 x_3^4 = 0$ at $p = 433$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p(\frac{1}{4}) \Gamma_p(\frac{1}{6}) \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{1}{4})$	$-\Gamma_p(\frac{1}{4}) \Gamma_p(\frac{1}{6}) \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{1}{4})$	$68+365\cdot 433+396\cdot 433^2+\dots$
$H^{1,1}$	$y_1^1 y_2^4 y_3^1 y_4^3$	p	p	433
	$y_1^2 y_2^2 y_3^2 y_4^2$	p	p	433
	$x_1^1 x_2^3 y_3^2 y_4^1 e_1 e_3$	$-p \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{2}{3})$	$-p \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{2}{3})$	433
	$x_1^1 x_2^2 x_3^4 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{4})$	$p \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{4})$	433
	$x_1^1 x_2^3 x_3^3 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{1}{12}) \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{3}{4})$	$p \Gamma_p(\frac{1}{12}) \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{3}{4})$	-433
	$x_1^1 x_2^4 x_3^4 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{1}{4})$	$p \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{1}{4})$	433
	$x_1^1 x_2^4 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{1}{6}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{2})$	$p \Gamma_p(\frac{1}{6}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{2})$	433
	$x_1^1 x_2^5 x_3^3 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{1}{12}) \Gamma_p(\frac{5}{6}) \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{3}{4})$	$p \Gamma_p(\frac{1}{12}) \Gamma_p(\frac{5}{6}) \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{3}{4})$	-433
	$x_1^1 x_2^5 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{1}{4}) \Gamma_p(\frac{5}{6}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{4})$	$p \Gamma_p(\frac{1}{4}) \Gamma_p(\frac{5}{6}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{4})$	$17\cdot 433+208\cdot 433^2+\dots$
	$x_1^2 x_3^1 y_3^2 y_4^1 e_1 e_3$	$-p \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{3})$	$-p \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{3})$	433
	$x_1^2 x_2^2 x_3^4 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{5}{12}) \Gamma_p(\frac{1}{6}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{3}{4})$	$p \Gamma_p(\frac{5}{12}) \Gamma_p(\frac{1}{6}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{3}{4})$	-433
	$x_1^2 x_2^3 x_3^4 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{4})$	$p \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{4})$	433
	$x_1^2 x_2^3 x_3^3 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{5}{12}) \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{3}{4})$	$p \Gamma_p(\frac{5}{12}) \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{3}{4})$	-433
	$x_1^2 x_2^3 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{1}{12}) \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{4})$	$p \Gamma_p(\frac{1}{12}) \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{4})$	-433
	$x_1^2 x_2^4 x_3^4 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{1}{2})$	$p \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{1}{2})$	433
	$x_1^2 x_2^5 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{1}{12}) \Gamma_p(\frac{5}{6}) \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{1}{4})$	$p \Gamma_p(\frac{1}{12}) \Gamma_p(\frac{5}{6}) \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{1}{4})$	-433
	$x_1^3 x_2^2 x_3^4 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{3}{4}) \Gamma_p(\frac{1}{6}) \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{3}{4})$	$p \Gamma_p(\frac{3}{4}) \Gamma_p(\frac{1}{6}) \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{3}{4})$	$51\cdot 433+140\cdot 433^2+\dots$
	$x_1^3 x_2^1 x_3^4 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{11}{12}) \Gamma_p(\frac{1}{6}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{4})$	$p \Gamma_p(\frac{11}{12}) \Gamma_p(\frac{1}{6}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{1}{4})$	-433
	$x_1^3 x_2^2 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{5}{6}) \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{1}{2})$	$p \Gamma_p(\frac{5}{6}) \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{1}{2})$	433
	$x_1^3 x_2^3 x_3^4 e_1 e_2 e_3 e_4$	$p \Gamma_p(\frac{11}{12}) \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{1}{4})$	$p \Gamma_p(\frac{11}{12}) \Gamma_p(\frac{1}{2}) \Gamma_p(\frac{1}{3}) \Gamma_p(\frac{1}{4})$	-433
$H^{2,0}$	$x_1^3 x_2^5 x_3^4 e_1 e_2 e_3 e_4$	$-p^2 \Gamma_p(\frac{3}{4}) \Gamma_p(\frac{5}{6}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{3}{4})$	$-p^2 \Gamma_p(\frac{3}{4}) \Gamma_p(\frac{5}{6}) \Gamma_p(\frac{2}{3}) \Gamma_p(\frac{3}{4})$	$121\cdot 433^2+\dots$
$H^{2,2}$	$y_1^2 y_2^5 y_3^2 y_4^4$	p^2	p^2	187489

Surface #8: $Q = (1, 2, 4, 1)$, $m = 8$

Equation: $x_0^8 + x_1^4 + x_2^2 + x_0 x_3^7 = 0$

$p = 449$

Table 8: Trace contributions for $x_0^8 + x_1^4 + x_2^2 + x_0 x_3^7 = 0$ at $p = 449$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{3}{28}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{7}\right)$	$-\Gamma_p\left(\frac{3}{28}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{7}\right)$	$285+169\cdot 449+128\cdot 449^2+\dots$
$H^{1,1}$	$y_1^3 y_2^3 y_3^1 y_4^3$	p	p	449
	$y_1^5 y_2^3 y_3^1 y_4^5$	p	p	449
	$x_2^3 x_3^1 y_1^4 y_4^2 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	449
	$x_1^1 x_2^3 x_3^1 x_4^7 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{7}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{7}\right)$	449
	$x_1^1 x_2^3 x_3^5 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{28}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{7}\right)$	$p\Gamma_p\left(\frac{1}{28}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{7}\right)$	$315\cdot 449+313\cdot 449^2+\dots$
	$x_2^2 x_2^3 x_3^6 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{6}{7}\right)$	$p\Gamma_p\left(\frac{7}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{6}{7}\right)$	449
	$x_2^2 x_2^3 x_3^4 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{28}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{7}\right)$	$p\Gamma_p\left(\frac{5}{28}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{7}\right)$	$402\cdot 449+132\cdot 449^2+\dots$
	$x_3^3 x_2^2 x_3^5 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{9}{28}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{7}\right)$	$p\Gamma_p\left(\frac{9}{28}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{7}\right)$	449
	$x_3^3 x_2^3 x_3^3 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{11}{28}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{7}\right)$	$p\Gamma_p\left(\frac{11}{28}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{7}\right)$	$349\cdot 449+309\cdot 449^2+\dots$
	$x_4^4 x_2^1 x_3^6 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{11}{28}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{6}{7}\right)$	$p\Gamma_p\left(\frac{11}{28}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{6}{7}\right)$	$364\cdot 449+183\cdot 449^2+\dots$
	$x_4^4 x_2^3 x_3^4 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{7}\right)$	$p\Gamma_p\left(\frac{3}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{7}\right)$	449
	$x_4^4 x_2^3 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{13}{28}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{7}\right)$	$p\Gamma_p\left(\frac{13}{28}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{7}\right)$	$18\cdot 449+438\cdot 449^2+\dots$
	$x_5^5 x_2^1 x_3^5 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{15}{28}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{7}\right)$	$p\Gamma_p\left(\frac{15}{28}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{7}\right)$	$25\cdot 449+115\cdot 449^2+\dots$
	$x_5^5 x_2^2 x_3^3 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{4}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{7}\right)$	$p\Gamma_p\left(\frac{4}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{7}\right)$	449
	$x_1^5 x_2^3 x_3^4 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{17}{28}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{7}\right)$	$p\Gamma_p\left(\frac{17}{28}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{7}\right)$	$206\cdot 449+287\cdot 449^2+\dots$
	$x_1^6 x_2^3 x_3^4 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{19}{28}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{7}\right)$	$p\Gamma_p\left(\frac{19}{28}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{7}\right)$	$220\cdot 449+237\cdot 449^2+\dots$
	$x_6^6 x_2^3 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{7}\right)$	$p\Gamma_p\left(\frac{5}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{7}\right)$	449
	$x_7^7 x_2^1 x_3^4 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{23}{28}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{7}\right)$	$p\Gamma_p\left(\frac{23}{28}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{7}\right)$	$363\cdot 449+415\cdot 449^2+\dots$
	$x_1^7 x_2^2 x_3^4 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{6}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{7}\right)$	$p\Gamma_p\left(\frac{6}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{7}\right)$	449
	$x_1^8 x_2^3 x_3^4 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{27}{28}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{7}\right)$	$p\Gamma_p\left(\frac{27}{28}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{7}\right)$	$258\cdot 449+413\cdot 449^2+\dots$
$H^{2,0}$	$x_1^8 x_2^3 x_3^6 x_4^1 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{25}{28}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{6}{7}\right)$	$-p^2\Gamma_p\left(\frac{25}{28}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{6}{7}\right)$	$167\cdot 449^2+\dots$
$H^{2,2}$	$y_1^7 y_2^3 y_3^1 y_4^7$	p^2	p^2	201601

Surface #9: $Q = (2, 1, 4, 1)$, $m = 8$

Equation: $x_0^4 + x_1^8 + x_2^2 + x_0 x_3^6 = 0$

$p = 769$

Table 9: Trace contributions for $x_0^4 + x_1^8 + x_2^2 + x_0 x_3^6 = 0$ at $p = 769$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{5}{24}\right)\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{6}\right)$	$-\Gamma_p\left(\frac{5}{24}\right)\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{6}\right)$	$683+137\cdot 769+395\cdot 769^2+\dots$
$H^{1,1}$	$y_1^1 y_2^5 y_3^1 y_4^4$	p	p	769
	$y_1^1 y_2^1 y_3^1 y_4^4$	p	p	769
	$x_1^1 x_2^4 x_3^1 x_4^6 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	769
	$x_1^1 x_2^5 x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{24}\right)\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{6}\right)$	$p\Gamma_p\left(\frac{1}{24}\right)\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{6}\right)$	$736\cdot 769+442\cdot 769^2+\dots$
	$x_1^1 x_2^6 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	769
	$x_1^1 x_2^7 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{7}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{7}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	769
	$x_1^2 x_2^1 x_3^1 y_4^3 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	769
	$x_1^2 x_2^3 x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{24}\right)\Gamma_p\left(\frac{3}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{6}\right)$	$p\Gamma_p\left(\frac{7}{24}\right)\Gamma_p\left(\frac{3}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{6}\right)$	$722\cdot 769+431\cdot 769^2+\dots$
	$x_1^2 x_2^4 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	769
	$x_1^2 x_2^5 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{3}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{3}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	769
	$x_1^2 x_2^6 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	769
	$x_1^2 x_2^7 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{11}{24}\right)\Gamma_p\left(\frac{7}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{6}\right)$	$p\Gamma_p\left(\frac{11}{24}\right)\Gamma_p\left(\frac{7}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{6}\right)$	$214\cdot 769+562\cdot 769^2+\dots$
	$x_1^3 x_2^1 x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{13}{24}\right)\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{6}\right)$	$p\Gamma_p\left(\frac{13}{24}\right)\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{6}\right)$	$442\cdot 769+409\cdot 769^2+\dots$
	$x_1^3 x_2^2 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	769
	$x_1^3 x_2^3 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{3}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{3}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	769
	$x_1^3 x_2^4 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	769
	$x_1^3 x_2^5 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{17}{24}\right)\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{6}\right)$	$p\Gamma_p\left(\frac{17}{24}\right)\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{6}\right)$	$589\cdot 769+210\cdot 769^2+\dots$
	$x_1^4 x_2^1 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{8}\right)\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{7}{8}\right)\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	769
	$x_1^4 x_2^2 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	769
	$x_1^4 x_2^3 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{23}{24}\right)\Gamma_p\left(\frac{3}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{6}\right)$	$p\Gamma_p\left(\frac{23}{24}\right)\Gamma_p\left(\frac{3}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{6}\right)$	$233\cdot 769+471\cdot 769^2+\dots$
$H^{2,0}$	$x_1^4 x_2^7 x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{19}{24}\right)\Gamma_p\left(\frac{7}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{6}\right)$	$-p^2\Gamma_p\left(\frac{19}{24}\right)\Gamma_p\left(\frac{7}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{6}\right)$	$152\cdot 769^2+\dots$
$H^{2,2}$	$y_1^3 y_2^7 y_3^1 y_4^6$	p^2	p^2	591361

Surface #12: $Q = (2, 1, 2, 3)$, $m = 8$

Equation: $x_0^4 + x_1^8 + x_2^4 + x_0 x_3^2 = 0$

$p = 257$

Table 12: Trace contributions for $x_0^4 + x_1^8 + x_2^4 + x_0 x_3^2 = 0$ at $p = 257$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{8}\right) \Gamma_p\left(\frac{1}{8}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	$-\Gamma_p\left(\frac{1}{8}\right) \Gamma_p\left(\frac{1}{8}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	$197+134\cdot 257+209\cdot 257^2+\dots$
$H^{1,1}$	$y_1^1 y_2^5 y_3^1 y_4^2$	p	p	257
	$y_1^2 y_2^5 y_3^2 y_4^2$	p	p	257
	$y_1^2 y_2^6 y_3^2 y_4^1$	p	p	257
	$y_1^3 y_2^3 y_3^3 y_4^1$	p	p	257
	$x_1^1 x_2^3 y_3^4 y_4^1 e_1 e_3$	$-p \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{3}{4}\right)$	$-p \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{3}{4}\right)$	257
	$x_1^1 x_2^3 x_3^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	257
	$x_1^1 x_2^4 x_3^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	257
	$x_1^1 x_2^5 x_3^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{5}{8}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{5}{8}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	257
	$x_1^1 x_2^6 x_3^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{3}{8}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{3}{8}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	257
	$x_1^1 x_2^7 x_3^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{8}\right) \Gamma_p\left(\frac{1}{8}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{1}{8}\right) \Gamma_p\left(\frac{1}{8}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	257
	$x_1^2 x_2^4 y_3^4 e_1 e_3$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	257
	$x_1^2 x_2^5 x_3^4 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{3}{8}\right) \Gamma_p\left(\frac{3}{8}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{3}{8}\right) \Gamma_p\left(\frac{3}{8}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	$242\cdot 257+218\cdot 257^2+\dots$
	$x_1^2 x_2^6 x_3^4 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{5}{8}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{5}{8}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	257
	$x_1^2 x_2^7 x_3^4 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{7}{8}\right) \Gamma_p\left(\frac{1}{8}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{7}{8}\right) \Gamma_p\left(\frac{1}{8}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	257
	$x_1^3 x_2^4 y_3^4 e_1 e_3$	$-p \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{4}\right)$	$-p \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{4}\right)$	257
	$x_1^3 x_2^5 x_3^4 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{5}{8}\right) \Gamma_p\left(\frac{1}{8}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{5}{8}\right) \Gamma_p\left(\frac{1}{8}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	257
	$x_1^3 x_2^6 x_3^4 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{5}{8}\right) \Gamma_p\left(\frac{3}{8}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{5}{8}\right) \Gamma_p\left(\frac{3}{8}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	257
	$x_1^3 x_2^7 x_3^4 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{7}{8}\right) \Gamma_p\left(\frac{1}{8}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{7}{8}\right) \Gamma_p\left(\frac{1}{8}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	$137\cdot 257+115\cdot 257^2+\dots$
	$x_1^4 x_2^1 x_3^4 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{8}\right) \Gamma_p\left(\frac{1}{8}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{1}{8}\right) \Gamma_p\left(\frac{1}{8}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	257
	$x_1^4 x_2^2 x_3^4 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{8}\right) \Gamma_p\left(\frac{1}{8}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{1}{8}\right) \Gamma_p\left(\frac{1}{8}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	257
$H^{2,0}$	$x_1^4 x_2^7 x_3^4 e_1 e_2 e_3 e_4$	$-p^2 \Gamma_p\left(\frac{7}{8}\right) \Gamma_p\left(\frac{7}{8}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	$-p^2 \Gamma_p\left(\frac{7}{8}\right) \Gamma_p\left(\frac{7}{8}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	$227\cdot 257^2+\dots$
$H^{2,2}$	$y_1^3 y_2^7 y_3^2 y_4^2$	p^2	p^2	66049

Surface #13: $Q = (1, 1, 3, 4)$, $m = 9$

Equation: $x_0^9 + x_1^9 + x_2^3 + x_0 x_3^2 = 0$

$p = 487$

Table 13: Trace contributions for $x_0^9 + x_1^9 + x_2^3 + x_0 x_3^2 = 0$ at $p = 487$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{18}\right) \Gamma_p\left(\frac{1}{9}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$-\Gamma_p\left(\frac{1}{18}\right) \Gamma_p\left(\frac{1}{9}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$380+472\cdot 487+49\cdot 487^2+\dots$
$H^{1,1}$	$y_1^2 y_2^2 y_3^2 y_4^2$	p	p	487
	$y_1^4 y_2^4 y_3^4 y_4^4$	p	p	487
	$y_1^5 y_2^5 y_3^5 y_4^1$	p	p	487
	$y_1^7 y_2^7 y_3^7 y_4^1$	p	p	487
	$x_1^1 x_2^3 x_3^2 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{1}\right)$	$p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{1}\right)$	487
	$x_1^1 x_2^6 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{3}{3}\right) \Gamma_p\left(\frac{4}{3}\right) \Gamma_p\left(\frac{1}{1}\right)$	$p \Gamma_p\left(\frac{3}{3}\right) \Gamma_p\left(\frac{4}{3}\right) \Gamma_p\left(\frac{1}{1}\right)$	487
	$x_1^1 x_2^7 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{18}\right) \Gamma_p\left(\frac{7}{9}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{1}{18}\right) \Gamma_p\left(\frac{7}{9}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$95\cdot 487+32\cdot 487^2+\dots$
	$x_1^2 x_2^6 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$255\cdot 487+326\cdot 487^2+\dots$
	$x_1^3 x_2^3 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{5}{18}\right) \Gamma_p\left(\frac{5}{9}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{5}{18}\right) \Gamma_p\left(\frac{5}{9}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$155\cdot 487+172\cdot 487^2+\dots$
	$x_1^3 x_2^8 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{5}{18}\right) \Gamma_p\left(\frac{8}{9}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{5}{18}\right) \Gamma_p\left(\frac{8}{9}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$300\cdot 487+317\cdot 487^2+\dots$
	$x_1^4 x_2^4 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{7}{18}\right) \Gamma_p\left(\frac{4}{9}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{7}{18}\right) \Gamma_p\left(\frac{4}{9}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$267\cdot 487+187\cdot 487^2+\dots$
	$x_1^4 x_2^7 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{7}{18}\right) \Gamma_p\left(\frac{7}{9}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{7}{18}\right) \Gamma_p\left(\frac{7}{9}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$218\cdot 487+372\cdot 487^2+\dots$
	$x_1^5 x_2^3 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	-487
	$x_1^5 x_2^6 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{3}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{3}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	-487
	$x_1^6 x_2^2 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{11}{18}\right) \Gamma_p\left(\frac{2}{9}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{11}{18}\right) \Gamma_p\left(\frac{2}{9}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$105\cdot 487+149\cdot 487^2+\dots$
	$x_1^6 x_2^5 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{11}{18}\right) \Gamma_p\left(\frac{5}{9}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{11}{18}\right) \Gamma_p\left(\frac{5}{9}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$228\cdot 487+232\cdot 487^2+\dots$
	$x_1^7 x_2^2 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{13}{18}\right) \Gamma_p\left(\frac{1}{9}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{13}{18}\right) \Gamma_p\left(\frac{1}{9}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$125\cdot 487+267\cdot 487^2+\dots$
	$x_1^7 x_2^4 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{13}{18}\right) \Gamma_p\left(\frac{4}{9}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{13}{18}\right) \Gamma_p\left(\frac{4}{9}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$22\cdot 487+362\cdot 487^2+\dots$
	$x_1^8 x_2^3 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$233\cdot 487+160\cdot 487^2+\dots$
	$x_1^9 x_2^2 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{17}{18}\right) \Gamma_p\left(\frac{2}{9}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{17}{18}\right) \Gamma_p\left(\frac{2}{9}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$446\cdot 487+423\cdot 487^2+\dots$
$H^{2,0}$	$x_1^9 x_2^8 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$-p^2 \Gamma_p\left(\frac{17}{18}\right) \Gamma_p\left(\frac{8}{9}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$-p^2 \Gamma_p\left(\frac{17}{18}\right) \Gamma_p\left(\frac{8}{9}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$355\cdot 487^2+\dots$
$H^{2,2}$	$y_1^8 y_2^8 y_3^2 y_4^2$	p^2	p^2	237169

Surface #14: $Q = (1, 3, 3, 2)$, $m = 9$

Equation: $x_0^9 + x_1^3 + x_2^3 + x_0 x_3^4 = 0$

$p = 1297$

Table 14: Trace contributions for $x_0^9 + x_1^3 + x_2^3 + x_0 x_3^4 = 0$ at $p = 1297$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{12}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{4}\right)$	$-\Gamma_p\left(\frac{1}{12}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{4}\right)$	$1247+1127 \cdot 1297+700 \cdot 1297^2+\dots$
$H^{1,1}$	$y_1^2 y_2^2 y_3^2 y_4^2$	p	p	1297
	$y_1^4 y_2^2 y_3^2 y_4^4$	p	p	1297
	$y_1^5 y_2^2 y_3^2 y_4^1$	p	p	1297
	$y_1^7 y_2^1 y_3^1 y_4^3$	p	p	1297
	$x_2^2 x_3^2 y_1^3 y_4^2 e_2 e_3$	$-p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	$-p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	1297
	$x_1^2 x_3^2 y_1^6 y_4^2 e_2 e_3$	$-p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	$-p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	1297
	$x_2^2 x_3^1 y_1^3 y_4^3 e_2 e_3$	$-p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	$-p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	1297
	$x_2^2 x_3^1 y_1^6 y_4^2 e_2 e_3$	$-p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	$-p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	1297
	$x_1^1 x_2^2 x_3^4 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{4}\right)$	$p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{4}\right)$	1297
	$x_1^1 x_2^2 x_3^4 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{4}\right)$	$p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{4}\right)$	1297
	$x_2^2 x_3^2 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{4}\right)$	$p \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{4}\right)$	$365 \cdot 1297+887 \cdot 1297^2+\dots$
	$x_1^3 x_2^2 x_3^3 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{4}\right)$	$p \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{4}\right)$	1297
	$x_2^3 x_3^2 x_4^3 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{4}\right)$	$p \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{4}\right)$	1297
	$x_4^4 x_2^2 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{5}{12}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{4}\right)$	$p \Gamma_p\left(\frac{5}{12}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{4}\right)$	$830 \cdot 1297+1085 \cdot 1297^2+\dots$
	$x_1^5 x_2^2 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{4}\right)$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{4}\right)$	1297
	$x_1^5 x_2^2 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	1297
	$x_6^6 x_2^2 x_3^3 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{7}{12}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{3}{4}\right)$	$p \Gamma_p\left(\frac{7}{12}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{3}{4}\right)$	$636 \cdot 1297+1248 \cdot 1297^2+\dots$
	$x_7^7 x_2^2 x_3^4 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{4}\right)$	$p \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{4}\right)$	1297
	$x_1^7 x_2^2 x_3^4 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{4}\right)$	$p \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{4}\right)$	1297
	$x_1^8 x_2^2 x_3^4 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$931 \cdot 1297+409 \cdot 1297^2+\dots$
$H^{2,0}$	$x_1^9 x_2^2 x_3^4 e_1 e_2 e_3 e_4$	$-p^2 \Gamma_p\left(\frac{11}{12}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{3}{4}\right)$	$-p^2 \Gamma_p\left(\frac{11}{12}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{3}{4}\right)$	$856 \cdot 1297^2+\dots$
$H^{2,2}$	$y_1^8 y_2^2 y_3^2 y_4^4$	p^2	p^2	1682209

Surface #15: $Q = (3, 1, 3, 2)$, $m = 9$

Equation: $x_0^3 + x_1^9 + x_2^3 + x_0x_3^3 = 0$

$p = 487$

Table 15: Trace contributions for $x_0^3 + x_1^9 + x_2^3 + x_0x_3^3 = 0$ at $p = 487$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{2}{9}\right)\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	$-\Gamma_p\left(\frac{2}{9}\right)\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	$4+19\cdot 487+412\cdot 487^2+\dots$
$H^{1,1}$	$y_1^1 y_2^4 y_3^1 y_4^3$	p	p	487
	$y_1^1 y_2^3 y_3^1 y_4^4$	p	p	487
	$y_1^2 y_2^2 y_3^2 y_4^2$	p	p	487
	$y_1^2 y_2^5 y_3^1 y_4^1$	p	p	487
	$x_1^1 x_2^3 y_3^3 y_4^2 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	487
	$x_1^1 x_2^6 y_3^1 y_4^1 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	487
	$x_1^1 x_2^3 x_3^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{9}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{9}\right)\Gamma_p\left(\frac{1}{3}\right)$	487
	$x_1^1 x_2^2 x_3^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{5}{9}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{5}{9}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$382\cdot 487+101\cdot 487^2+\dots$
	$x_1^1 x_2^6 x_3^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	487
	$x_1^1 x_2^3 x_3^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{9}\right)\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{2}{9}\right)\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	487
	$x_1^1 x_2^8 x_3^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{8}{9}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{8}{9}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	487
	$x_1^2 x_2^3 y_3^3 y_4^2 e_1 e_3$	$-p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	$-p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	487
	$x_1^2 x_2^6 y_3^1 y_4^1 e_1 e_3$	$-p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	$-p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	487
	$x_1^2 x_2^3 x_3^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{4}{9}\right)\Gamma_p\left(\frac{2}{9}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{4}{9}\right)\Gamma_p\left(\frac{2}{9}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$276\cdot 487+61\cdot 487^2+\dots$
	$x_1^2 x_2^6 x_3^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{4}{9}\right)\Gamma_p\left(\frac{4}{9}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{4}{9}\right)\Gamma_p\left(\frac{4}{9}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	487
	$x_1^2 x_2^3 x_3^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{4}{9}\right)\Gamma_p\left(\frac{5}{9}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{4}{9}\right)\Gamma_p\left(\frac{5}{9}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	487
	$x_1^2 x_2^6 x_3^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{8}{9}\right)\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{8}{9}\right)\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	$30\cdot 487+108\cdot 487^2+\dots$
	$x_1^3 x_2^1 x_3^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{8}{9}\right)\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{8}{9}\right)\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	487
	$x_1^3 x_2^2 x_3^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{9}\right)\Gamma_p\left(\frac{2}{9}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{7}{9}\right)\Gamma_p\left(\frac{2}{9}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	487
	$x_1^3 x_2^4 x_3^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{8}{9}\right)\Gamma_p\left(\frac{4}{9}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{8}{9}\right)\Gamma_p\left(\frac{4}{9}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	$269\cdot 487+168\cdot 487^2+\dots$
$H^{2,0}$	$x_1^3 x_2^8 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{7}{9}\right)\Gamma_p\left(\frac{8}{9}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$-p^2\Gamma_p\left(\frac{7}{9}\right)\Gamma_p\left(\frac{8}{9}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$122\cdot 487^2+\dots$
$H^{2,2}$	$y_1^2 y_2^8 y_3^2 y_4^3$	p^2	p^2	237169

Surface #16: $Q = (1, 1, 5, 3)$, $m = 10$

Equation: $x_0^{10} + x_1^{10} + x_2^2 + x_0 x_3^3 = 0$

$p = 601$

Table 16: Trace contributions for $x_0^{10} + x_1^{10} + x_2^2 + x_0 x_3^3 = 0$ at $p = 601$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{15}\right)\Gamma_p\left(\frac{1}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$-\Gamma_p\left(\frac{1}{15}\right)\Gamma_p\left(\frac{1}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$271+282\cdot 601+393\cdot 601^2+\dots$
$H^{1,1}$	$y_1^3 y_2^3 y_3^1 y_4^3$	p	p	601
	$y_1^5 y_2^5 y_3^2 y_4^2$	p	p	601
	$y_1^7 y_2^7 y_3^1 y_4^1$	p	p	601
	$x_1^5 x_2^3 x_3^3 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{1}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{1}\right)$	601
	$x_1^8 x_2^3 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{30}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{1}{30}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$577\cdot 601+183\cdot 601^2+\dots$
	$x_1^2 x_2^7 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{15}\right)\Gamma_p\left(\frac{1}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{1}{15}\right)\Gamma_p\left(\frac{1}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$118\cdot 601+418\cdot 601^2+\dots$
	$x_1^3 x_2^6 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{30}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{7}{30}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$577\cdot 601+183\cdot 601^2+\dots$
	$x_1^9 x_2^3 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{4}{15}\right)\Gamma_p\left(\frac{9}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{4}{15}\right)\Gamma_p\left(\frac{9}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$551\cdot 601+334\cdot 601^2+\dots$
	$x_1^4 x_2^3 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	601
	$x_1^4 x_2^8 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{11}{30}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{11}{30}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$25\cdot 601+417\cdot 601^2+\dots$
	$x_1^5 x_2^4 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{13}{30}\right)\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{13}{30}\right)\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$577\cdot 601+183\cdot 601^2+\dots$
	$x_1^3 x_2^7 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{15}\right)\Gamma_p\left(\frac{7}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{7}{15}\right)\Gamma_p\left(\frac{7}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$451\cdot 601+88\cdot 601^2+\dots$
	$x_1^6 x_2^3 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{8}{15}\right)\Gamma_p\left(\frac{3}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{8}{15}\right)\Gamma_p\left(\frac{3}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$4\cdot 601+383\cdot 601^2+\dots$
	$x_1^6 x_2^6 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{17}{30}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{17}{30}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$25\cdot 601+417\cdot 601^2+\dots$
	$x_1^7 x_2^2 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{19}{30}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{19}{30}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$577\cdot 601+183\cdot 601^2+\dots$
	$x_1^5 x_2^3 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	601
	$x_1^8 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{11}{15}\right)\Gamma_p\left(\frac{1}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{11}{15}\right)\Gamma_p\left(\frac{1}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$12\cdot 601+453\cdot 601^2+\dots$
	$x_1^8 x_2^4 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{23}{30}\right)\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{23}{30}\right)\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$25\cdot 601+417\cdot 601^2+\dots$
	$x_1^9 x_2^3 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{13}{15}\right)\Gamma_p\left(\frac{3}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{13}{15}\right)\Gamma_p\left(\frac{3}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$382\cdot 601+57\cdot 601^2+\dots$
	$x_1^{10} x_2^2 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{29}{30}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{29}{30}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$25\cdot 601+417\cdot 601^2+\dots$
$H^{2,0}$	$x_1^{10} x_2^9 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{14}{15}\right)\Gamma_p\left(\frac{9}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$-p^2\Gamma_p\left(\frac{14}{15}\right)\Gamma_p\left(\frac{9}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$275\cdot 601^2+\dots$
$H^{2,2}$	$y_1^9 y_2^9 y_3^1 y_4^3$	p^2	p^2	361201

Surface #17: $Q = (2, 1, 5, 2)$, $m = 10$

Equation: $x_0^5 + x_1^{10} + x_2^2 + x_0 x_3^4 = 0$

$p = 401$

Table 17: Trace contributions for $x_0^5 + x_1^{10} + x_2^2 + x_0 x_3^4 = 0$ at $p = 401$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{3}{20}\right)\Gamma_p\left(\frac{1}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$-\Gamma_p\left(\frac{3}{20}\right)\Gamma_p\left(\frac{1}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$46+154\cdot 401+158\cdot 401^2+\dots$
$H^{1,1}$	$y_1^2 y_2^1 y_3^1 y_4^2$	p	p	401
	$y_1^3 y_2^3 y_3^1 y_4^3$	p	p	401
	$x_1^1 x_2^4 y_2^5 y_3^1 e_1 e_4$	$-p\Gamma_p\left(\frac{1}{1}\right)$	$-p\Gamma_p\left(\frac{1}{1}\right)$	401
	$x_1^1 x_2^5 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{1}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{1}\right)$	401
	$x_1^1 x_2^7 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{20}\right)\Gamma_p\left(\frac{7}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$p\Gamma_p\left(\frac{1}{20}\right)\Gamma_p\left(\frac{7}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$43\cdot 401+175\cdot 401^2+\dots$
	$x_1^1 x_2^9 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{10}\right)\Gamma_p\left(\frac{9}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{10}\right)\Gamma_p\left(\frac{9}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	401
	$x_1^2 x_2^3 y_2^5 y_3^1 e_1 e_4$	$-p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{3}{4}\right)$	$-p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{3}{4}\right)$	401
	$x_1^2 x_2^5 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	401
	$x_1^2 x_2^7 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{10}\right)\Gamma_p\left(\frac{7}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{3}{10}\right)\Gamma_p\left(\frac{7}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	401
	$x_1^2 x_2^9 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{20}\right)\Gamma_p\left(\frac{9}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$p\Gamma_p\left(\frac{1}{20}\right)\Gamma_p\left(\frac{9}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$191\cdot 401+387\cdot 401^2+\dots$
	$x_1^3 x_2^4 y_2^5 y_3^1 e_1 e_4$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	401
	$x_1^3 x_2^5 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{9}{20}\right)\Gamma_p\left(\frac{3}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$p\Gamma_p\left(\frac{9}{20}\right)\Gamma_p\left(\frac{3}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$149\cdot 401+190\cdot 401^2+\dots$
	$x_1^3 x_2^7 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	401
	$x_1^3 x_2^9 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{11}{20}\right)\Gamma_p\left(\frac{7}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$p\Gamma_p\left(\frac{11}{20}\right)\Gamma_p\left(\frac{7}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$218\cdot 401+144\cdot 401^2+\dots$
	$x_1^4 x_2^4 y_2^5 y_3^1 e_1 e_4$	$-p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{4}\right)$	$-p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{4}\right)$	401
	$x_1^4 x_2^5 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{13}{20}\right)\Gamma_p\left(\frac{1}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$p\Gamma_p\left(\frac{13}{20}\right)\Gamma_p\left(\frac{1}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$21\cdot 401+350\cdot 401^2+\dots$
	$x_1^4 x_2^7 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{10}\right)\Gamma_p\left(\frac{3}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{7}{10}\right)\Gamma_p\left(\frac{3}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	401
	$x_1^4 x_2^9 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	401
	$x_1^5 x_2^1 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{9}{10}\right)\Gamma_p\left(\frac{1}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{9}{10}\right)\Gamma_p\left(\frac{1}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	401
	$x_1^5 x_2^3 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{19}{20}\right)\Gamma_p\left(\frac{3}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$p\Gamma_p\left(\frac{19}{20}\right)\Gamma_p\left(\frac{3}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$28\cdot 401+259\cdot 401^2+\dots$
$H^{2,0}$	$x_1^9 x_2^1 x_3^4 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{17}{20}\right)\Gamma_p\left(\frac{9}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$-p^2\Gamma_p\left(\frac{17}{20}\right)\Gamma_p\left(\frac{9}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$340\cdot 401^2+\dots$
$H^{2,2}$	$y_1^4 y_2^9 y_3^1 y_4^4$	p^2	p^2	160801

Surface #18: $Q = (2, 2, 5, 1)$, $m = 10$

Equation: $x_0^5 + x_1^5 + x_2^2 + x_0x_3^8 = 0$

$p = 401$

Table 18: Trace contributions for $x_0^5 + x_1^5 + x_2^2 + x_0x_3^8 = 0$ at $p = 401$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{7}{40}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{8}\right)$	$-\Gamma_p\left(\frac{7}{40}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{8}\right)$	$28+21\cdot 401+363\cdot 401^2+\dots$
$H^{1,1}$	$y_1^2 y_2^2 y_3^1 y_4^6$	p	p	401
	$y_1^3 y_2^3 y_3^1 y_4^3$	p	p	401
	$x_1^1 x_2^3 x_3^1 x_4^7 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{40}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{8}\right)$	$p\Gamma_p\left(\frac{1}{40}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{8}\right)$	$345\cdot 401+127\cdot 401^2+\dots$
	$x_1^4 x_2^1 y_3^1 y_4^4 e_1 e_2$	$-p\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{4}{5}\right)$	$-p\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{4}{5}\right)$	401
	$x_1^1 x_2^2 x_3^1 x_4^7 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{40}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{8}\right)$	$p\Gamma_p\left(\frac{3}{40}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{8}\right)$	$228\cdot 401+99\cdot 401^2+\dots$
	$x_1^2 x_2^2 x_3^1 x_4^7 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{40}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{8}\right)$	$p\Gamma_p\left(\frac{3}{40}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{8}\right)$	$346\cdot 401+168\cdot 401^2+\dots$
	$x_1^2 x_2^3 y_3^1 y_4^4 e_1 e_2$	$-p\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{3}{5}\right)$	$-p\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{3}{5}\right)$	401
	$x_1^2 x_2^3 x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{11}{40}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{8}\right)$	$p\Gamma_p\left(\frac{11}{40}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{8}\right)$	$64\cdot 401+107\cdot 401^2+\dots$
	$x_1^2 x_2^4 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{13}{40}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{8}\right)$	$p\Gamma_p\left(\frac{13}{40}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{8}\right)$	$297\cdot 401+252\cdot 401^2+\dots$
	$x_1^3 x_2^1 x_3^1 x_4^7 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{17}{40}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{8}\right)$	$p\Gamma_p\left(\frac{17}{40}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{8}\right)$	$377\cdot 401+86\cdot 401^2+\dots$
	$x_1^3 x_2^2 y_3^1 y_4^4 e_1 e_2$	$-p\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{2}{5}\right)$	$-p\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{2}{5}\right)$	401
	$x_1^3 x_2^2 x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{19}{40}\right)\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{8}\right)$	$p\Gamma_p\left(\frac{19}{40}\right)\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{8}\right)$	$223\cdot 401+160\cdot 401^2+\dots$
	$x_1^3 x_2^3 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{21}{40}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{8}\right)$	$p\Gamma_p\left(\frac{21}{40}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{8}\right)$	$205\cdot 401+257\cdot 401^2+\dots$
	$x_1^3 x_2^4 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{23}{40}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{8}\right)$	$p\Gamma_p\left(\frac{23}{40}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{8}\right)$	$284\cdot 401+43\cdot 401^2+\dots$
	$x_1^4 x_2^1 y_3^1 y_4^4 e_1 e_2$	$-p\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{5}\right)$	$-p\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{5}\right)$	401
	$x_1^4 x_2^1 x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{27}{40}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{8}\right)$	$p\Gamma_p\left(\frac{27}{40}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{8}\right)$	$374\cdot 401+211\cdot 401^2+\dots$
	$x_1^4 x_2^2 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{29}{40}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{8}\right)$	$p\Gamma_p\left(\frac{29}{40}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{8}\right)$	$94\cdot 401+300\cdot 401^2+\dots$
	$x_1^4 x_2^3 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{31}{40}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{8}\right)$	$p\Gamma_p\left(\frac{31}{40}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{8}\right)$	$226\cdot 401+271\cdot 401^2+\dots$
	$x_1^5 x_2^1 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{37}{40}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{8}\right)$	$p\Gamma_p\left(\frac{37}{40}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{8}\right)$	$350\cdot 401+67\cdot 401^2+\dots$
	$x_1^5 x_2^2 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{39}{40}\right)\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{8}\right)$	$p\Gamma_p\left(\frac{39}{40}\right)\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{8}\right)$	$179\cdot 401+244\cdot 401^2+\dots$
$H^{2,0}$	$x_1^5 x_2^4 x_3^1 x_4^7 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{33}{40}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{8}\right)$	$-p^2\Gamma_p\left(\frac{33}{40}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{8}\right)$	$43\cdot 401^2+\dots$
$H^{2,2}$	$y_1^4 y_2^4 y_3^1 y_4^8$	p^2	p^2	160801

Surface #19: $Q = (5, 2, 2, 1)$, $m = 10$

Equation: $x_0^2 + x_1^5 + x_2^5 + x_0x_3^5 = 0$

$p = 251$

Table 19: Trace contributions for $x_0^2 + x_1^5 + x_2^5 + x_0x_3^5 = 0$ at $p = 251$.

[illegible]

Surface #20: $Q = (1, 4, 6, 1)$, $m = 12$

Equation: $x_0^{12} + x_1^3 + x_2^2 + x_0 x_3^{11} = 0$

$p = 2377$

Table 20: Trace contributions for $x_0^{12} + x_1^3 + x_2^2 + x_0 x_3^{11} = 0$ at $p = 2377$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{5}{66}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{11}\right)$	$-\Gamma_p\left(\frac{5}{66}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{11}\right)$	$2000+1978\cdot 2377+1766\cdot 2377^2+\dots$
$H^{1,1}$	$y_1^5 y_2^2 y_3^1 y_4^5$	p	p	2377
	$y_1^1 y_2^2 y_3^1 y_4^7$	p	p	2377
	$x_1^1 x_2^2 x_3^1 x_4^9 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{66}\right)\Gamma_p\left(\frac{2}{33}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{9}{11}\right)$	$p\Gamma_p\left(\frac{1}{66}\right)\Gamma_p\left(\frac{2}{33}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{9}{11}\right)$	$228\cdot 2377+2005\cdot 2377^2+\dots$
	$x_1^2 x_2^2 x_3^1 x_4^8 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{66}\right)\Gamma_p\left(\frac{2}{33}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{8}{11}\right)$	$p\Gamma_p\left(\frac{7}{66}\right)\Gamma_p\left(\frac{2}{33}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{8}{11}\right)$	$1141\cdot 2377+2278\cdot 2377^2+\dots$
	$x_1^3 x_2^2 x_3^1 x_4^7 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{13}{66}\right)\Gamma_p\left(\frac{2}{33}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{11}\right)$	$p\Gamma_p\left(\frac{13}{66}\right)\Gamma_p\left(\frac{2}{33}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{11}\right)$	$1571\cdot 2377+1804\cdot 2377^2+\dots$
	$x_1^4 x_2^1 x_3^1 x_4^{10} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{17}{66}\right)\Gamma_p\left(\frac{1}{33}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{10}{11}\right)$	$p\Gamma_p\left(\frac{17}{66}\right)\Gamma_p\left(\frac{1}{33}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{10}{11}\right)$	$2022\cdot 2377+1785\cdot 2377^2+\dots$
	$x_1^4 x_2^2 x_3^1 x_4^6 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{19}{66}\right)\Gamma_p\left(\frac{2}{33}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{6}{11}\right)$	$p\Gamma_p\left(\frac{19}{66}\right)\Gamma_p\left(\frac{2}{33}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{6}{11}\right)$	$1402\cdot 2377+740\cdot 2377^2+\dots$
	$x_1^5 x_2^1 x_3^1 x_4^9 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{23}{66}\right)\Gamma_p\left(\frac{1}{33}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{9}{11}\right)$	$p\Gamma_p\left(\frac{23}{66}\right)\Gamma_p\left(\frac{1}{33}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{9}{11}\right)$	$976\cdot 2377+2075\cdot 2377^2+\dots$
	$x_1^5 x_2^2 x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{25}{66}\right)\Gamma_p\left(\frac{2}{33}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{11}\right)$	$p\Gamma_p\left(\frac{25}{66}\right)\Gamma_p\left(\frac{2}{33}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{11}\right)$	$767\cdot 2377+1431\cdot 2377^2+\dots$
	$x_1^6 x_2^1 x_3^1 x_4^8 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{29}{66}\right)\Gamma_p\left(\frac{1}{33}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{8}{11}\right)$	$p\Gamma_p\left(\frac{29}{66}\right)\Gamma_p\left(\frac{1}{33}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{8}{11}\right)$	$2371\cdot 2377+1090\cdot 2377^2+\dots$
	$x_1^6 x_2^2 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{31}{66}\right)\Gamma_p\left(\frac{2}{33}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{11}\right)$	$p\Gamma_p\left(\frac{31}{66}\right)\Gamma_p\left(\frac{2}{33}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{11}\right)$	$1884\cdot 2377+1199\cdot 2377^2+\dots$
	$x_1^7 x_2^1 x_3^1 x_4^7 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{35}{66}\right)\Gamma_p\left(\frac{1}{33}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{11}\right)$	$p\Gamma_p\left(\frac{35}{66}\right)\Gamma_p\left(\frac{1}{33}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{11}\right)$	$135\cdot 2377+2180\cdot 2377^2+\dots$
	$x_1^7 x_2^2 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{37}{66}\right)\Gamma_p\left(\frac{2}{33}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{11}\right)$	$p\Gamma_p\left(\frac{37}{66}\right)\Gamma_p\left(\frac{2}{33}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{11}\right)$	$396\cdot 2377+1092\cdot 2377^2+\dots$
	$x_1^8 x_2^1 x_3^1 x_4^6 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{41}{66}\right)\Gamma_p\left(\frac{1}{33}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{6}{11}\right)$	$p\Gamma_p\left(\frac{41}{66}\right)\Gamma_p\left(\frac{1}{33}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{6}{11}\right)$	$344\cdot 2377+889\cdot 2377^2+\dots$
	$x_1^8 x_2^2 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{43}{66}\right)\Gamma_p\left(\frac{2}{33}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{11}\right)$	$p\Gamma_p\left(\frac{43}{66}\right)\Gamma_p\left(\frac{2}{33}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{11}\right)$	$1264\cdot 2377+1752\cdot 2377^2+\dots$
	$x_1^9 x_2^1 x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{47}{66}\right)\Gamma_p\left(\frac{1}{33}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{11}\right)$	$p\Gamma_p\left(\frac{47}{66}\right)\Gamma_p\left(\frac{1}{33}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{11}\right)$	$2043\cdot 2377+150\cdot 2377^2+\dots$
	$x_1^9 x_2^2 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{53}{66}\right)\Gamma_p\left(\frac{2}{33}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{11}\right)$	$p\Gamma_p\left(\frac{53}{66}\right)\Gamma_p\left(\frac{2}{33}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{11}\right)$	$154\cdot 2377+152\cdot 2377^2+\dots$
	$x_1^{10} x_2^1 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{59}{66}\right)\Gamma_p\left(\frac{1}{33}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{11}\right)$	$p\Gamma_p\left(\frac{59}{66}\right)\Gamma_p\left(\frac{1}{33}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{11}\right)$	$2203\cdot 2377+2184\cdot 2377^2+\dots$
	$x_1^{11} x_2^1 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{59}{66}\right)\Gamma_p\left(\frac{1}{33}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{11}\right)$	$p\Gamma_p\left(\frac{59}{66}\right)\Gamma_p\left(\frac{1}{33}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{11}\right)$	$25\cdot 2377+2150\cdot 2377^2+\dots$
	$x_1^{12} x_2^1 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{65}{66}\right)\Gamma_p\left(\frac{1}{33}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{11}\right)$	$p\Gamma_p\left(\frac{65}{66}\right)\Gamma_p\left(\frac{1}{33}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{11}\right)$	$490\cdot 2377+2165\cdot 2377^2+\dots$
$H^{2,0}$	$x_1^{12} x_2^2 x_3^1 x_4^{10} e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{61}{66}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{10}{11}\right)$	$-p^2\Gamma_p\left(\frac{61}{66}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{10}{11}\right)$	$2005\cdot 2377^2+\dots$
$H^{2,2}$	$y_1^{11} y_2^2 y_3^1 y_4^{11}$	p^2	p^2	5650129

Surface #21: $Q = (4, 1, 6, 1)$, $m = 12$

Equation: $x_0^3 + x_1^{12} + x_2^2 + x_0 x_3^8 = 0$

$p = 577$

Table 21: Trace contributions for $x_0^3 + x_1^{12} + x_2^2 + x_0 x_3^8 = 0$ at $p = 577$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{7}{24}\right)\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{8}\right)$	$-\Gamma_p\left(\frac{7}{24}\right)\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{8}\right)$	$34+553\cdot 577+175\cdot 577^2+\dots$
$H^{1,1}$	$y_1^1 y_2^7 y_3^1 y_4^5$	p	p	577
	$y_1^2 y_2^5 y_3^1 y_4^4$	p	p	577
	$x_1^1 x_2^6 x_3^1 x_4^8 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{1}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{1}\right)$	577
	$x_1^1 x_2^7 x_3^1 x_4^7 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{24}\right)\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{8}\right)$	$p\Gamma_p\left(\frac{1}{24}\right)\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{8}\right)$	$288\cdot 577+131\cdot 577^2+\dots$
	$x_1^1 x_2^8 x_3^1 x_4^6 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$p\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	577
	$x_1^1 x_2^9 x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{8}\right)$	$p\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{8}\right)$	577
	$x_1^1 x_2^{10} x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^1 x_2^{11} x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{24}\right)\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{8}\right)$	$p\Gamma_p\left(\frac{5}{24}\right)\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{8}\right)$	$539\cdot 577+503\cdot 577^2+\dots$
	$x_1^2 x_2^3 x_3^1 x_4^6 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{8}\right)$	$p\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{8}\right)$	577
	$x_1^2 x_2^4 x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$p\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	577
	$x_1^2 x_2^5 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{11}{24}\right)\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{8}\right)$	$p\Gamma_p\left(\frac{11}{24}\right)\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{8}\right)$	$138\cdot 577+361\cdot 577^2+\dots$
	$x_1^2 x_2^6 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^2 x_2^7 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{13}{24}\right)\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{8}\right)$	$p\Gamma_p\left(\frac{13}{24}\right)\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{8}\right)$	$46\cdot 577+143\cdot 577^2+\dots$
	$x_1^2 x_2^8 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$p\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	577
	$x_1^2 x_2^9 x_3^1 x_4^0 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{8}\right)$	$p\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{8}\right)$	577
	$x_1^3 x_2^2 x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{19}{24}\right)\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{8}\right)$	$p\Gamma_p\left(\frac{19}{24}\right)\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{8}\right)$	$167\cdot 577+347\cdot 577^2+\dots$
	$x_1^3 x_2^3 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^3 x_2^4 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{8}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{8}\right)$	$p\Gamma_p\left(\frac{7}{8}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{8}\right)$	577
	$x_1^3 x_2^5 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$p\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	577
	$x_1^3 x_2^6 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{23}{24}\right)\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{8}\right)$	$p\Gamma_p\left(\frac{23}{24}\right)\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{8}\right)$	$575\cdot 577+50\cdot 577^2+\dots$
$H^{2,0}$	$x_1^3 x_2^{11} x_3^7 x_4^1 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{17}{24}\right)\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{8}\right)$	$-p^2\Gamma_p\left(\frac{17}{24}\right)\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{8}\right)$	$17\cdot 577^2+\dots$
$H^{2,2}$	$y_1^2 y_2^{11} y_3^1 y_4^8$	p^2	p^2	332929

Surface #22: $Q = (6, 1, 4, 1)$, $m = 12$

Equation: $x_0^2 + x_1^{12} + x_2^3 + x_0 x_3^6 = 0$

$p = 433$

Table 22: Trace contributions for $x_0^2 + x_1^{12} + x_2^3 + x_0 x_3^6 = 0$ at $p = 433$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{6}\right)$	$-\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{6}\right)$	$68+365\cdot 433+396\cdot 433^2+\dots$
$H^{1,1}$	$y_1^1 y_2^5 y_3^2 y_4^3$	p	p	433
	$y_1^1 y_2^2 y_3^3 y_4^4$	p	p	433
	$x_1^1 x_2^4 x_3^2 x_4^6 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{1}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{1}\right)$	433
	$x_1^1 x_2^5 x_3^2 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{5}{6}\right)$	$p\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{5}{6}\right)$	$17\cdot 433+208\cdot 433^2+\dots$
	$x_1^1 x_2^5 x_3^2 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	433
	$x_1^1 x_2^7 x_3^2 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	-433
	$x_1^1 x_2^8 x_3^2 x_4^6 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{1}\right)$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{1}\right)$	433
	$x_1^1 x_2^8 x_3^2 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	433
	$x_1^1 x_2^9 x_3^2 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{6}\right)$	$p\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{6}\right)$	-433
	$x_1^1 x_2^9 x_3^2 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{5}{6}\right)$	$p\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{5}{6}\right)$	-433
	$x_1^1 x_2^{10} x_3^2 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	433
	$x_1^1 x_2^{11} x_3^2 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	-433
	$x_1^2 x_2^1 x_3^2 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	-433
	$x_1^2 x_2^2 x_3^2 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	433
	$x_1^2 x_2^3 x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{5}{6}\right)$	$p\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{5}{6}\right)$	-433
	$x_1^2 x_2^3 x_3^2 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{6}\right)$	$p\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{6}\right)$	-433
	$x_1^2 x_2^4 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	433
	$x_1^2 x_2^5 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	-433
	$x_1^2 x_2^6 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	433
	$x_1^2 x_2^7 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{6}\right)$	$p\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{6}\right)$	$51\cdot 433+140\cdot 433^2+\dots$
$H^{2,0}$	$x_1^2 x_2^{11} x_3^2 x_4^5 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{5}{6}\right)$	$-p^2\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{5}{6}\right)$	$121\cdot 433^2+\dots$
$H^{2,2}$	$y_1^1 y_2^{11} y_3^2 y_4^6$	p^2	p^2	187489

Surface #23: $Q = (2, 3, 6, 1)$, $m = 12$

Equation: $x_0^6 + x_1^4 + x_2^2 + x_0 x_3^{10} = 0$

$p = 3361$

Table 23: Trace contributions for $x_0^6 + x_1^4 + x_2^2 + x_0 x_3^{10} = 0$ at $p = 3361$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{3}{20}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{10}\right)$	$-\Gamma_p\left(\frac{3}{20}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{10}\right)$	$247+3286\cdot 3361+2101\cdot 3361^2+\dots$
$H^{1,1}$	$y_1^1 y_2^3 y_3^1 y_4^6$	p	p	3361
	$y_1^3 y_2^1 y_3^1 y_4^8$	p	p	3361
	$y_1^3 y_2^3 y_3^1 y_4^3$	p	p	3361
	$y_1^5 y_2^1 y_3^1 y_4^5$	p	p	3361
	$x_2^2 x_3^1 y_1^1 y_4^7 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	3361
	$x_2^2 x_3^1 y_1^1 y_4^4 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	3361
	$x_1^1 x_2^2 x_3^1 x_4^{10} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{10}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{10}\right)$	3361
	$x_1^1 x_2^3 x_3^1 x_4^7 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{20}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{10}\right)$	$p\Gamma_p\left(\frac{1}{20}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{10}\right)$	$3104\cdot 3361+1806\cdot 3361^2+\dots$
	$x_2^2 x_2^2 x_3^1 x_4^8 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{5}\right)$	$p\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{5}\right)$	3361
	$x_2^2 x_2^2 x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	3361
	$x_1^3 x_3^1 y_2^2 y_4^5 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	3361
	$x_2^3 x_1^1 x_3^1 x_4^9 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{20}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{9}{10}\right)$	$p\Gamma_p\left(\frac{7}{20}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{9}{10}\right)$	$1153\cdot 3361+1655\cdot 3361^2+\dots$
	$x_2^3 x_2^2 x_3^1 x_4^6 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{5}\right)$	$p\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{5}\right)$	3361
	$x_2^3 x_3^2 x_1^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{9}{20}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{10}\right)$	$p\Gamma_p\left(\frac{9}{20}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{10}\right)$	$1326\cdot 3361+302\cdot 3361^2+\dots$
	$x_1^4 x_2^1 x_3^1 x_4^7 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{11}{20}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{10}\right)$	$p\Gamma_p\left(\frac{11}{20}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{10}\right)$	$621\cdot 3361+890\cdot 3361^2+\dots$
	$x_1^4 x_2^2 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{5}\right)$	$p\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{5}\right)$	3361
	$x_1^4 x_2^3 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{13}{20}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{10}\right)$	$p\Gamma_p\left(\frac{13}{20}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{10}\right)$	$2778\cdot 3361+1865\cdot 3361^2+\dots$
	$x_2^5 x_2^1 x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	3361
	$x_1^5 x_2^2 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{5}\right)$	$p\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{5}\right)$	3361
	$x_1^6 x_2^1 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{19}{20}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{10}\right)$	$p\Gamma_p\left(\frac{19}{20}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{10}\right)$	$1177\cdot 3361+1841\cdot 3361^2+\dots$
$H^{2,0}$	$x_1^6 x_2^3 x_3^1 x_4^9 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{17}{20}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{9}{10}\right)$	$-p^2\Gamma_p\left(\frac{17}{20}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{9}{10}\right)$	$2980\cdot 3361^2+\dots$
$H^{2,2}$	$y_1^5 y_2^3 y_3^1 y_4^{10}$	p^2	p^2	11296321

Surface #24: $Q = (3, 2, 6, 1)$, $m = 12$

Equation: $x_0^4 + x_1^6 + x_2^2 + x_0x_3^9 = 0$

$p = 433$

Table 24: Trace contributions for $x_0^4 + x_1^6 + x_2^2 + x_0x_3^9 = 0$ at $p = 433$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{2}{9}\right)\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{9}\right)$	$-\Gamma_p\left(\frac{2}{9}\right)\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{9}\right)$	$10+385\cdot 433+307\cdot 433^2+\dots$
$H^{1,1}$	$y_1^1 y_2^3 y_3^1 y_4^7$	p	p	433
	$y_1^1 y_2^5 y_3^1 y_4^4$	p	p	433
	$y_1^3 y_2^1 y_3^1 y_4^6$	p	p	433
	$y_1^3 y_2^3 y_3^1 y_4^3$	p	p	433
	$x_2^1 x_3^1 y_1^2 y_4^5 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	433
	$x_1^1 x_2^1 x_3^1 x_4^9 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{1}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{1}\right)$	433
	$x_1^1 x_2^4 x_3^1 x_4^7 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{18}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{9}\right)$	$p\Gamma_p\left(\frac{1}{18}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{9}\right)$	433
	$x_1^1 x_2^5 x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{9}\right)$	$p\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{9}\right)$	$21\cdot 433+223\cdot 433^2+\dots$
	$x_2^1 x_3^1 y_1^2 y_4^6 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	433
	$x_2^1 x_3^1 y_1^4 y_4^3 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	433
	$x_1^2 x_2^1 x_3^1 x_4^8 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{18}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{8}{9}\right)$	$p\Gamma_p\left(\frac{5}{18}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{8}{9}\right)$	433
	$x_1^2 x_2^3 x_3^1 x_4^6 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	433
	$x_1^2 x_2^4 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{18}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{9}\right)$	$p\Gamma_p\left(\frac{7}{18}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{9}\right)$	433
	$x_1^2 x_2^5 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{4}{9}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{9}\right)$	$p\Gamma_p\left(\frac{4}{9}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{9}\right)$	$196\cdot 433+26\cdot 433^2+\dots$
	$x_1^3 x_2^1 x_3^1 x_4^7 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{9}\right)\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{9}\right)$	$p\Gamma_p\left(\frac{5}{9}\right)\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{9}\right)$	$95\cdot 433+281\cdot 433^2+\dots$
	$x_1^3 x_2^2 x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{11}{18}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{9}\right)$	$p\Gamma_p\left(\frac{11}{18}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{9}\right)$	433
	$x_1^3 x_2^3 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	433
	$x_1^3 x_2^4 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{13}{18}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{9}\right)$	$p\Gamma_p\left(\frac{13}{18}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{9}\right)$	433
	$x_1^4 x_2^1 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{8}{9}\right)\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{9}\right)$	$p\Gamma_p\left(\frac{8}{9}\right)\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{9}\right)$	$165\cdot 433+330\cdot 433^2+\dots$
	$x_1^4 x_2^2 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{17}{18}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{9}\right)$	$p\Gamma_p\left(\frac{17}{18}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{9}\right)$	433
$H^{2,0}$	$x_1^4 x_2^5 x_3^1 x_4^8 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{7}{9}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{8}{9}\right)$	$-p^2\Gamma_p\left(\frac{7}{9}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{8}{9}\right)$	$130\cdot 433^2+\dots$
$H^{2,2}$	$y_1^3 y_2^5 y_3^1 y_4^9$	p^2	p^2	187489

Surface #25: $Q = (6, 1, 2, 3)$, $m = 12$

Equation: $x_0^2 + x_1^{12} + x_2^6 + x_0 x_3^2 = 0$

$p = 577$

Table 25: Trace contributions for $x_0^2 + x_1^{12} + x_2^6 + x_0 x_3^2 = 0$ at $p = 577$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{12}\right) \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{2}\right)$	$-\Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{12}\right) \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{2}\right)$	$485+298\cdot 577+181\cdot 577^2+\dots$
$H^{1,1}$	$y_1^1 y_2^3 y_3^3 y_4^2$	p	p	577
	$y_1^1 y_2^5 y_3^3 y_4^1$	p	p	577
	$y_1^1 y_2^7 y_3^1 y_4^2$	p	p	577
	$y_1^1 y_2^9 y_3^3 y_4^1$	p	p	577
	$x_1^1 x_2^4 y_3^4 y_4^1 e_1 e_4$	$-p \Gamma_p\left(\frac{1}{1}\right)$	$-p \Gamma_p\left(\frac{1}{1}\right)$	577
	$x_1^1 x_2^8 y_3^2 e_1 e_4$	$-p \Gamma_p\left(\frac{1}{1}\right)$	$-p \Gamma_p\left(\frac{1}{1}\right)$	577
	$x_1^1 x_3^6 y_2^1 e_1 e_3$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^1 x_2^5 x_3^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{1}\right)$	$p \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{1}\right)$	577
	$x_1^1 x_2^4 x_3^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{1}\right)$	$p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{1}\right)$	577
	$x_1^1 x_2^5 x_3^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{5}{12}\right) \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{5}{12}\right) \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{2}\right)$	$301\cdot 577+545\cdot 577^2+\dots$
	$x_1^1 x_2^3 x_3^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{1}\right)$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{1}\right)$	577
	$x_1^1 x_2^4 x_3^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{1}\right) \Gamma_p\left(\frac{7}{12}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{1}{1}\right) \Gamma_p\left(\frac{7}{12}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^1 x_2^8 x_3^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{1}\right)$	$p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{1}\right)$	577
	$x_1^1 x_2^9 x_3^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^1 x_2^{10} x_3^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{1}\right)$	$p \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{1}\right)$	577
	$x_1^1 x_2^{11} x_3^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{1}\right) \Gamma_p\left(\frac{11}{12}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{1}{1}\right) \Gamma_p\left(\frac{11}{12}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^2 x_2^4 x_3^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{12}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{12}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^2 x_2^3 x_3^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^2 x_2^5 x_3^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{5}{12}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{5}{12}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^2 x_2^7 x_3^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{7}{12}\right) \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{7}{12}\right) \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{2}\right)$	$554\cdot 577+495\cdot 577^2+\dots$
$H^{2,0}$	$x_1^2 x_2^{11} x_3^1 e_1 e_2 e_3 e_4$	$-p^2 \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{11}{12}\right) \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{2}\right)$	$-p^2 \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{11}{12}\right) \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{2}\right)$	$508\cdot 577^2+\dots$
$H^{2,2}$	$y_1^1 y_2^{11} y_3^5 y_4^2$	p^2	p^2	332929

Surface #26: $Q = (6, 1, 3, 2)$, $m = 12$

Equation: $x_0^2 + x_1^{12} + x_2^4 + x_0 x_3^3 = 0$

$p = 577$

Table 26: Trace contributions for $x_0^2 + x_1^{12} + x_2^4 + x_0 x_3^3 = 0$ at $p = 577$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{3}\right)$	$-\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{3}\right)$	$22+73\cdot 577+361\cdot 577^2+\dots$
$H^{1,1}$	$y_1^1 y_2^3 y_3^3 y_4^2$	p	p	577
	$y_1^1 y_2^3 y_3^1 y_4^3$	p	p	577
	$y_1^1 y_2^3 y_3^3 y_4^1$	p	p	577
	$y_1^1 y_2^9 y_3^1 y_4^2$	p	p	577
	$x_1^1 x_2^3 y_2^6 y_3^2 e_1 e_4$	$-p\Gamma_p\left(\frac{1}{4}\right)$	$-p\Gamma_p\left(\frac{1}{4}\right)$	577
	$x_1^1 x_2^3 y_2^4 y_4^2 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^1 x_2^3 y_2^8 y_4^1 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^1 x_2^3 x_3^3 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{4}\right)$	$p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{4}\right)$	577
	$x_1^1 x_2^3 x_3^3 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{2}{3}\right)$	$363\cdot 577+487\cdot 577^2+\dots$
	$x_1^1 x_2^3 x_3^3 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	577
	$x_1^1 x_2^3 x_3^3 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{3}\right)$	$294\cdot 577+309\cdot 577^2+\dots$
	$x_1^1 x_2^3 x_3^3 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$363\cdot 577+487\cdot 577^2+\dots$
	$x_1^1 x_2^3 x_3^3 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^1 x_2^3 x_3^3 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$213\cdot 577+89\cdot 577^2+\dots$
	$x_1^1 x_2^3 x_3^3 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{2}{3}\right)$	$363\cdot 577+487\cdot 577^2+\dots$
	$x_1^2 x_2^3 x_3^3 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{3}\right)$	$213\cdot 577+89\cdot 577^2+\dots$
	$x_1^2 x_2^3 x_3^3 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{3}\right)\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{5}{3}\right)\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$363\cdot 577+487\cdot 577^2+\dots$
	$x_1^2 x_2^3 x_3^3 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$213\cdot 577+89\cdot 577^2+\dots$
	$x_1^2 x_2^3 x_3^3 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{2}{3}\right)$	$210\cdot 577+142\cdot 577^2+\dots$
	$x_1^2 x_2^3 x_3^3 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{3}\right)$	$213\cdot 577+89\cdot 577^2+\dots$
$H^{2,0}$	$x_1^2 x_2^{11} x_3^3 x_4^2 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{2}{3}\right)$	$-p^2\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{2}{3}\right)$	$341\cdot 577^2+\dots$
$H^{2,2}$	$y_1^1 y_2^{11} y_3^3 y_4^3$	p^2	p^2	332929

Surface #27: $Q = (6, 2, 3, 1)$, $m = 12$

Equation: $x_0^2 + x_1^6 + x_2^4 + x_0x_3^6 = 0$

$p = 577$

Table 27: Trace contributions for $x_0^2 + x_1^6 + x_2^4 + x_0x_3^6 = 0$ at $p = 577$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{6}\right)$	$-\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{6}\right)$	$70+205\cdot 577+34\cdot 577^2+\dots$
$H^{1,1}$	$y_1^1 y_2^1 y_3^3 y_4^4$	p	p	577
	$y_1^1 y_2^3 y_3^1 y_4^5$	p	p	577
	$y_1^1 y_2^3 y_3^3 y_4^2$	p	p	577
	$y_1^1 y_2^5 y_3^1 y_4^3$	p	p	577
	$x_1^1 x_2^2 y_3^4 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^1 x_3^2 y_2^4 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^1 x_2^3 x_3^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{5}{6}\right)$	$p\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{5}{6}\right)$	$213\cdot 577+89\cdot 577^2+\dots$
	$x_1^1 x_2^3 y_3^2 e_1 e_2$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^1 x_2^3 x_3^6 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^1 x_3^3 x_3^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^1 x_2^3 x_3^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$363\cdot 577+487\cdot 577^2+\dots$
	$x_1^1 x_2^4 x_3^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{6}\right)$	$p\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{6}\right)$	$363\cdot 577+487\cdot 577^2+\dots$
	$x_1^1 x_2^4 x_3^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{5}{6}\right)$	$p\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{5}{6}\right)$	$66\cdot 577+466\cdot 577^2+\dots$
	$x_1^1 x_2^5 x_3^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$213\cdot 577+89\cdot 577^2+\dots$
	$x_1^2 x_2^1 x_3^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$363\cdot 577+487\cdot 577^2+\dots$
	$x_1^2 x_2^3 x_3^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{6}\right)$	$p\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{6}\right)$	$306\cdot 577+348\cdot 577^2+\dots$
	$x_1^2 x_2^3 x_3^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{5}{6}\right)$	$p\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{5}{6}\right)$	$213\cdot 577+89\cdot 577^2+\dots$
	$x_1^2 x_2^3 x_3^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$213\cdot 577+89\cdot 577^2+\dots$
	$x_1^2 x_2^3 x_3^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^2 x_2^3 x_3^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{6}\right)$	$p\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{6}\right)$	$363\cdot 577+487\cdot 577^2+\dots$
$H^{2,0}$	$x_1^2 x_2^3 x_3^5 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{5}{6}\right)$	$-p^2\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{5}{6}\right)$	$305\cdot 577^2+\dots$
$H^{2,2}$	$y_1^1 y_2^5 y_3^3 y_4^6$	p^2	p^2	332929

Surface #28: $Q = (2, 1, 4, 5)$, $m = 12$

Equation: $x_0^6 + x_1^{12} + x_2^3 + x_0 x_3^2 = 0$

$p = 433$

Table 28: Trace contributions for $x_0^6 + x_1^{12} + x_2^3 + x_0 x_3^2 = 0$ at $p = 433$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$68+365\cdot 433+396\cdot 433^2+\dots$
$H^{1,1}$	$y_1^1 y_2^7 y_3^1 y_4^2$	p	p	433
	$y_1^2 y_2^5 y_3^2 y_4^2$	p	p	433
	$y_1^2 y_2^8 y_3^2 y_4^1$	p	p	433
	$y_1^4 y_2^4 y_3^1 y_4^2$	p	p	433
	$y_1^4 y_2^0 y_3^1 y_4^1$	p	p	433
	$y_1^5 y_2^5 y_3^2 y_4^1$	p	p	433
	$x_1^1 x_2^4 x_3^2 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{1}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{1}\right)$	433
	$x_1^1 x_2^8 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{1}\right)$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{1}\right)$	433
	$x_1^1 x_2^9 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	-433
	$x_1^2 x_2^3 y_2^6 y_4^1 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	433
	$x_1^2 x_2^5 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	-433
	$x_1^2 x_2^{11} x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	-433
	$x_1^3 x_2^3 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$17\cdot 433+208\cdot 433^2+\dots$
	$x_1^3 x_2^9 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	-433
	$x_1^4 x_2^1 x_3^6 y_4^1 e_1 e_3$	$-p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	$-p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	433
	$x_1^4 x_2^3 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	-433
	$x_1^4 x_2^7 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$51\cdot 433+140\cdot 433^2+\dots$
	$x_1^5 x_2^1 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	-433
	$x_1^5 x_2^5 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	-433
	$x_1^6 x_2^3 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	-433
$H^{2,0}$	$x_1^6 x_2^{11} x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p^2\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$121\cdot 433^2+\dots$
$H^{2,2}$	$y_1^5 y_2^{11} y_3^2 y_4^2$	p^2	p^2	187489

Surface #29: $Q = (3, 4, 4, 1)$, $m = 12$

Equation: $x_0^4 + x_1^3 + x_2^3 + x_0x_3^9 = 0$

$p = 1297$

Table 29: Trace contributions for $x_0^4 + x_1^3 + x_2^3 + x_0x_3^9 = 0$ at $p = 1297$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{2}{9}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{9}\right)$	$-\Gamma_p\left(\frac{2}{9}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{9}\right)$	$383+451\cdot 1297+395\cdot 1297^2+\dots$
$H^{1,1}$	$y_1^1 y_2^2 y_3^2 y_4^4$	p	p	1297
	$y_1^2 y_2^2 y_3^1 y_4^8$	p	p	1297
	$y_1^2 y_2^2 y_3^2 y_4^2$	p	p	1297
	$y_1^3 y_2^1 y_3^1 y_4^6$	p	p	1297
	$x_1^2 x_2^2 y_1^1 y_4^7 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	1297
	$x_1^2 x_2^2 y_1^2 y_4^5 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	1297
	$x_1^2 x_2^2 y_1^3 y_4^3 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	1297
	$x_1^2 x_2^1 y_1^1 y_4^7 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	1297
	$x_1^2 x_2^1 y_1^2 y_4^5 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	1297
	$x_1^2 x_2^1 y_1^3 y_4^3 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	1297
	$x_1^1 x_2^1 x_3^2 x_4^9 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	1297
	$x_1^1 x_2^2 x_3^1 x_4^9 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	1297
	$x_1^1 x_2^2 x_3^2 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{5}{6}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{5}{6}\right)$	$660\cdot 1297+731\cdot 1297^2+\dots$
	$x_1^2 x_2^1 x_3^2 x_4^6 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{5}{6}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{5}{6}\right)$	1297
	$x_1^2 x_2^2 x_3^1 x_4^6 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{5}{6}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{5}{6}\right)$	1297
	$x_1^2 x_2^2 x_3^2 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{5}{6}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{5}{6}\right)$	$1184\cdot 1297+789\cdot 1297^2+\dots$
	$x_1^3 x_2^1 x_3^1 x_4^7 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{5}{6}\right)$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{5}{6}\right)$	$1033\cdot 1297+1287\cdot 1297^2+\dots$
	$x_1^3 x_2^1 x_3^2 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{5}{6}\right)$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{5}{6}\right)$	1297
	$x_1^3 x_2^2 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{5}{6}\right)$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{5}{6}\right)$	1297
	$x_1^4 x_2^1 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{5}{6}\right)$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{5}{6}\right)$	$564\cdot 1297+835\cdot 1297^2+\dots$
$H^{2,0}$	$x_1^4 x_2^2 x_3^2 x_4^8 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{7}{9}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{8}{9}\right)$	$-p^2\Gamma_p\left(\frac{7}{9}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{8}{9}\right)$	$1148\cdot 1297^2+\dots$
$H^{2,2}$	$y_1^3 y_2^2 y_3^2 y_4^9$	p^2	p^2	1682209

Surface #30: $Q = (4, 1, 3, 4)$, $m = 12$

Equation: $x_0^3 + x_1^{12} + x_2^4 + x_0 x_3^2 = 0$

$p = 577$

Table 30: Trace contributions for $x_0^3 + x_1^{12} + x_2^4 + x_0 x_3^2 = 0$ at $p = 577$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{12}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	$-\Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{12}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	$485+298\cdot 577+181\cdot 577^2+\dots$
$H^{1,1}$	$y_1^1 y_2^7 y_3^3 y_4^1$	p	p	577
	$y_1^1 y_2^{10} y_3^2 y_4^1$	p	p	577
	$y_1^2 y_2^2 y_3^2 y_4^2$	p	p	577
	$y_1^2 y_2^5 y_3^1 y_4^2$	p	p	577
	$x_1^1 x_2^4 y_3^3 y_4^1 e_1 e_4$	$-p \Gamma_p\left(\frac{1}{4}\right)$	$-p \Gamma_p\left(\frac{1}{4}\right)$	577
	$x_1^1 x_2^6 y_3^2 y_4^1 e_1 e_4$	$-p \Gamma_p\left(\frac{1}{4}\right)$	$-p \Gamma_p\left(\frac{1}{4}\right)$	577
	$x_1^1 x_2^9 y_3^1 y_4^1 e_1 e_4$	$-p \Gamma_p\left(\frac{1}{4}\right)$	$-p \Gamma_p\left(\frac{1}{4}\right)$	577
	$x_1^1 x_2^3 x_3^3 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^1 x_2^6 x_3^2 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^1 x_2^7 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{12}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{12}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	$554\cdot 577+495\cdot 577^2+\dots$
	$x_1^1 x_2^9 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^1 x_2^{10} x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^2 x_2^4 y_3^3 y_4^1 e_1 e_4$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^2 x_2^6 y_3^2 y_4^1 e_1 e_4$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^2 x_2^9 y_3^1 y_4^1 e_1 e_4$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^2 x_2^3 x_3^3 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^2 x_2^6 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^2 x_2^9 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^3 x_2^2 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^3 x_2^5 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{5}{12}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{5}{12}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	$301\cdot 577+545\cdot 577^2+\dots$
$H^{2,0}$	$x_1^3 x_2^{11} x_3^3 x_4^1 e_1 e_2 e_3 e_4$	$-p^2 \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{11}{12}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	$-p^2 \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{11}{12}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	$508\cdot 577^2+\dots$
$H^{2,2}$	$y_1^2 y_2^{11} y_3^3 y_4^2$	p^2	p^2	332929

Surface #31: $Q = (4, 3, 4, 1)$, $m = 12$

Equation: $x_0^3 + x_1^4 + x_2^3 + x_0 x_3^8 = 0$

$p = 577$

Table 31: Trace contributions for $x_0^3 + x_1^4 + x_2^3 + x_0 x_3^8 = 0$ at $p = 577$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{7}{24}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{8}\right)$	$-\Gamma_p\left(\frac{7}{24}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{8}\right)$	$34+553\cdot 577+175\cdot 577^2+\dots$
$H^{1,1}$	$y_1^1 y_2^2 y_3^1 y_4^1$	p	p	577
	$y_1^1 y_2^2 y_3^1 y_4^5$	p	p	577
	$y_1^2 y_2^1 y_3^2 y_4^4$	p	p	577
	$y_1^2 y_2^2 y_3^2 y_4^4$	p	p	577
	$x_1^1 x_2^2 y_3^1 y_4^6 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	577
	$x_1^1 x_2^2 y_3^2 y_4^4 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	577
	$x_1^1 x_2^2 y_3^3 y_4^2 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	577
	$x_1^1 x_2^2 x_3^2 x_4^6 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{3}{4}\right)$	$p\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{3}{4}\right)$	577
	$x_1^1 x_2^2 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{24}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{3}{8}\right)$	$p\Gamma_p\left(\frac{1}{24}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{3}{8}\right)$	$288\cdot 577+131\cdot 577^2+\dots$
	$x_1^1 x_2^2 x_3^2 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{24}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{3}{8}\right)$	$p\Gamma_p\left(\frac{5}{24}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{3}{8}\right)$	$539\cdot 577+503\cdot 577^2+\dots$
	$x_1^2 x_2^1 y_3^1 y_4^6 e_1 e_3$	$-p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	$-p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	577
	$x_1^2 x_2^1 y_3^2 y_4^4 e_1 e_3$	$-p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	$-p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	577
	$x_1^2 x_2^1 y_3^3 y_4^2 e_1 e_3$	$-p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	$-p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	577
	$x_1^2 x_2^1 x_3^2 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{11}{24}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{5}{8}\right)$	$p\Gamma_p\left(\frac{11}{24}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{5}{8}\right)$	$138\cdot 577+361\cdot 577^2+\dots$
	$x_1^2 x_2^2 x_3^1 x_4^6 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{3}{8}\right)$	$p\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{3}{8}\right)$	577
	$x_1^2 x_2^2 x_3^2 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{13}{24}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{3}{8}\right)$	$p\Gamma_p\left(\frac{13}{24}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{3}{8}\right)$	577
	$x_1^2 x_2^2 x_3^3 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{19}{24}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{5}{8}\right)$	$p\Gamma_p\left(\frac{19}{24}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{5}{8}\right)$	$46\cdot 577+143\cdot 577^2+\dots$
	$x_1^3 x_2^1 x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{23}{24}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{8}\right)$	$p\Gamma_p\left(\frac{23}{24}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{8}\right)$	$167\cdot 577+347\cdot 577^2+\dots$
	$x_1^3 x_2^1 x_3^2 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{23}{24}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{8}\right)$	$p\Gamma_p\left(\frac{23}{24}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{8}\right)$	$575\cdot 577+50\cdot 577^2+\dots$
	$x_1^3 x_2^2 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{4}\right)$	$p\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{4}\right)$	577
$H^{2,0}$	$x_1^3 x_2^3 x_3^2 x_4^7 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{17}{24}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{7}{8}\right)$	$-p^2\Gamma_p\left(\frac{17}{24}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{7}{8}\right)$	$17\cdot 577^2+\dots$
$H^{2,2}$	$y_1^2 y_2^3 y_3^2 y_4^8$	p^2	p^2	332929

Surface #32: $Q = (2, 2, 3, 5)$, $m = 12$

Equation: $x_0^6 + x_1^6 + x_2^4 + x_0 x_3^2 = 0$

$p = 577$

Table 32: Trace contributions for $x_0^6 + x_1^6 + x_2^4 + x_0 x_3^2 = 0$ at $p = 577$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{12}\right) \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	$-\Gamma_p\left(\frac{1}{12}\right) \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	$485+298\cdot 577+181\cdot 577^2+\dots$
$H^{1,1}$	$y_1^1 y_2^1 y_3^2 y_4^2$	p	p	577
	$y_1^1 y_2^2 y_3^2 y_4^2$	p	p	577
	$y_1^3 y_2^3 y_3^1 y_4^2$	p	p	577
	$y_1^3 y_2^3 y_3^3 y_4^1$	p	p	577
	$y_1^4 y_2^4 y_3^2 y_4^1$	p	p	577
	$y_1^5 y_2^5 y_3^1 y_4^1$	p	p	577
	$x_1^1 x_2^2 x_3^2 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^1 x_2^4 x_3^3 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{12}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{1}{12}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^1 x_2^2 y_3^2 y_4^1 e_1 e_2$	$-p \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{5}{6}\right)$	$-p \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{5}{6}\right)$	577
	$x_1^2 x_2^3 x_3^3 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^2 x_2^4 y_3^2 y_4^1 e_1 e_2$	$-p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	$-p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	577
	$x_1^3 x_2^2 x_3^3 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{5}{12}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{5}{12}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^3 x_2^2 y_3^2 y_4^1 e_1 e_2$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^3 x_2^5 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{5}{12}\right) \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{5}{12}\right) \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	$301\cdot 577+545\cdot 577^2+\dots$
	$x_1^4 x_2^1 x_3^3 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{7}{12}\right) \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{7}{12}\right) \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	$554\cdot 577+495\cdot 577^2+\dots$
	$x_1^4 x_2^2 y_3^2 y_4^1 e_1 e_2$	$-p \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	$-p \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	577
	$x_1^4 x_2^2 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{7}{12}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{7}{12}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^5 x_2^1 y_3^2 y_4^1 e_1 e_2$	$-p \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{6}\right)$	$-p \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{6}\right)$	577
	$x_1^5 x_2^3 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^6 x_2^2 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{11}{12}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{11}{12}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	577
$H^{2,0}$	$x_1^6 x_2^5 x_3^3 x_4^1 e_1 e_2 e_3 e_4$	$-p^2 \Gamma_p\left(\frac{11}{12}\right) \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	$-p^2 \Gamma_p\left(\frac{11}{12}\right) \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right)$	$508\cdot 577^2+\dots$
$H^{2,2}$	$y_1^5 y_2^5 y_3^2 y_4^2$	p^2	p^2	332929

Surface #33: $Q = (3, 2, 4, 3)$, $m = 12$

Equation: $x_0^4 + x_1^6 + x_2^3 + x_0x_3^3 = 0$

$p = 433$

Table 33: Trace contributions for $x_0^4 + x_1^6 + x_2^3 + x_0x_3^3 = 0$ at $p = 433$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	$-\Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	$4+431\cdot 433+107\cdot 433^2+\dots$
$H^{1,1}$	$y_1^1 y_2^5 y_3^2 y_4^1$	p	p	433
	$y_1^2 y_2^2 y_3^2 y_4^2$	p	p	433
	$y_1^2 y_2^4 y_3^1 y_4^2$	p	p	433
	$y_1^3 y_2^1 y_3^1 y_4^3$	p	p	433
	$x_2^2 x_3^2 y_1^2 y_4^2 e_2 e_3$	$-p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	$-p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	433
	$x_2^4 x_3^1 y_1^2 y_4^2 e_2 e_3$	$-p \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	$-p \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	433
	$x_1^1 x_4^3 y_2^2 y_3^1 e_1 e_4$	$-p \Gamma_p\left(\frac{1}{3}\right)$	$-p \Gamma_p\left(\frac{1}{3}\right)$	433
	$x_1^1 x_4^3 y_2^4 y_3^1 e_1 e_4$	$-p \Gamma_p\left(\frac{1}{3}\right)$	$-p \Gamma_p\left(\frac{1}{3}\right)$	433
	$x_1^1 x_2^2 x_3^2 x_4^3 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	$p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	433
	$x_1^1 x_2^4 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	$p \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	433
	$x_1^1 x_2^5 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	$p \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	433
	$x_1^2 x_4^2 y_2^2 y_3^2 e_1 e_4$	$-p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	$-p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	433
	$x_1^2 x_4^4 y_2^1 y_3^2 e_1 e_4$	$-p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	$-p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	433
	$x_1^2 x_2^2 x_3^2 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	$p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	433
	$x_1^2 x_2^4 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	$p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	433
	$x_1^3 x_4^1 y_2^2 y_3^2 e_1 e_4$	$-p \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	$-p \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	433
	$x_1^3 x_4^1 y_2^4 y_3^2 e_1 e_4$	$-p \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	$-p \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	433
	$x_1^3 x_2^2 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	$p \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	433
	$x_1^3 x_2^4 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	$p \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	433
	$x_1^4 x_2^1 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	$p \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	433
$H^{2,0}$	$x_1^4 x_2^5 x_3^2 x_4^2 e_1 e_2 e_3 e_4$	$-p^2 \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	$-p^2 \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	$325\cdot 433^2+\dots$
$H^{2,2}$	$y_1^3 y_2^5 y_3^2 y_4^3$	p^2	p^2	187489

Surface #34: $Q = (4, 3, 3, 2)$, $m = 12$

Equation: $x_0^3 + x_1^4 + x_2^4 + x_0x_3^4 = 0$

$p = 193$

Table 34: Trace contributions for $x_0^3 + x_1^4 + x_2^4 + x_0x_3^4 = 0$ at $p = 193$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{4}\right)$	$-\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{4}\right)$	$3+192\cdot 193+63\cdot 193^2+\dots$
$H^{1,1}$	$y_1^1 y_2^2 y_3^2 y_4^3$	p	p	193
	$y_1^1 y_2^3 y_3^3 y_4^4$	p	p	193
	$y_1^2 y_2^1 y_3^1 y_4^4$	p	p	193
	$y_1^2 y_2^2 y_3^2 y_4^2$	p	p	193
	$x_2^1 x_3^1 y_1^1 y_4^3 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{3}{4}\right)$	$-p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{3}{4}\right)$	193
	$x_2^1 x_3^2 y_1^2 y_4^2 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{3}{4}\right)$	$-p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{3}{4}\right)$	193
	$x_2^2 x_3^2 y_1^1 y_4^3 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	193
	$x_2^2 x_3^2 y_1^2 y_4^2 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	193
	$x_2^3 x_3^1 y_1^1 y_4^3 e_2 e_3$	$-p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{4}\right)$	$-p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{4}\right)$	193
	$x_2^3 x_3^1 y_1^2 y_4^2 e_2 e_3$	$-p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{4}\right)$	$-p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{4}\right)$	193
	$x_1^1 x_4^2 y_2^2 y_3^2 e_1 e_4$	$-p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{4}\right)$	$-p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{4}\right)$	193
	$x_1^1 x_2^3 x_3^3 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{4}\right)$	$p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{4}\right)$	193
	$x_1^1 x_2^3 x_3^4 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	193
	$x_1^1 x_2^3 x_3^4 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{4}\right)$	$p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{4}\right)$	193
	$x_1^1 x_2^3 x_3^4 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{4}\right)$	$p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{4}\right)$	193
	$x_1^2 x_2^2 y_3^2 y_4^2 e_1 e_4$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	193
	$x_1^2 x_2^3 x_3^3 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)$	193
	$x_1^2 x_2^3 x_3^4 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	193
	$x_1^2 x_2^3 x_3^4 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)$	193
	$x_1^3 x_2^1 x_3^4 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{3}{4}\right)$	$p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{3}{4}\right)$	193
$H^{2,0}$	$x_1^3 x_2^3 x_3^3 x_4^4 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{3}{4}\right)$	$-p^2\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{3}{4}\right)$	$129\cdot 193^2+\dots$
$H^{2,2}$	$y_1^2 y_2^3 y_3^3 y_4^4$	p^2	p^2	37249

Surface #35: $Q = (2, 1, 7, 4)$, $m = 14$

Equation: $x_0^7 + x_1^{14} + x_2^2 + x_0 x_3^3 = 0$

$p = 3529$

Table 35: Trace contributions for $x_0^7 + x_1^{14} + x_2^2 + x_0 x_3^3 = 0$ at $p = 3529$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{2}{21}\right)\Gamma_p\left(\frac{1}{14}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$-\Gamma_p\left(\frac{2}{21}\right)\Gamma_p\left(\frac{1}{14}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$1995+1575\cdot 3529+957\cdot 3529^2+\dots$
$H^{1,1}$	$y_1^2 y_2^9 y_3^1 y_4^2$	p	p	3529
	$y_1^3 y_2^9 y_3^1 y_4^3$	p	p	3529
	$y_1^4 y_2^{11} y_3^1 y_4^1$	p	p	3529
	$y_1^5 y_2^5 y_3^1 y_4^2$	p	p	3529
	$x_1^1 x_4^1 y_2^1 y_3^1 e_1 e_4$	$-p\Gamma_p\left(\frac{1}{1}\right)$	$-p\Gamma_p\left(\frac{1}{1}\right)$	3529
	$x_1^1 x_2^7 x_3^3 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{1}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{1}\right)$	3529
	$x_1^1 x_2^{11} x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{21}\right)\Gamma_p\left(\frac{11}{14}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{1}{21}\right)\Gamma_p\left(\frac{11}{14}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$1088\cdot 3529+1654\cdot 3529^2+\dots$
	$x_1^2 x_2^9 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{4}{21}\right)\Gamma_p\left(\frac{9}{14}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{4}{21}\right)\Gamma_p\left(\frac{9}{14}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$1237\cdot 3529+3219\cdot 3529^2+\dots$
	$x_1^2 x_2^{13} x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{21}\right)\Gamma_p\left(\frac{13}{14}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{5}{21}\right)\Gamma_p\left(\frac{13}{14}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$288\cdot 3529+1668\cdot 3529^2+\dots$
	$x_1^3 x_2^2 y_2^1 y_3^1 e_1 e_4$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	3529
	$x_1^3 x_2^7 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	3529
	$x_1^4 x_2^{11} x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{8}{21}\right)\Gamma_p\left(\frac{11}{14}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{8}{21}\right)\Gamma_p\left(\frac{11}{14}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$2021\cdot 3529+2903\cdot 3529^2+\dots$
	$x_1^4 x_2^5 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{10}{21}\right)\Gamma_p\left(\frac{5}{14}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{10}{21}\right)\Gamma_p\left(\frac{5}{14}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$2961\cdot 3529+1352\cdot 3529^2+\dots$
	$x_1^4 x_2^9 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{11}{21}\right)\Gamma_p\left(\frac{9}{14}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{11}{21}\right)\Gamma_p\left(\frac{9}{14}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$1050\cdot 3529+2097\cdot 3529^2+\dots$
	$x_1^5 x_4^1 y_2^1 y_3^1 e_1 e_4$	$-p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	$-p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	3529
	$x_1^5 x_2^3 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{13}{21}\right)\Gamma_p\left(\frac{3}{14}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{13}{21}\right)\Gamma_p\left(\frac{3}{14}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$2277\cdot 3529+90\cdot 3529^2+\dots$
	$x_1^5 x_2^7 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	3529
	$x_1^6 x_2^{11} x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{16}{21}\right)\Gamma_p\left(\frac{11}{14}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{16}{21}\right)\Gamma_p\left(\frac{11}{14}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$870\cdot 3529+2889\cdot 3529^2+\dots$
	$x_1^6 x_2^5 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{17}{21}\right)\Gamma_p\left(\frac{5}{14}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{17}{21}\right)\Gamma_p\left(\frac{5}{14}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$2385\cdot 3529+229\cdot 3529^2+\dots$
	$x_1^7 x_2^3 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{20}{21}\right)\Gamma_p\left(\frac{3}{14}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{20}{21}\right)\Gamma_p\left(\frac{3}{14}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$1891\cdot 3529+138\cdot 3529^2+\dots$
$H^{2,0}$	$x_1^7 x_2^{13} x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{19}{21}\right)\Gamma_p\left(\frac{13}{14}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$-p^2\Gamma_p\left(\frac{19}{21}\right)\Gamma_p\left(\frac{13}{14}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$444\cdot 3529^2+\dots$
$H^{2,2}$	$y_1^6 y_2^{13} y_3^1 y_4^3$	p^2	p^2	12453841

Surface #36: $Q = (2, 2, 7, 3)$, $m = 14$

Equation: $x_0^7 + x_1^7 + x_2^2 + x_0 x_3^4 = 0$

$p = 3137$

Table 36: Trace contributions for $x_0^7 + x_1^7 + x_2^2 + x_0 x_3^4 = 0$ at $p = 3137$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{3}{28}\right)\Gamma_p\left(\frac{1}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$-\Gamma_p\left(\frac{3}{28}\right)\Gamma_p\left(\frac{1}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$1554+1208\cdot 3137+2528\cdot 3137^2+\dots$
$H^{1,1}$	$y_1^2 y_2^2 y_3^1 y_4^4$	p	p	3137
	$y_1^3 y_2^3 y_3^1 y_4^4$	p	p	3137
	$y_1^4 y_2^4 y_3^1 y_4^2$	p	p	3137
	$y_1^5 y_2^5 y_3^1 y_4^1$	p	p	3137
	$x_1^1 x_2^3 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{28}\right)\Gamma_p\left(\frac{5}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$p\Gamma_p\left(\frac{1}{28}\right)\Gamma_p\left(\frac{5}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$1633\cdot 3137+835\cdot 3137^2+\dots$
	$x_1^1 x_2^6 y_3^2 y_4^1 e_1 e_2$	$-p\Gamma_p\left(\frac{1}{7}\right)\Gamma_p\left(\frac{6}{7}\right)$	$-p\Gamma_p\left(\frac{1}{7}\right)\Gamma_p\left(\frac{6}{7}\right)$	3137
	$x_1^2 x_2^4 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{28}\right)\Gamma_p\left(\frac{4}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$p\Gamma_p\left(\frac{5}{28}\right)\Gamma_p\left(\frac{4}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$1245\cdot 3137+2361\cdot 3137^2+\dots$
	$x_1^2 x_2^5 y_3^2 y_4^1 e_1 e_2$	$-p\Gamma_p\left(\frac{2}{7}\right)\Gamma_p\left(\frac{5}{7}\right)$	$-p\Gamma_p\left(\frac{2}{7}\right)\Gamma_p\left(\frac{5}{7}\right)$	3137
	$x_1^3 x_2^3 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{9}{28}\right)\Gamma_p\left(\frac{3}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$p\Gamma_p\left(\frac{9}{28}\right)\Gamma_p\left(\frac{3}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$427\cdot 3137+1395\cdot 3137^2+\dots$
	$x_1^3 x_2^4 y_3^2 y_4^1 e_1 e_2$	$-p\Gamma_p\left(\frac{3}{7}\right)\Gamma_p\left(\frac{4}{7}\right)$	$-p\Gamma_p\left(\frac{3}{7}\right)\Gamma_p\left(\frac{4}{7}\right)$	3137
	$x_1^3 x_2^6 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{11}{28}\right)\Gamma_p\left(\frac{6}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$p\Gamma_p\left(\frac{11}{28}\right)\Gamma_p\left(\frac{6}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$1730\cdot 3137+794\cdot 3137^2+\dots$
	$x_1^4 x_2^2 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{13}{28}\right)\Gamma_p\left(\frac{2}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$p\Gamma_p\left(\frac{13}{28}\right)\Gamma_p\left(\frac{2}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$2859\cdot 3137+1608\cdot 3137^2+\dots$
	$x_1^4 x_2^3 y_3^2 y_4^1 e_1 e_2$	$-p\Gamma_p\left(\frac{4}{7}\right)\Gamma_p\left(\frac{3}{7}\right)$	$-p\Gamma_p\left(\frac{4}{7}\right)\Gamma_p\left(\frac{3}{7}\right)$	3137
	$x_1^4 x_2^5 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{15}{28}\right)\Gamma_p\left(\frac{5}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$p\Gamma_p\left(\frac{15}{28}\right)\Gamma_p\left(\frac{5}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$2065\cdot 3137+35\cdot 3137^2+\dots$
	$x_1^5 x_2^1 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{17}{28}\right)\Gamma_p\left(\frac{1}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$p\Gamma_p\left(\frac{17}{28}\right)\Gamma_p\left(\frac{1}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$1282\cdot 3137+2793\cdot 3137^2+\dots$
	$x_1^5 x_2^2 y_3^2 y_4^1 e_1 e_2$	$-p\Gamma_p\left(\frac{5}{7}\right)\Gamma_p\left(\frac{2}{7}\right)$	$-p\Gamma_p\left(\frac{5}{7}\right)\Gamma_p\left(\frac{2}{7}\right)$	3137
	$x_1^5 x_2^4 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{19}{28}\right)\Gamma_p\left(\frac{4}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$p\Gamma_p\left(\frac{19}{28}\right)\Gamma_p\left(\frac{4}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$1844\cdot 3137+2960\cdot 3137^2+\dots$
	$x_1^6 x_2^1 y_3^2 y_4^1 e_1 e_2$	$-p\Gamma_p\left(\frac{6}{7}\right)\Gamma_p\left(\frac{1}{7}\right)$	$-p\Gamma_p\left(\frac{6}{7}\right)\Gamma_p\left(\frac{1}{7}\right)$	3137
	$x_1^6 x_2^3 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{23}{28}\right)\Gamma_p\left(\frac{3}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$p\Gamma_p\left(\frac{23}{28}\right)\Gamma_p\left(\frac{3}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$320\cdot 3137+2331\cdot 3137^2+\dots$
	$x_1^7 x_2^2 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{27}{28}\right)\Gamma_p\left(\frac{2}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$p\Gamma_p\left(\frac{27}{28}\right)\Gamma_p\left(\frac{2}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$1070\cdot 3137+3016\cdot 3137^2+\dots$
$H^{2,0}$	$x_1^7 x_2^6 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{25}{28}\right)\Gamma_p\left(\frac{6}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$-p^2\Gamma_p\left(\frac{25}{28}\right)\Gamma_p\left(\frac{6}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$1298\cdot 3137^2+\dots$
$H^{2,2}$	$y_1^6 y_2^6 y_3^1 y_4^4$	p^2	p^2	9840769

Surface #37: $Q = (3, 1, 5, 6)$, $m = 15$

Equation: $x_0^5 + x_1^{15} + x_2^3 + x_0 x_3^2 = 0$

$p = 1801$

Table 37: Trace contributions for $x_0^5 + x_1^{15} + x_2^3 + x_0 x_3^2 = 0$ at $p = 1801$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{10}\right)\Gamma_p\left(\frac{1}{15}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-\Gamma_p\left(\frac{1}{10}\right)\Gamma_p\left(\frac{1}{15}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$72+845\cdot 1801+1550\cdot 1801^2+\dots$
$H^{1,1}$	$y_1^1 y_2^{11} y_3^2 y_4^1$	p	p	1801
	$y_1^2 y_2^2 y_3^2 y_4^2$	p	p	1801
	$y_1^2 y_2^7 y_3^1 y_4^2$	p	p	1801
	$y_1^3 y_2^8 y_3^2 y_4^1$	p	p	1801
	$y_1^3 y_2^{13} y_3^1 y_4^1$	p	p	1801
	$y_1^4 y_2^4 y_3^2 y_4^2$	p	p	1801
	$x_1^1 x_2^2 y_3^5 y_4^2 e_1 e_4$	$-p\Gamma_p\left(\frac{1}{1}\right)$	$-p\Gamma_p\left(\frac{1}{1}\right)$	1801
	$x_1^1 x_2^2 y_3^{10} y_4^1 e_1 e_4$	$-p\Gamma_p\left(\frac{1}{1}\right)$	$-p\Gamma_p\left(\frac{1}{1}\right)$	1801
	$x_1^1 x_2^3 x_3^2 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{1}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{1}\right)$	1801
	$x_1^1 x_2^{10} x_3^2 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{1}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{1}\right)$	1801
	$x_1^1 x_2^{11} x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{10}\right)\Gamma_p\left(\frac{11}{15}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{10}\right)\Gamma_p\left(\frac{11}{15}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$1026\cdot 1801+1383\cdot 1801^2+\dots$
	$x_1^2 x_2^8 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{10}\right)\Gamma_p\left(\frac{8}{15}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{3}{10}\right)\Gamma_p\left(\frac{8}{15}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$903\cdot 1801+410\cdot 1801^2+\dots$
	$x_1^2 x_2^{13} x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{10}\right)\Gamma_p\left(\frac{13}{15}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{3}{10}\right)\Gamma_p\left(\frac{13}{15}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$1416\cdot 1801+391\cdot 1801^2+\dots$
	$x_1^3 x_2^5 y_3^2 y_4^1 e_1 e_4$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	1801
	$x_1^3 x_2^5 y_3^{10} y_4^1 e_1 e_4$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	1801
	$x_1^3 x_2^{10} x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	1801
	$x_1^3 x_2^{10} x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	1801
	$x_1^4 x_2^2 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{10}\right)\Gamma_p\left(\frac{2}{15}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{7}{10}\right)\Gamma_p\left(\frac{2}{15}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$1525\cdot 1801+1463\cdot 1801^2+\dots$
	$x_1^4 x_2^7 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{10}\right)\Gamma_p\left(\frac{7}{15}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{7}{10}\right)\Gamma_p\left(\frac{7}{15}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$1081\cdot 1801+438\cdot 1801^2+\dots$
	$x_1^5 x_2^4 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{9}{10}\right)\Gamma_p\left(\frac{4}{15}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{9}{10}\right)\Gamma_p\left(\frac{4}{15}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$409\cdot 1801+1590\cdot 1801^2+\dots$
$H^{2,0}$	$x_1^5 x_2^{14} x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{9}{10}\right)\Gamma_p\left(\frac{14}{15}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p^2\Gamma_p\left(\frac{9}{10}\right)\Gamma_p\left(\frac{14}{15}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$1776\cdot 1801^2+\dots$
$H^{2,2}$	$y_1^4 y_2^{14} y_3^2 y_4^2$	p^2	p^2	3243601

Surface #38: $Q = (3, 5, 5, 2)$, $m = 15$

Equation: $x_0^5 + x_1^3 + x_2^3 + x_0x_3^6 = 0$

$p = 271$

Table 38: Trace contributions for $x_0^5 + x_1^3 + x_2^3 + x_0x_3^6 = 0$ at $p = 271$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{6}\right)$	$-\Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{6}\right)$	$29+270 \cdot 271+27 \cdot 271^2+\dots$
$H^{1,1}$	$y_1^1 y_2^2 y_3^2 y_4^3$	p	p	271
	$y_1^2 y_2^2 y_3^3 y_4^6$	p	p	271
	$y_1^2 y_2^3 y_3^2 y_4^2$	p	p	271
	$y_1^3 y_2^1 y_3^1 y_4^5$	p	p	271
	$y_1^3 y_2^2 y_3^2 y_4^1$	p	p	271
	$y_1^4 y_2^1 y_3^1 y_4^4$	p	p	271
	$x_2^1 x_3^2 y_1^1 y_4^5 e_2 e_3$	$-p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	$-p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	271
	$x_2^1 x_3^2 y_1^2 y_4^4 e_2 e_3$	$-p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	$-p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	271
	$x_2^1 x_3^2 y_1^3 y_4^3 e_2 e_3$	$-p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	$-p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	271
	$x_2^1 x_3^2 y_1^4 y_4^2 e_2 e_3$	$-p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	$-p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	271
	$x_2^2 x_3^1 y_1^1 y_4^5 e_2 e_3$	$-p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	$-p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	271
	$x_2^2 x_3^1 y_1^2 y_4^4 e_2 e_3$	$-p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	$-p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	271
	$x_2^2 x_3^1 y_1^3 y_4^3 e_2 e_3$	$-p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	$-p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	271
	$x_2^2 x_3^1 y_1^4 y_4^2 e_2 e_3$	$-p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	$-p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	271
	$x_1^1 x_2^1 x_3^2 x_4^6 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	$p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	271
	$x_1^1 x_2^2 x_3^1 x_4^6 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	$p \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	271
	$x_1^2 x_2^2 x_3^2 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	$p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	271
	$x_1^3 x_2^1 x_3^2 x_4^3 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	-271
	$x_1^3 x_2^2 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	-271
	$x_1^4 x_2^1 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	$p \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	271
$H^{2,0}$	$x_1^5 x_2^2 x_3^2 x_4^5 e_1 e_2 e_3 e_4$	$-p^2 \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{5}{6}\right)$	$-p^2 \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{5}{6}\right)$	$243 \cdot 271^2+\dots$
$H^{2,2}$	$y_1^4 y_2^2 y_3^2 y_4^6$	p^2	p^2	73441

Surface #39: $Q = (5, 3, 5, 2)$, $m = 15$

Equation: $x_0^3 + x_1^5 + x_2^3 + x_0x_3^5 = 0$

$p = 1801$

Table 39: Trace contributions for $x_0^3 + x_1^5 + x_2^3 + x_0x_3^5 = 0$ at $p = 1801$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{4}{15}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{5}\right)$	$-\Gamma_p\left(\frac{4}{15}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{5}\right)$	$1688+1534\cdot 1801+563\cdot 1801^2+\dots$
$H^{1,1}$	$y_1^1 y_2^2 y_3^1 y_4^5$	p	p	1801
	$y_1^1 y_2^2 y_3^3 y_4^4$	p	p	1801
	$y_1^1 y_2^4 y_3^1 y_4^3$	p	p	1801
	$y_1^2 y_2^1 y_3^2 y_4^3$	p	p	1801
	$y_1^2 y_2^2 y_3^2 y_4^2$	p	p	1801
	$y_1^2 y_2^3 y_3^2 y_4^1$	p	p	1801
	$x_1^1 x_2^2 y_3^1 y_4^4 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	1801
	$x_1^1 x_2^2 y_3^2 y_4^3 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	1801
	$x_1^1 x_2^2 y_3^3 y_4^2 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	1801
	$x_1^1 x_2^3 y_3^1 y_4^4 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	1801
	$x_1^1 x_2^3 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{15}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{3}{5}\right)$	$p\Gamma_p\left(\frac{2}{15}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{3}{5}\right)$	$80\cdot 1801+1399\cdot 1801^2+\dots$
	$x_1^1 x_2^3 x_3^2 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{15}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{4}{5}\right)$	$p\Gamma_p\left(\frac{1}{15}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{4}{5}\right)$	$871\cdot 1801+954\cdot 1801^2+\dots$
	$x_1^2 x_3^1 y_2^1 y_4^4 e_1 e_3$	$-p\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{3}\right)$	$-p\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{3}\right)$	1801
	$x_1^2 x_3^1 y_2^2 y_4^3 e_1 e_3$	$-p\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{3}\right)$	$-p\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{3}\right)$	1801
	$x_1^2 x_3^1 y_2^3 y_4^2 e_1 e_3$	$-p\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{3}\right)$	$-p\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{3}\right)$	1801
	$x_1^2 x_3^1 y_2^4 y_4^1 e_1 e_3$	$-p\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{3}\right)$	$-p\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{3}\right)$	1801
	$x_1^2 x_2^2 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{8}{15}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{2}{5}\right)$	$p\Gamma_p\left(\frac{8}{15}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{2}{5}\right)$	$1117\cdot 1801+955\cdot 1801^2+\dots$
	$x_1^2 x_2^3 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{15}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{5}\right)$	$p\Gamma_p\left(\frac{7}{15}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{5}\right)$	$287\cdot 1801+1025\cdot 1801^2+\dots$
	$x_1^3 x_2^1 x_3^2 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{14}{15}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{5}\right)$	$p\Gamma_p\left(\frac{14}{15}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{5}\right)$	$641\cdot 1801+1173\cdot 1801^2+\dots$
	$x_1^3 x_2^2 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{13}{15}\right)\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{5}\right)$	$p\Gamma_p\left(\frac{13}{15}\right)\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{5}\right)$	$878\cdot 1801+877\cdot 1801^2+\dots$
$H^{2,0}$	$x_1^3 x_2^3 x_3^4 x_4^1 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{11}{15}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{4}{5}\right)$	$-p^2\Gamma_p\left(\frac{11}{15}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{4}{5}\right)$	$255\cdot 1801^2+\dots$
$H^{2,2}$	$y_1^2 y_2^4 y_3^2 y_4^5$	p^2	p^2	3243601

Surface #40: $Q = (3, 3, 5, 4)$, $m = 15$

Equation: $x_0^5 + x_1^5 + x_2^3 + x_0 x_3^3 = 0$

$p = 1801$

Table 40: Trace contributions for $x_0^5 + x_1^5 + x_2^3 + x_0 x_3^3 = 0$ at $p = 1801$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{2}{15}\right) \Gamma_p\left(\frac{1}{5}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	$-\Gamma_p\left(\frac{2}{15}\right) \Gamma_p\left(\frac{1}{5}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	$699+1290 \cdot 1801+1086 \cdot 1801^2+\dots$
$H^{1,1}$	$y_1^1 y_2^1 y_3^2 y_4^3$	p	p	1801
	$y_1^1 y_2^2 y_3^2 y_4^3$	p	p	1801
	$y_1^2 y_2^2 y_3^2 y_4^2$	p	p	1801
	$y_1^3 y_2^3 y_3^1 y_4^2$	p	p	1801
	$y_1^3 y_2^3 y_3^2 y_4^1$	p	p	1801
	$y_1^4 y_2^4 y_3^1 y_4^1$	p	p	1801
	$x_1^1 x_2^3 x_3^2 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{15}\right) \Gamma_p\left(\frac{3}{5}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	$p \Gamma_p\left(\frac{1}{15}\right) \Gamma_p\left(\frac{3}{5}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	$86 \cdot 1801+290 \cdot 1801^2+\dots$
	$x_1^1 x_2^4 y_3^2 y_4^2 e_1 e_2$	$-p \Gamma_p\left(\frac{1}{5}\right) \Gamma_p\left(\frac{4}{5}\right)$	$-p \Gamma_p\left(\frac{1}{5}\right) \Gamma_p\left(\frac{4}{5}\right)$	1801
	$x_1^1 x_2^4 y_3^2 y_4^1 e_1 e_2$	$-p \Gamma_p\left(\frac{1}{5}\right) \Gamma_p\left(\frac{4}{5}\right)$	$-p \Gamma_p\left(\frac{1}{5}\right) \Gamma_p\left(\frac{4}{5}\right)$	1801
	$x_1^2 x_2^2 x_3^2 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{4}{15}\right) \Gamma_p\left(\frac{2}{5}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	$p \Gamma_p\left(\frac{4}{15}\right) \Gamma_p\left(\frac{2}{5}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	$964 \cdot 1801+554 \cdot 1801^2+\dots$
	$x_1^2 x_2^3 y_3^1 y_4^2 e_1 e_2$	$-p \Gamma_p\left(\frac{2}{5}\right) \Gamma_p\left(\frac{3}{5}\right)$	$-p \Gamma_p\left(\frac{2}{5}\right) \Gamma_p\left(\frac{3}{5}\right)$	1801
	$x_1^2 x_2^3 y_3^2 y_4^1 e_1 e_2$	$-p \Gamma_p\left(\frac{2}{5}\right) \Gamma_p\left(\frac{3}{5}\right)$	$-p \Gamma_p\left(\frac{2}{5}\right) \Gamma_p\left(\frac{3}{5}\right)$	1801
	$x_1^3 x_2^1 x_3^2 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{7}{15}\right) \Gamma_p\left(\frac{1}{5}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	$p \Gamma_p\left(\frac{7}{15}\right) \Gamma_p\left(\frac{1}{5}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	$1782 \cdot 1801+1020 \cdot 1801^2+\dots$
	$x_1^3 x_2^2 y_3^1 y_4^2 e_1 e_2$	$-p \Gamma_p\left(\frac{3}{5}\right) \Gamma_p\left(\frac{2}{5}\right)$	$-p \Gamma_p\left(\frac{3}{5}\right) \Gamma_p\left(\frac{2}{5}\right)$	1801
	$x_1^3 x_2^2 y_3^2 y_4^1 e_1 e_2$	$-p \Gamma_p\left(\frac{3}{5}\right) \Gamma_p\left(\frac{2}{5}\right)$	$-p \Gamma_p\left(\frac{3}{5}\right) \Gamma_p\left(\frac{2}{5}\right)$	1801
	$x_1^3 x_2^4 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{8}{15}\right) \Gamma_p\left(\frac{4}{5}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	$p \Gamma_p\left(\frac{8}{15}\right) \Gamma_p\left(\frac{4}{5}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	$1327 \cdot 1801+1543 \cdot 1801^2+\dots$
	$x_1^4 x_2^1 y_3^2 y_4^2 e_1 e_2$	$-p \Gamma_p\left(\frac{4}{5}\right) \Gamma_p\left(\frac{1}{5}\right)$	$-p \Gamma_p\left(\frac{4}{5}\right) \Gamma_p\left(\frac{1}{5}\right)$	1801
	$x_1^4 x_2^1 y_3^2 y_4^1 e_1 e_2$	$-p \Gamma_p\left(\frac{4}{5}\right) \Gamma_p\left(\frac{1}{5}\right)$	$-p \Gamma_p\left(\frac{4}{5}\right) \Gamma_p\left(\frac{1}{5}\right)$	1801
	$x_1^4 x_2^3 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{11}{15}\right) \Gamma_p\left(\frac{3}{5}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	$p \Gamma_p\left(\frac{11}{15}\right) \Gamma_p\left(\frac{3}{5}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	$312 \cdot 1801+693 \cdot 1801^2+\dots$
	$x_1^5 x_2^2 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{14}{15}\right) \Gamma_p\left(\frac{2}{5}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	$p \Gamma_p\left(\frac{14}{15}\right) \Gamma_p\left(\frac{2}{5}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	$1445 \cdot 1801+718 \cdot 1801^2+\dots$
$H^{2,0}$	$x_1^5 x_2^4 x_3^2 x_4^2 e_1 e_2 e_3 e_4$	$-p^2 \Gamma_p\left(\frac{13}{15}\right) \Gamma_p\left(\frac{4}{5}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	$-p^2 \Gamma_p\left(\frac{13}{15}\right) \Gamma_p\left(\frac{4}{5}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	$1296 \cdot 1801^2+\dots$
$H^{2,2}$	$y_1^4 y_2^4 y_3^2 y_4^3$	p^2	p^2	3243601

Surface #41: $Q = (1, 2, 8, 5)$, $m = 16$

Equation: $x_0^{16} + x_1^8 + x_2^2 + x_0 x_3^3 = 0$

$p = 769$

Table 41: Trace contributions for $x_0^{16} + x_1^8 + x_2^2 + x_0 x_3^3 = 0$ at $p = 769$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{24}\right) \Gamma_p\left(\frac{1}{8}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{3}\right)$	$-\Gamma_p\left(\frac{1}{24}\right) \Gamma_p\left(\frac{1}{8}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{3}\right)$	$683+137\cdot 769+395\cdot 769^2+\dots$
$H^{1,1}$	$y_1^3 y_2^3 y_3^1 y_4^3$	p	p	769
	$y_1^5 y_2^5 y_3^1 y_4^1$	p	p	769
	$y_1^7 y_2^7 y_3^1 y_4^1$	p	p	769
	$y_1^9 y_2^1 y_3^1 y_4^3$	p	p	769
	$y_1^{11} y_2^3 y_3^1 y_4^1$	p	p	769
	$y_1^{13} y_2^5 y_3^1 y_4^1$	p	p	769
	$x_2^4 x_3^1 y_1^8 y_4^2 e_2 e_3$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	769
	$x_1^1 x_2^4 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{3}\right)$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{3}\right)$	769
	$x_2^7 x_3^6 x_1^1 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{12}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{3}\right)$	$p \Gamma_p\left(\frac{1}{12}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{3}\right)$	769
	$x_4^4 x_5^1 x_3^1 x_7^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{5}{24}\right) \Gamma_p\left(\frac{5}{8}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{3}\right)$	$p \Gamma_p\left(\frac{5}{24}\right) \Gamma_p\left(\frac{5}{8}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{3}\right)$	$736\cdot 769+442\cdot 769^2+\dots$
	$x_5^5 x_7^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{7}{24}\right) \Gamma_p\left(\frac{7}{8}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{3}\right)$	$p \Gamma_p\left(\frac{7}{24}\right) \Gamma_p\left(\frac{7}{8}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{3}\right)$	$214\cdot 769+562\cdot 769^2+\dots$
	$x_6^6 x_4^1 x_3^1 x_7^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{3}\right)$	$p \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{3}\right)$	769
	$x_7^7 x_6^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{5}{12}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{3}\right)$	$p \Gamma_p\left(\frac{5}{12}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{3}\right)$	769
	$x_8^8 x_3^1 x_7^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{11}{24}\right) \Gamma_p\left(\frac{11}{8}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{3}\right)$	$p \Gamma_p\left(\frac{11}{24}\right) \Gamma_p\left(\frac{11}{8}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{3}\right)$	$722\cdot 769+431\cdot 769^2+\dots$
	$x_9^9 x_5^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{13}{24}\right) \Gamma_p\left(\frac{13}{8}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{3}\right)$	$p \Gamma_p\left(\frac{13}{24}\right) \Gamma_p\left(\frac{13}{8}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{3}\right)$	$589\cdot 769+210\cdot 769^2+\dots$
	$x_{10}^{10} x_2^2 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{7}{12}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{3}\right)$	$p \Gamma_p\left(\frac{7}{12}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{3}\right)$	769
	$x_{11}^{11} x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{3}\right)$	$p \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{3}\right)$	769
	$x_{12}^{12} x_1^1 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{17}{24}\right) \Gamma_p\left(\frac{1}{8}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{3}\right)$	$p \Gamma_p\left(\frac{17}{24}\right) \Gamma_p\left(\frac{1}{8}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{3}\right)$	$442\cdot 769+409\cdot 769^2+\dots$
	$x_{13}^{13} x_2^3 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{19}{24}\right) \Gamma_p\left(\frac{3}{8}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{3}\right)$	$p \Gamma_p\left(\frac{19}{24}\right) \Gamma_p\left(\frac{3}{8}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{3}\right)$	$233\cdot 769+471\cdot 769^2+\dots$
	$x_{14}^{14} x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{11}{12}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{3}\right)$	$p \Gamma_p\left(\frac{11}{12}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{3}\right)$	769
$H^{2,0}$	$x_1^{16} x_2^2 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$-p^2 \Gamma_p\left(\frac{23}{24}\right) \Gamma_p\left(\frac{7}{8}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{3}\right)$	$-p^2 \Gamma_p\left(\frac{23}{24}\right) \Gamma_p\left(\frac{7}{8}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{3}\right)$	$152\cdot 769^2+\dots$
$H^{2,2}$	$y_1^{15} y_2^7 y_3^1 y_4^3$	p^2	p^2	591361

Surface #42: $Q = (1, 4, 8, 3)$, $m = 16$

Equation: $x_0^{16} + x_1^4 + x_2^2 + x_0 x_3^5 = 0$

$p = 641$

Table 42: Trace contributions for $x_0^{16} + x_1^4 + x_2^2 + x_0 x_3^5 = 0$ at $p = 641$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{20}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{5}\right)$	$-\Gamma_p\left(\frac{1}{20}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{5}\right)$	$277+452\cdot 641+463\cdot 641^2+\dots$
$H^{1,1}$	$y_1^3 y_2^3 y_3^3 y_4^3$	p	p	641
	$y_1^5 y_2^5 y_3^5 y_4^5$	p	p	641
	$y_1^7 y_2^7 y_3^7 y_4^7$	p	p	641
	$y_1^9 y_2^9 y_3^9 y_4^9$	p	p	641
	$y_1^{11} y_2^{11} y_3^{11} y_4^{11}$	p	p	641
	$y_1^{13} y_2^{13} y_3^{13} y_4^{13}$	p	p	641
	$x_2^2 x_3^1 y_1^4 y_4^1 e_2 e_3$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	641
	$x_2^2 x_3^1 y_1^8 y_4^1 e_2 e_3$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	641
	$x_2^2 x_3^1 y_1^{12} y_4^1 e_2 e_3$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	641
	$x_1^1 x_2^2 x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	641
	$x_1^3 x_2^3 x_3^3 x_4^3 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{3}{20}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{3}{5}\right)$	$p \Gamma_p\left(\frac{3}{20}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{3}{5}\right)$	$35\cdot 641+521\cdot 641^2+\dots$
	$x_1^4 x_2^2 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{5}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{4}{5}\right)$	$p \Gamma_p\left(\frac{1}{5}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{4}{5}\right)$	641
	$x_1^6 x_2^2 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{7}{20}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{5}\right)$	$p \Gamma_p\left(\frac{7}{20}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{5}\right)$	$522\cdot 641+235\cdot 641^2+\dots$
	$x_1^7 x_2^2 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{2}{5}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{3}{5}\right)$	$p \Gamma_p\left(\frac{2}{5}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{3}{5}\right)$	641
	$x_1^8 x_2^1 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{9}{20}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{4}{5}\right)$	$p \Gamma_p\left(\frac{9}{20}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{4}{5}\right)$	$42\cdot 641+12\cdot 641^2+\dots$
	$x_1^9 x_2^3 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{11}{20}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{5}\right)$	$p \Gamma_p\left(\frac{11}{20}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{5}\right)$	$290\cdot 641+634\cdot 641^2+\dots$
	$x_1^{10} x_2^2 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{3}{5}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{5}\right)$	$p \Gamma_p\left(\frac{3}{5}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{5}\right)$	641
	$x_1^{11} x_2^1 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{13}{20}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{3}{5}\right)$	$p \Gamma_p\left(\frac{13}{20}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{3}{5}\right)$	$237\cdot 641+168\cdot 641^2+\dots$
	$x_1^{13} x_2^2 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{4}{5}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{5}\right)$	$p \Gamma_p\left(\frac{4}{5}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{5}\right)$	641
	$x_1^{14} x_2^1 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{17}{20}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{5}\right)$	$p \Gamma_p\left(\frac{17}{20}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{5}\right)$	$348\cdot 641+167\cdot 641^2+\dots$
$H^{2,0}$	$x_1^{16} x_2^2 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$-p^2 \Gamma_p\left(\frac{19}{20}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{4}{5}\right)$	$-p^2 \Gamma_p\left(\frac{19}{20}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{4}{5}\right)$	$361\cdot 641^2+\dots$
$H^{2,2}$	$y_1^{15} y_2^3 y_3^1 y_4^5$	p^2	p^2	410881

Surface #43: $Q = (4, 1, 8, 3)$, $m = 16$

Equation: $x_0^4 + x_1^{16} + x_2^2 + x_0 x_3^4 = 0$

$p = 7681$

Table 43: Trace contributions for $x_0^4 + x_1^{16} + x_2^2 + x_0 x_3^4 = 0$ at $p = 7681$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{3}{16}\right)\Gamma_p\left(\frac{1}{16}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$-\Gamma_p\left(\frac{3}{16}\right)\Gamma_p\left(\frac{1}{16}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$2041+4149\cdot 7681+2933\cdot 7681^2+\dots$
$H^{1,1}$	$y_1^1 y_2^5 y_3^1 y_4^4$	p	p	7681
	$y_1^1 y_2^9 y_3^1 y_4^3$	p	p	7681
	$y_1^1 y_2^{13} y_3^1 y_4^2$	p	p	7681
	$y_1^3 y_2^3 y_3^1 y_4^3$	p	p	7681
	$y_1^3 y_2^7 y_3^1 y_4^2$	p	p	7681
	$y_1^3 y_2^{11} y_3^1 y_4^1$	p	p	7681
	$x_1^1 x_2^8 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{1}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{1}\right)$	7681
	$x_1^1 x_2^{11} x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{16}\right)\Gamma_p\left(\frac{11}{16}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$p\Gamma_p\left(\frac{1}{16}\right)\Gamma_p\left(\frac{11}{16}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$636\cdot 7681+5006\cdot 7681^2+\dots$
	$x_1^1 x_2^{14} x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{7}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{7}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	7681
	$x_1^2 x_3^1 y_2^4 y_4^3 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	7681
	$x_1^2 x_3^1 y_2^8 y_4^2 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	7681
	$x_1^2 x_3^1 y_2^{12} y_4^1 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	7681
	$x_1^2 x_2^7 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{16}\right)\Gamma_p\left(\frac{7}{16}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$p\Gamma_p\left(\frac{5}{16}\right)\Gamma_p\left(\frac{7}{16}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$823\cdot 7681+568\cdot 7681^2+\dots$
	$x_1^2 x_2^{10} x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{8}\right)\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{3}{8}\right)\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	7681
	$x_1^2 x_2^{13} x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{16}\right)\Gamma_p\left(\frac{13}{16}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$p\Gamma_p\left(\frac{7}{16}\right)\Gamma_p\left(\frac{13}{16}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$702\cdot 7681+5894\cdot 7681^2+\dots$
	$x_1^3 x_2^3 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{9}{16}\right)\Gamma_p\left(\frac{3}{16}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$p\Gamma_p\left(\frac{9}{16}\right)\Gamma_p\left(\frac{3}{16}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$1499\cdot 7681+4122\cdot 7681^2+\dots$
	$x_1^3 x_2^6 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{3}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{3}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	7681
	$x_1^3 x_2^9 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{11}{16}\right)\Gamma_p\left(\frac{9}{16}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$p\Gamma_p\left(\frac{11}{16}\right)\Gamma_p\left(\frac{9}{16}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$28\cdot 7681+102\cdot 7681^2+\dots$
	$x_1^4 x_2^2 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{8}\right)\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{7}{8}\right)\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	7681
	$x_1^4 x_2^5 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{15}{16}\right)\Gamma_p\left(\frac{5}{16}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$p\Gamma_p\left(\frac{15}{16}\right)\Gamma_p\left(\frac{5}{16}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$7524\cdot 7681+329\cdot 7681^2+\dots$
$H^{2,0}$	$x_1^4 x_2^{15} x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{13}{16}\right)\Gamma_p\left(\frac{15}{16}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$-p^2\Gamma_p\left(\frac{13}{16}\right)\Gamma_p\left(\frac{15}{16}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$4087\cdot 7681^2+\dots$
$H^{2,2}$	$y_1^3 y_2^{15} y_3^1 y_4^4$	p^2	p^2	58997761

Surface #44: $Q = (2, 6, 9, 1)$, $m = 18$

Equation: $x_0^9 + x_1^3 + x_2^2 + x_0 x_3^{16} = 0$

$p = 2593$

Table 44: Trace contributions for $x_0^9 + x_1^3 + x_2^2 + x_0 x_3^{16} = 0$ at $p = 2593$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{5}{48}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{16}\right)$	$-\Gamma_p\left(\frac{5}{48}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{16}\right)$	$2381+1047\cdot 2593+1386\cdot 2593^2+\dots$
$H^{1,1}$	$y_1^2 y_2^2 y_3^1 y_4^{10}$	p	p	2593
	$y_1^4 y_2^1 y_3^1 y_4^{12}$	p	p	2593
	$y_1^5 y_2^2 y_3^1 y_4^5$	p	p	2593
	$y_1^7 y_2^1 y_3^1 y_4^7$	p	p	2593
	$x_1^1 x_2^2 x_3^1 x_4^{13} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{48}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{13}{16}\right)$	$p\Gamma_p\left(\frac{1}{48}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{13}{16}\right)$	$2490\cdot 2593+2034\cdot 2593^2+\dots$
	$x_1^2 x_2^2 x_3^1 x_4^{11} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{48}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{11}{16}\right)$	$p\Gamma_p\left(\frac{7}{48}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{11}{16}\right)$	$634\cdot 2593+903\cdot 2593^2+\dots$
	$x_1^3 x_2^1 x_3^1 x_4^{15} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{11}{48}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{15}{16}\right)$	$p\Gamma_p\left(\frac{11}{48}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{15}{16}\right)$	$2060\cdot 2593+1299\cdot 2593^2+\dots$
	$x_1^3 x_2^2 y_3^1 y_4^8 e_1 e_2$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	2593
	$x_1^3 x_2^2 x_3^1 x_4^9 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{13}{48}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{9}{16}\right)$	$p\Gamma_p\left(\frac{13}{48}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{9}{16}\right)$	$244\cdot 2593+1015\cdot 2593^2+\dots$
	$x_1^4 x_2^1 x_3^1 x_4^{13} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{19}{48}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{13}{16}\right)$	$p\Gamma_p\left(\frac{19}{48}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{13}{16}\right)$	$271\cdot 2593+642\cdot 2593^2+\dots$
	$x_1^4 x_2^2 x_3^1 x_4^7 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{19}{48}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{16}\right)$	$p\Gamma_p\left(\frac{19}{48}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{16}\right)$	$1786\cdot 2593+1796\cdot 2593^2+\dots$
	$x_1^5 x_2^1 x_3^1 x_4^{11} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{23}{48}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{11}{16}\right)$	$p\Gamma_p\left(\frac{23}{48}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{11}{16}\right)$	$1774\cdot 2593+1818\cdot 2593^2+\dots$
	$x_1^5 x_2^2 x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{25}{48}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{16}\right)$	$p\Gamma_p\left(\frac{25}{48}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{16}\right)$	$858\cdot 2593+1671\cdot 2593^2+\dots$
	$x_1^6 x_2^1 y_3^1 y_4^8 e_1 e_2$	$-p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	$-p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	2593
	$x_1^6 x_2^1 x_3^1 x_4^9 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{29}{48}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{9}{16}\right)$	$p\Gamma_p\left(\frac{29}{48}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{9}{16}\right)$	$196\cdot 2593+1471\cdot 2593^2+\dots$
	$x_1^6 x_2^2 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{31}{48}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{16}\right)$	$p\Gamma_p\left(\frac{31}{48}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{16}\right)$	$2172\cdot 2593+2116\cdot 2593^2+\dots$
	$x_1^7 x_2^1 x_3^1 x_4^7 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{35}{48}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{16}\right)$	$p\Gamma_p\left(\frac{35}{48}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{16}\right)$	$627\cdot 2593+2365\cdot 2593^2+\dots$
	$x_1^7 x_2^2 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{37}{48}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{16}\right)$	$p\Gamma_p\left(\frac{37}{48}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{16}\right)$	$180\cdot 2593+2166\cdot 2593^2+\dots$
	$x_1^8 x_2^1 x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{41}{48}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{16}\right)$	$p\Gamma_p\left(\frac{41}{48}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{16}\right)$	$2229\cdot 2593+1037\cdot 2593^2+\dots$
	$x_1^9 x_2^1 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{47}{48}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{16}\right)$	$p\Gamma_p\left(\frac{47}{48}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{16}\right)$	$1586\cdot 2593+1452\cdot 2593^2+\dots$
$H^{2,0}$	$x_1^9 x_2^2 x_3^1 x_4^{15} e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{43}{48}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{15}{16}\right)$	$-p^2\Gamma_p\left(\frac{43}{48}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{15}{16}\right)$	$159\cdot 2593^2+\dots$
$H^{2,2}$	$y_1^8 y_2^2 y_3^1 y_4^{16}$	p^2	p^2	6723649

Surface #45: $Q = (6, 1, 9, 2)$, $m = 18$

Equation: $x_0^3 + x_1^{18} + x_2^2 + x_0 x_3^6 = 0$

$p = 1297$

Table 45: Trace contributions for $x_0^3 + x_1^{18} + x_2^2 + x_0 x_3^6 = 0$ at $p = 1297$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{5}{18}\right)\Gamma_p\left(\frac{1}{18}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{6}\right)$	$-\Gamma_p\left(\frac{5}{18}\right)\Gamma_p\left(\frac{1}{18}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{6}\right)$	$383+451\cdot 1297+395\cdot 1297^2+\dots$
$H^{1,1}$	$y_1^1 y_2^1 y_3^1 y_4^5$	p	p	1297
	$y_1^1 y_2^1 y_3^1 y_4^3$	p	p	1297
	$y_1^2 y_2^5 y_3^1 y_4^4$	p	p	1297
	$y_1^2 y_2^{11} y_3^1 y_4^2$	p	p	1297
	$x_1^1 x_2^6 y_3^9 y_4^1 e_1 e_4$	$-p\Gamma_p\left(\frac{1}{1}\right)$	$-p\Gamma_p\left(\frac{1}{1}\right)$	1297
	$x_1^1 x_2^9 y_3^6 y_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{1}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{1}\right)$	1297
	$x_1^1 x_2^{11} x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{18}\right)\Gamma_p\left(\frac{11}{18}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{6}\right)$	$p\Gamma_p\left(\frac{1}{18}\right)\Gamma_p\left(\frac{11}{18}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{6}\right)$	$1184\cdot 1297+789\cdot 1297^2+\dots$
	$x_1^1 x_2^{13} x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{13}{18}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{13}{18}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$104\cdot 1297+452\cdot 1297^2+\dots$
	$x_1^1 x_2^{15} x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	1297
	$x_1^1 x_2^{17} x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{17}{18}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{17}{18}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$440\cdot 1297+640\cdot 1297^2+\dots$
	$x_1^2 x_2^3 y_3^9 y_4^1 e_1 e_4$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	1297
	$x_1^2 x_2^5 x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{18}\right)\Gamma_p\left(\frac{5}{18}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{6}\right)$	$p\Gamma_p\left(\frac{7}{18}\right)\Gamma_p\left(\frac{5}{18}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{6}\right)$	$660\cdot 1297+731\cdot 1297^2+\dots$
	$x_1^2 x_2^7 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{4}{9}\right)\Gamma_p\left(\frac{7}{18}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{4}{9}\right)\Gamma_p\left(\frac{7}{18}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$347\cdot 1297+451\cdot 1297^2+\dots$
	$x_1^2 x_2^9 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	1297
	$x_1^2 x_2^{11} x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{9}\right)\Gamma_p\left(\frac{11}{18}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{5}{9}\right)\Gamma_p\left(\frac{11}{18}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$1069\cdot 1297+146\cdot 1297^2+\dots$
	$x_1^2 x_2^{13} x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{11}{18}\right)\Gamma_p\left(\frac{13}{18}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{6}\right)$	$p\Gamma_p\left(\frac{11}{18}\right)\Gamma_p\left(\frac{13}{18}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{6}\right)$	$564\cdot 1297+835\cdot 1297^2+\dots$
	$x_1^3 x_2^1 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{9}\right)\Gamma_p\left(\frac{1}{18}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{7}{9}\right)\Gamma_p\left(\frac{1}{18}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$846\cdot 1297+393\cdot 1297^2+\dots$
	$x_1^3 x_2^3 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	1297
	$x_1^3 x_2^5 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{8}{9}\right)\Gamma_p\left(\frac{5}{18}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{8}{9}\right)\Gamma_p\left(\frac{5}{18}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$1085\cdot 1297+509\cdot 1297^2+\dots$
	$x_1^3 x_2^7 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{17}{18}\right)\Gamma_p\left(\frac{7}{18}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{6}\right)$	$p\Gamma_p\left(\frac{17}{18}\right)\Gamma_p\left(\frac{7}{18}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{6}\right)$	$1033\cdot 1297+1287\cdot 1297^2+\dots$
$H^{2,0}$	$x_1^3 x_2^{17} x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{13}{18}\right)\Gamma_p\left(\frac{17}{18}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{6}\right)$	$-p^2\Gamma_p\left(\frac{13}{18}\right)\Gamma_p\left(\frac{17}{18}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{6}\right)$	$1148\cdot 1297^2+\dots$
$H^{2,2}$	$y_1^2 y_2^{17} y_3^1 y_4^6$	p^2	p^2	1682209

Surface #46: $Q = (6, 2, 9, 1)$, $m = 18$

Equation: $x_0^3 + x_1^9 + x_2^2 + x_0x_3^{12} = 0$

$p = 1297$

Table 46: Trace contributions for $x_0^3 + x_1^9 + x_2^2 + x_0x_3^{12} = 0$ at $p = 1297$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{11}{36}\right)\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{12}\right)$	$-\Gamma_p\left(\frac{11}{36}\right)\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{12}\right)$	$190+127\cdot 1297+430\cdot 1297^2+\dots$
$H^{1,1}$	$y_1^1 y_2^4 y_3^1 y_4^9$	p	p	1297
	$y_1^1 y_2^2 y_3^1 y_4^5$	p	p	1297
	$y_1^2 y_2^2 y_3^1 y_4^8$	p	p	1297
	$y_1^2 y_2^5 y_3^1 y_4^4$	p	p	1297
	$x_1^1 x_2^3 x_3^1 x_4^{11} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{36}\right)\Gamma_p\left(\frac{5}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{11}{12}\right)$	$p\Gamma_p\left(\frac{1}{36}\right)\Gamma_p\left(\frac{5}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{11}{12}\right)$	$767\cdot 1297+1045\cdot 1297^2+\dots$
	$x_1^1 x_2^6 y_3^1 y_4^6 e_1 e_2$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	1297
	$x_1^1 x_2^6 x_3^9 x_4^1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$p\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	1297
	$x_1^1 x_2^7 x_3^7 x_4^1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{36}\right)\Gamma_p\left(\frac{7}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{12}\right)$	$p\Gamma_p\left(\frac{5}{36}\right)\Gamma_p\left(\frac{7}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{12}\right)$	$102\cdot 1297+848\cdot 1297^2+\dots$
	$x_1^1 x_2^8 x_3^3 x_4^1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{36}\right)\Gamma_p\left(\frac{8}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{12}\right)$	$p\Gamma_p\left(\frac{7}{36}\right)\Gamma_p\left(\frac{8}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{12}\right)$	$1024\cdot 1297+196\cdot 1297^2+\dots$
	$x_1^2 x_2^5 x_3^1 x_4^{11} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{13}{36}\right)\Gamma_p\left(\frac{5}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{11}{12}\right)$	$p\Gamma_p\left(\frac{13}{36}\right)\Gamma_p\left(\frac{5}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{11}{12}\right)$	$672\cdot 1297+876\cdot 1297^2+\dots$
	$x_1^2 x_2^3 y_3^1 y_4^6 e_1 e_2$	$-p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	$-p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	1297
	$x_1^2 x_2^3 x_3^9 x_4^1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$p\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	1297
	$x_1^2 x_2^4 x_3^7 x_4^1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{17}{36}\right)\Gamma_p\left(\frac{7}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{12}\right)$	$p\Gamma_p\left(\frac{17}{36}\right)\Gamma_p\left(\frac{7}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{12}\right)$	$434\cdot 1297+509\cdot 1297^2+\dots$
	$x_1^2 x_2^5 x_3^5 x_4^1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{19}{36}\right)\Gamma_p\left(\frac{5}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{12}\right)$	$p\Gamma_p\left(\frac{19}{36}\right)\Gamma_p\left(\frac{5}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{12}\right)$	$260\cdot 1297+439\cdot 1297^2+\dots$
	$x_1^2 x_2^6 x_3^3 x_4^1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$p\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	1297
	$x_1^2 x_2^7 x_3^3 x_4^1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{23}{36}\right)\Gamma_p\left(\frac{7}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{12}\right)$	$p\Gamma_p\left(\frac{23}{36}\right)\Gamma_p\left(\frac{7}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{12}\right)$	$276\cdot 1297+1213\cdot 1297^2+\dots$
	$x_1^3 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{31}{36}\right)\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{12}\right)$	$p\Gamma_p\left(\frac{31}{36}\right)\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{12}\right)$	$19\cdot 1297+294\cdot 1297^2+\dots$
	$x_1^3 x_2^2 x_3^5 x_4^1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{31}{36}\right)\Gamma_p\left(\frac{2}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{12}\right)$	$p\Gamma_p\left(\frac{31}{36}\right)\Gamma_p\left(\frac{2}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{12}\right)$	$1208\cdot 1297+828\cdot 1297^2+\dots$
	$x_1^3 x_2^3 x_3^3 x_4^1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$p\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	1297
	$x_1^3 x_2^4 x_3^1 x_4^1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{35}{36}\right)\Gamma_p\left(\frac{4}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{12}\right)$	$p\Gamma_p\left(\frac{35}{36}\right)\Gamma_p\left(\frac{4}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{12}\right)$	$301\cdot 1297+1257\cdot 1297^2+\dots$
$H^{2,0}$	$x_1^3 x_2^8 x_3^1 x_4^{11} e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{25}{36}\right)\Gamma_p\left(\frac{8}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{11}{12}\right)$	$-p^2\Gamma_p\left(\frac{25}{36}\right)\Gamma_p\left(\frac{8}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{11}{12}\right)$	$1140\cdot 1297^2+\dots$
$H^{2,2}$	$y_1^2 y_2^8 y_3^1 y_4^{12}$	p^2	p^2	1682209

Surface #47: $Q = (9, 2, 6, 1)$, $m = 18$

Equation: $x_0^2 + x_1^9 + x_2^3 + x_0x_3^9 = 0$

$p = 487$

Table 47: Trace contributions for $x_0^2 + x_1^9 + x_2^3 + x_0x_3^9 = 0$ at $p = 487$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{4}{9}\right)\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{9}\right)$	$-\Gamma_p\left(\frac{4}{9}\right)\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{9}\right)$	$441+179\cdot 487+338\cdot 487^2+\dots$
$H^{1,1}$	$y_1^1 y_2^2 y_3^2 y_4^6$	p	p	487
	$y_1^1 y_2^4 y_3^4 y_4^4$	p	p	487
	$y_1^1 y_2^5 y_3^2 y_4^3$	p	p	487
	$y_1^1 y_2^7 y_3^1 y_4^4$	p	p	487
	$x_2^3 x_3^2 y_1^1 y_4^2 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	487
	$x_2^6 x_3^1 y_1^1 y_4^5 e_2 e_3$	$-p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	$-p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	487
	$x_1^1 x_2^3 x_3^9 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{9}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{9}\right)$	487
	$x_1^1 x_2^4 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{4}{9}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{7}{9}\right)$	$p\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{4}{9}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{7}{9}\right)$	$232\cdot 487+160\cdot 487^2+\dots$
	$x_1^1 x_2^5 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{9}\right)\Gamma_p\left(\frac{5}{9}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{8}{9}\right)$	$p\Gamma_p\left(\frac{2}{9}\right)\Gamma_p\left(\frac{5}{9}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{8}{9}\right)$	$115\cdot 487+389\cdot 487^2+\dots$
	$x_1^1 x_2^6 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{9}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{9}\right)$	487
	$x_1^1 x_2^7 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{7}{9}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{9}\right)$	$p\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{7}{9}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{9}\right)$	487
	$x_1^1 x_2^8 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{9}\right)\Gamma_p\left(\frac{8}{9}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{9}\right)$	$p\Gamma_p\left(\frac{2}{9}\right)\Gamma_p\left(\frac{8}{9}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{9}\right)$	$142\cdot 487+163\cdot 487^2+\dots$
	$x_1^1 x_2^9 x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{9}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{9}\right)$	$232\cdot 487+160\cdot 487^2+\dots$
	$x_1^1 x_2^8 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{9}\right)\Gamma_p\left(\frac{8}{9}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{9}\right)$	$p\Gamma_p\left(\frac{2}{9}\right)\Gamma_p\left(\frac{8}{9}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{9}\right)$	$254\cdot 487+326\cdot 487^2+\dots$
	$x_1^2 x_2^1 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{4}{9}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{7}{9}\right)$	$p\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{4}{9}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{7}{9}\right)$	$232\cdot 487+160\cdot 487^2+\dots$
	$x_1^2 x_2^2 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{9}\right)\Gamma_p\left(\frac{5}{9}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{8}{9}\right)$	$p\Gamma_p\left(\frac{2}{9}\right)\Gamma_p\left(\frac{5}{9}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{8}{9}\right)$	$254\cdot 487+326\cdot 487^2+\dots$
	$x_1^2 x_2^3 x_3^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{9}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{9}\right)$	$463\cdot 487+421\cdot 487^2+\dots$
	$x_1^2 x_2^4 x_3^1 x_4^6 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{4}{9}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{7}{9}\right)$	$p\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{4}{9}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{7}{9}\right)$	487
	$x_1^2 x_2^5 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{9}\right)\Gamma_p\left(\frac{5}{9}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{8}{9}\right)$	$p\Gamma_p\left(\frac{2}{9}\right)\Gamma_p\left(\frac{5}{9}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{8}{9}\right)$	$72\cdot 487+328\cdot 487^2+\dots$
	$x_1^2 x_2^6 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{9}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{9}\right)$	$254\cdot 487+326\cdot 487^2+\dots$
$H^{2,0}$	$x_1^2 x_2^8 x_3^2 x_4^8 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{5}{9}\right)\Gamma_p\left(\frac{8}{9}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{8}{9}\right)$	$-p^2\Gamma_p\left(\frac{5}{9}\right)\Gamma_p\left(\frac{8}{9}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{8}{9}\right)$	$307\cdot 487^2+\dots$
$H^{2,2}$	$y_1^1 y_2^8 y_3^2 y_4^9$	p^2	p^2	237169

Surface #48: $Q = (3, 1, 9, 5)$, $m = 18$

Equation: $x_0^6 + x_1^{18} + x_2^2 + x_0 x_3^3 = 0$

$p = 1297$

Table 48: Trace contributions for $x_0^6 + x_1^{18} + x_2^2 + x_0 x_3^3 = 0$ at $p = 1297$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{1}{18}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$-\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{1}{18}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$1065+544\cdot 1297+1271\cdot 1297^2+\dots$
$H^{1,1}$	$y_1^1 y_2^7 y_3^1 y_4^3$	p	p	1297
	$y_1^1 y_2^{13} y_3^1 y_4^2$	p	p	1297
	$y_1^3 y_2^3 y_3^1 y_4^3$	p	p	1297
	$y_1^3 y_2^9 y_3^1 y_4^2$	p	p	1297
	$y_1^3 y_2^{15} y_3^1 y_4^1$	p	p	1297
	$y_1^5 y_2^5 y_3^1 y_4^2$	p	p	1297
	$y_1^5 y_2^{11} y_3^1 y_4^1$	p	p	1297
	$x_1^1 x_2^9 x_3^3 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{1}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{1}\right)$	1297
	$x_1^1 x_2^{14} x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{18}\right)\Gamma_p\left(\frac{7}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{1}{18}\right)\Gamma_p\left(\frac{7}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$846\cdot 1297+393\cdot 1297^2+\dots$
	$x_2^2 x_1^{11} x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{9}\right)\Gamma_p\left(\frac{11}{18}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{2}{9}\right)\Gamma_p\left(\frac{11}{18}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$1121\cdot 1297+432\cdot 1297^2+\dots$
	$x_2^2 x_1^{16} x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{18}\right)\Gamma_p\left(\frac{8}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{5}{18}\right)\Gamma_p\left(\frac{8}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$1085\cdot 1297+509\cdot 1297^2+\dots$
	$x_3^3 x_1^1 y_2^6 y_4^2 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	1297
	$x_3^3 x_1^1 y_2^{12} y_4^1 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	1297
	$x_3^3 x_2^8 x_1^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{18}\right)\Gamma_p\left(\frac{4}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{7}{18}\right)\Gamma_p\left(\frac{4}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$347\cdot 1297+451\cdot 1297^2+\dots$
	$x_3^3 x_2^{13} x_1^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{4}{9}\right)\Gamma_p\left(\frac{13}{18}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{4}{9}\right)\Gamma_p\left(\frac{13}{18}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$291\cdot 1297+702\cdot 1297^2+\dots$
	$x_4^4 x_2^5 x_3^3 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{9}\right)\Gamma_p\left(\frac{5}{18}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{5}{9}\right)\Gamma_p\left(\frac{5}{18}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$156\cdot 1297+1237\cdot 1297^2+\dots$
	$x_4^4 x_2^{10} x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{11}{18}\right)\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{11}{18}\right)\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$1069\cdot 1297+146\cdot 1297^2+\dots$
	$x_5^5 x_2^2 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{13}{18}\right)\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{13}{18}\right)\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$104\cdot 1297+452\cdot 1297^2+\dots$
	$x_1^5 x_2^7 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{9}\right)\Gamma_p\left(\frac{7}{18}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{7}{9}\right)\Gamma_p\left(\frac{7}{18}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$479\cdot 1297+957\cdot 1297^2+\dots$
	$x_1^6 x_2^4 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{17}{18}\right)\Gamma_p\left(\frac{2}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{17}{18}\right)\Gamma_p\left(\frac{2}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$440\cdot 1297+640\cdot 1297^2+\dots$
$H^{2,0}$	$x_1^6 x_2^{17} x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{8}{9}\right)\Gamma_p\left(\frac{17}{18}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$-p^2\Gamma_p\left(\frac{8}{9}\right)\Gamma_p\left(\frac{17}{18}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$587\cdot 1297^2+\dots$
$H^{2,2}$	$y_1^5 y_2^{17} y_3^1 y_4^3$	p^2	p^2	1682209

Surface #49: $Q = (2, 3, 9, 4)$, $m = 18$

Equation: $x_0^9 + x_1^6 + x_2^2 + x_0x_3^4 = 0$

$p = 433$

Table 49: Trace contributions for $x_0^9 + x_1^6 + x_2^2 + x_0x_3^4 = 0$ at $p = 433$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$-\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$365+67\cdot 433+36\cdot 433^2+\dots$
$H^{1,1}$	$y_1^2 y_2^5 y_3^1 y_4^2$	p	p	433
	$y_1^3 y_2^3 y_3^1 y_4^3$	p	p	433
	$y_1^4 y_2^1 y_3^1 y_4^4$	p	p	433
	$y_1^5 y_2^5 y_3^1 y_4^1$	p	p	433
	$y_1^6 y_2^3 y_3^1 y_4^2$	p	p	433
	$y_1^7 y_2^1 y_3^1 y_4^3$	p	p	433
	$x_2^3 x_3^1 y_1^3 y_4^3 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	433
	$x_2^3 x_3^1 y_1^6 y_4^2 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	433
	$x_1^1 x_4^3 y_2^3 y_3^1 e_1 e_4$	$-p\Gamma_p\left(\frac{1}{4}\right)$	$-p\Gamma_p\left(\frac{1}{4}\right)$	433
	$x_1^1 x_3^3 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	433
	$x_1^2 x_2^5 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	433
	$x_1^3 x_4^3 y_2^3 y_3^1 e_1 e_4$	$-p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{3}{4}\right)$	$-p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{3}{4}\right)$	433
	$x_1^3 x_2^3 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	433
	$x_1^4 x_2^5 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$p\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$416\cdot 433+224\cdot 433^2+\dots$
	$x_1^5 x_4^3 y_2^3 y_3^1 e_1 e_4$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	433
	$x_1^5 x_2^3 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	433
	$x_1^6 x_2^1 x_3^3 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$p\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$382\cdot 433+292\cdot 433^2+\dots$
	$x_1^7 x_4^3 y_2^3 y_3^1 e_1 e_4$	$-p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{4}\right)$	$-p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{4}\right)$	433
	$x_1^7 x_2^3 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	433
	$x_1^8 x_2^1 x_3^3 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	433
$H^{2,0}$	$x_1^9 x_2^5 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$-p^2\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$312\cdot 433^2+\dots$
$H^{2,2}$	$y_1^8 y_2^5 y_3^1 y_4^4$	p^2	p^2	187489

Surface #50: $Q = (4, 5, 10, 1)$, $m = 20$

Equation: $x_0^5 + x_1^4 + x_2^2 + x_0 x_3^{16} = 0$

$p = 641$

Table 50: Trace contributions for $x_0^5 + x_1^4 + x_2^2 + x_0 x_3^{16} = 0$ at $p = 641$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{3}{16}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{16}\right)$	$-\Gamma_p\left(\frac{3}{16}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{16}\right)$	$251+509\cdot 641+311\cdot 641^2+\dots$
$H^{1,1}$	$y_1^1 y_2^3 y_3^1 y_4^9$	p	p	641
	$y_1^2 y_2^3 y_3^1 y_4^{14}$	p	p	641
	$y_1^2 y_2^3 y_3^1 y_4^6$	p	p	641
	$y_1^3 y_2^1 y_3^1 y_4^{11}$	p	p	641
	$y_1^3 y_2^3 y_3^1 y_4^3$	p	p	641
	$y_1^4 y_2^1 y_3^1 y_4^8$	p	p	641
	$x_2^2 x_3^1 y_1^1 y_4^{13} e_2 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	641
	$x_2^2 x_3^1 y_1^2 y_4^{10} e_2 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	641
	$x_2^2 x_3^1 y_1^3 y_4^7 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	641
	$x_2^2 x_3^1 y_1^4 y_4^4 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	641
	$x_1^1 x_2^2 x_3^1 x_4^{16} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{16}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{16}\right)$	641
	$x_1^1 x_2^3 x_3^1 x_4^{11} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{16}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{11}{16}\right)$	$p\Gamma_p\left(\frac{1}{16}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{11}{16}\right)$	$446\cdot 641+489\cdot 641^2+\dots$
	$x_1^2 x_2^2 x_3^1 x_4^{12} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	641
	$x_1^2 x_2^3 x_3^1 x_4^7 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{16}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{16}\right)$	$p\Gamma_p\left(\frac{5}{16}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{16}\right)$	$302\cdot 641+93\cdot 641^2+\dots$
	$x_1^3 x_2^1 x_3^1 x_4^{13} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{16}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{13}{16}\right)$	$p\Gamma_p\left(\frac{7}{16}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{13}{16}\right)$	$592\cdot 641+459\cdot 641^2+\dots$
	$x_1^3 x_2^2 x_3^1 x_4^8 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	641
	$x_1^3 x_2^3 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{9}{16}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{16}\right)$	$p\Gamma_p\left(\frac{9}{16}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{16}\right)$	$484\cdot 641+69\cdot 641^2+\dots$
	$x_1^4 x_2^1 x_3^1 x_4^9 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{11}{16}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{9}{16}\right)$	$p\Gamma_p\left(\frac{11}{16}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{9}{16}\right)$	$537\cdot 641+513\cdot 641^2+\dots$
	$x_1^4 x_2^2 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	641
	$x_1^5 x_2^1 x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{15}{16}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{16}\right)$	$p\Gamma_p\left(\frac{15}{16}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{16}\right)$	$332\cdot 641+459\cdot 641^2+\dots$
$H^{2,0}$	$x_1^5 x_2^3 x_3^1 x_4^{15} e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{13}{16}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{15}{16}\right)$	$-p^2\Gamma_p\left(\frac{13}{16}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{15}{16}\right)$	$166\cdot 641^2+\dots$
$H^{2,2}$	$y_1^4 y_2^3 y_3^1 y_4^{16}$	p^2	p^2	410881

Surface #51: $Q = (5, 4, 10, 1)$, $m = 20$

Equation: $x_0^4 + x_1^5 + x_2^2 + x_0 x_3^{15} = 0$

$p = 601$

Table 51: Trace contributions for $x_0^4 + x_1^5 + x_2^2 + x_0 x_3^{15} = 0$ at $p = 601$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{7}{30}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{15}\right)$	$-\Gamma_p\left(\frac{7}{30}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{15}\right)$	$419+58\cdot 601+344\cdot 601^2+\dots$
$H^{1,1}$	$y_1^1 y_2^2 y_3^1 y_4^{13}$	p	p	601
	$y_1^1 y_2^3 y_3^1 y_4^{10}$	p	p	601
	$y_1^1 y_2^4 y_3^1 y_4^7$	p	p	601
	$y_1^3 y_2^1 y_3^1 y_4^9$	p	p	601
	$y_1^3 y_2^2 y_3^1 y_4^6$	p	p	601
	$y_1^3 y_2^3 y_3^1 y_4^3$	p	p	601
	$x_1^1 x_2^3 x_3^1 x_4^{13} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{30}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{13}{15}\right)$	$p\Gamma_p\left(\frac{1}{30}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{13}{15}\right)$	$38\cdot 601+93\cdot 601^2+\dots$
	$x_1^1 x_2^4 x_3^1 x_4^9 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{10}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{5}\right)$	$p\Gamma_p\left(\frac{1}{10}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{5}\right)$	$432\cdot 601+213\cdot 601^2+\dots$
	$x_1^2 x_2^1 y_3^1 y_4^{12} e_1 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	601
	$x_1^2 x_2^2 y_3^1 y_4^9 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	601
	$x_1^2 x_2^3 y_3^1 y_4^6 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	601
	$x_1^2 x_2^4 y_3^1 y_4^3 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	601
	$x_1^2 x_2^3 x_3^1 x_4^{12} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{10}\right)\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{15}\right)$	$p\Gamma_p\left(\frac{3}{10}\right)\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{15}\right)$	$423\cdot 601+34\cdot 601^2+\dots$
	$x_1^2 x_2^4 x_3^1 x_4^8 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{13}{30}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{8}{15}\right)$	$p\Gamma_p\left(\frac{13}{30}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{8}{15}\right)$	$148\cdot 601+408\cdot 601^2+\dots$
	$x_1^2 x_2^4 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{13}{30}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{15}\right)$	$p\Gamma_p\left(\frac{13}{30}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{15}\right)$	$338\cdot 601+348\cdot 601^2+\dots$
	$x_1^3 x_2^1 x_3^1 x_4^{11} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{17}{30}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{11}{15}\right)$	$p\Gamma_p\left(\frac{17}{30}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{11}{15}\right)$	$585\cdot 601+316\cdot 601^2+\dots$
	$x_1^3 x_2^2 x_3^1 x_4^8 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{19}{30}\right)\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{15}\right)$	$p\Gamma_p\left(\frac{19}{30}\right)\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{15}\right)$	$467\cdot 601+527\cdot 601^2+\dots$
	$x_1^3 x_2^3 x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{9}{10}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{15}\right)$	$p\Gamma_p\left(\frac{9}{10}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{15}\right)$	$314\cdot 601+436\cdot 601^2+\dots$
	$x_1^4 x_2^1 x_3^1 x_4^6 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{19}{10}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{5}\right)$	$p\Gamma_p\left(\frac{19}{10}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{5}\right)$	$32\cdot 601+517\cdot 601^2+\dots$
	$x_1^4 x_2^2 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{29}{30}\right)\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{15}\right)$	$p\Gamma_p\left(\frac{29}{30}\right)\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{15}\right)$	$174\cdot 601+507\cdot 601^2+\dots$
$H^{2,0}$	$x_1^4 x_2^4 x_3^1 x_4^{14} e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{23}{30}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{14}{15}\right)$	$-p^2\Gamma_p\left(\frac{23}{30}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{14}{15}\right)$	$459\cdot 601^2+\dots$
$H^{2,2}$	$y_1^3 y_2^4 y_3^1 y_4^{15}$	p^2	p^2	361201

Surface #52: $Q = (10, 1, 4, 5)$, $m = 20$

Equation: $x_0^2 + x_1^{20} + x_2^5 + x_0 x_3^2 = 0$

$p = 401$

Table 52: Trace contributions for $x_0^2 + x_1^{20} + x_2^5 + x_0 x_3^2 = 0$ at $p = 401$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{20}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{20}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)$	$46+154\cdot 401+158\cdot 401^2+\dots$
$H^{1,1}$	$y_1^1 y_2^3 y_3^3 y_4^2$	p	p	401
	$y_1^1 y_2^3 y_3^2 y_4^2$	p	p	401
	$y_1^1 y_2^9 y_3^4 y_4^1$	p	p	401
	$y_1^1 y_2^{11} y_3^1 y_4^2$	p	p	401
	$y_1^1 y_2^{13} y_3^3 y_4^1$	p	p	401
	$y_1^1 y_2^{17} y_3^2 y_4^1$	p	p	401
	$x_1^1 x_2^4 y_3^4 y_4^1 e_1 e_4$	$-p\Gamma_p\left(\frac{1}{1}\right)$	$-p\Gamma_p\left(\frac{1}{1}\right)$	401
	$x_1^1 x_2^8 y_3^3 e_1 e_4$	$-p\Gamma_p\left(\frac{1}{1}\right)$	$-p\Gamma_p\left(\frac{1}{1}\right)$	401
	$x_1^1 x_2^2 y_3^{12} y_4^1 e_1 e_4$	$-p\Gamma_p\left(\frac{1}{1}\right)$	$-p\Gamma_p\left(\frac{1}{1}\right)$	401
	$x_1^1 x_2^2 y_3^{16} y_4^1 e_1 e_4$	$-p\Gamma_p\left(\frac{1}{1}\right)$	$-p\Gamma_p\left(\frac{1}{1}\right)$	401
	$x_1^1 x_2^4 x_3^4 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{1}\right)$	$p\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{1}\right)$	401
	$x_1^1 x_2^8 x_3^3 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{1}{1}\right)$	$p\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{1}{1}\right)$	401
	$x_1^1 x_2^9 x_3^4 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{1}\right)\Gamma_p\left(\frac{9}{20}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{1}\right)\Gamma_p\left(\frac{9}{20}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{2}\right)$	$191\cdot 401+387\cdot 401^2+\dots$
	$x_1^1 x_2^{12} x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{1}\right)$	$p\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{1}\right)$	401
	$x_1^1 x_2^{13} x_3^3 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{13}{20}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{13}{20}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{1}{2}\right)$	$28\cdot 401+259\cdot 401^2+\dots$
	$x_1^1 x_2^{16} x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{1}\right)$	$p\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{1}\right)$	401
	$x_1^1 x_2^{17} x_3^4 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{17}{20}\right)\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{17}{20}\right)\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{2}\right)$	$218\cdot 401+144\cdot 401^2+\dots$
	$x_1^2 x_2^3 x_3^3 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{3}{20}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{3}{20}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{1}{2}\right)$	$149\cdot 401+190\cdot 401^2+\dots$
	$x_1^2 x_2^7 x_3^4 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{7}{20}\right)\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{7}{20}\right)\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{2}\right)$	$43\cdot 401+175\cdot 401^2+\dots$
	$x_1^2 x_2^{11} x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{11}{20}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{11}{20}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)$	$21\cdot 401+350\cdot 401^2+\dots$
$H^{2,0}$	$x_1^2 x_2^{19} x_3^4 x_4^1 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{19}{20}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p^2\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{19}{20}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{2}\right)$	$340\cdot 401^2+\dots$
$H^{2,2}$	$y_1^1 y_2^{19} y_3^4 y_4^2$	p^2	p^2	160801

Surface #53: $Q = (10, 4, 5, 1)$, $m = 20$

Equation: $x_0^2 + x_1^5 + x_2^4 + x_0x_3^{10} = 0$

$p = 401$

Table 53: Trace contributions for $x_0^2 + x_1^5 + x_2^4 + x_0x_3^{10} = 0$ at $p = 401$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{9}{20}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{10}\right)$	$-\Gamma_p\left(\frac{9}{20}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{10}\right)$	$270+354\cdot 401+386\cdot 401^2+\dots$
$H^{1,1}$	$y_1^1 y_2^1 y_3^3 y_4^6$	p	p	401
	$y_1^1 y_2^2 y_3^1 y_4^9$	p	p	401
	$y_1^1 y_2^2 y_3^3 y_4^4$	p	p	401
	$y_1^1 y_2^3 y_3^1 y_4^7$	p	p	401
	$y_1^1 y_2^3 y_3^3 y_4^2$	p	p	401
	$y_1^1 y_2^4 y_3^1 y_4^5$	p	p	401
	$x_1^1 x_2^2 y_3^1 y_4^8 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	401
	$x_1^1 x_2^3 y_2^1 y_4^6 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	401
	$x_1^1 x_2^3 y_2^3 y_4^4 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	401
	$x_1^1 x_2^3 y_2^4 y_4^2 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	401
	$x_1^1 x_2^3 x_3^1 x_4^7 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{20}\right)\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{7}{10}\right)$	$p\Gamma_p\left(\frac{3}{20}\right)\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{7}{10}\right)$	$73\cdot 401+354\cdot 401^2+\dots$
	$x_1^1 x_2^3 x_3^2 x_4^8 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{10}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{4}{5}\right)$	$p\Gamma_p\left(\frac{1}{10}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{4}{5}\right)$	$72\cdot 401+109\cdot 401^2+\dots$
	$x_1^1 x_2^3 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{20}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{3}{10}\right)$	$p\Gamma_p\left(\frac{7}{20}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{3}{10}\right)$	$302\cdot 401+273\cdot 401^2+\dots$
	$x_1^1 x_2^4 x_3^1 x_4^9 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{20}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{9}{10}\right)$	$p\Gamma_p\left(\frac{1}{20}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{9}{10}\right)$	$187\cdot 401+161\cdot 401^2+\dots$
	$x_1^1 x_2^4 x_3^2 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{10}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{5}\right)$	$p\Gamma_p\left(\frac{3}{10}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{5}\right)$	$318\cdot 401+269\cdot 401^2+\dots$
	$x_1^2 x_2^2 x_3^2 x_4^6 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{10}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{3}{5}\right)$	$p\Gamma_p\left(\frac{7}{10}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{3}{5}\right)$	$372\cdot 401+69\cdot 401^2+\dots$
	$x_1^2 x_2^3 x_3^1 x_4^7 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{13}{20}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{10}\right)$	$p\Gamma_p\left(\frac{13}{20}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{10}\right)$	$193\cdot 401+140\cdot 401^2+\dots$
	$x_1^2 x_2^3 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{13}{20}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{10}\right)$	$p\Gamma_p\left(\frac{13}{20}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{10}\right)$	$81\cdot 401+386\cdot 401^2+\dots$
	$x_1^2 x_2^3 x_3^2 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{9}{10}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{5}\right)$	$p\Gamma_p\left(\frac{9}{10}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{5}\right)$	$39\cdot 401+353\cdot 401^2+\dots$
	$x_1^2 x_2^3 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{17}{20}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{3}{10}\right)$	$p\Gamma_p\left(\frac{17}{20}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{3}{10}\right)$	$11\cdot 401+51\cdot 401^2+\dots$
$H^{2,0}$	$x_1^2 x_2^4 x_3^3 x_4^9 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{11}{20}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{9}{10}\right)$	$-p^2\Gamma_p\left(\frac{11}{20}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{9}{10}\right)$	$251\cdot 401^2+\dots$
$H^{2,2}$	$y_1^1 y_2^4 y_3^3 y_4^{10}$	p^2	p^2	160801

Surface #54: $Q = (2, 5, 10, 3)$, $m = 20$

Equation: $x_0^{10} + x_1^4 + x_2^2 + x_0 x_3^6 = 0$

$p = 3361$

Table 54: Trace contributions for $x_0^{10} + x_1^4 + x_2^2 + x_0 x_3^6 = 0$ at $p = 3361$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{6}\right)$	$-\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{6}\right)$	$447+820\cdot 3361+2911\cdot 3361^2+\dots$
$H^{1,1}$	$y_1^1 y_2^3 y_3^1 y_4^4$	p	p	3361
	$y_1^3 y_2^1 y_3^1 y_4^6$	p	p	3361
	$y_1^3 y_2^3 y_3^1 y_4^3$	p	p	3361
	$y_1^5 y_2^1 y_3^1 y_4^5$	p	p	3361
	$y_1^1 y_2^3 y_3^3 y_4^2$	p	p	3361
	$y_1^7 y_2^1 y_3^1 y_4^4$	p	p	3361
	$y_1^7 y_2^3 y_3^1 y_4^1$	p	p	3361
	$y_1^9 y_2^1 y_3^1 y_4^3$	p	p	3361
	$x_2^1 x_3^1 y_1^2 y_2^5 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	3361
	$x_2^1 x_3^1 y_1^4 y_2^4 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	3361
	$x_2^1 x_3^1 y_1^6 y_2^4 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	3361
	$x_2^1 x_3^1 y_1^8 y_2^4 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	3361
	$x_1^1 x_2^1 x_3^1 x_4^6 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	3361
	$x_1^3 x_2^3 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	3361
	$x_1^4 x_2^2 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	3361
	$x_1^1 x_3^1 y_2^2 y_3^3 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	3361
	$x_1^5 x_2^1 x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{6}\right)$	$p\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{6}\right)$	$201\cdot 3361+1917\cdot 3361^2+\dots$
	$x_1^6 x_2^3 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{6}\right)$	$p\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{6}\right)$	$2341\cdot 3361+2487\cdot 3361^2+\dots$
	$x_1^7 x_2^2 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	3361
	$x_1^8 x_2^1 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	3361
$H^{2,0}$	$x_1^{10} x_2^3 x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{6}\right)$	$-p^2\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{6}\right)$	$2767\cdot 3361^2+\dots$
$H^{2,2}$	$y_1^9 y_2^3 y_3^1 y_4^6$	p^2	p^2	11296321

Surface #55: $Q = (5, 2, 10, 3)$, $m = 20$

Equation: $x_0^4 + x_1^{10} + x_2^2 + x_0x_3^5 = 0$

$p = 401$

Table 55: Trace contributions for $x_0^4 + x_1^{10} + x_2^2 + x_0x_3^5 = 0$ at $p = 401$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{5}\right)$	$-\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{5}\right)$	$56+94\cdot 401+205\cdot 401^2+\dots$
$H^{1,1}$	$y_1^1 y_2^3 y_3^1 y_4^5$	p	p	401
	$y_1^1 y_2^5 y_3^1 y_4^4$	p	p	401
	$y_1^1 y_2^7 y_3^1 y_4^3$	p	p	401
	$y_1^1 y_2^9 y_3^1 y_4^2$	p	p	401
	$y_1^3 y_2^1 y_3^1 y_4^4$	p	p	401
	$y_1^3 y_2^3 y_3^1 y_4^3$	p	p	401
	$y_1^3 y_2^5 y_3^1 y_4^2$	p	p	401
	$y_1^3 y_2^7 y_3^1 y_4^1$	p	p	401
	$x_2^5 x_3^1 y_1^2 y_3^3 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	401
	$x_1^1 x_2^5 x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{1}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{1}\right)$	401
	$x_1^1 x_2^8 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{10}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{5}\right)$	$p\Gamma_p\left(\frac{1}{10}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{5}\right)$	$72\cdot 401+109\cdot 401^2+\dots$
	$x_1^2 x_3^1 y_2^2 y_4^4 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	401
	$x_1^2 x_3^3 y_2^4 y_4^3 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	401
	$x_1^2 x_3^1 y_2^6 y_4^2 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	401
	$x_1^2 x_3^1 y_2^8 y_4^1 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	401
	$x_1^2 x_2^4 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{10}\right)\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{5}\right)$	$p\Gamma_p\left(\frac{3}{10}\right)\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{5}\right)$	$318\cdot 401+269\cdot 401^2+\dots$
	$x_1^2 x_2^7 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{5}\right)$	$p\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{5}\right)$	$275\cdot 401+305\cdot 401^2+\dots$
	$x_1^3 x_2^3 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{3}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{5}\right)$	$p\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{3}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{5}\right)$	$35\cdot 401+69\cdot 401^2+\dots$
	$x_1^3 x_2^6 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{10}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{5}\right)$	$p\Gamma_p\left(\frac{7}{10}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{5}\right)$	$372\cdot 401+69\cdot 401^2+\dots$
	$x_1^4 x_2^1 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{9}{10}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{5}\right)$	$p\Gamma_p\left(\frac{9}{10}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{5}\right)$	$39\cdot 401+353\cdot 401^2+\dots$
$H^{2,0}$	$x_1^4 x_2^9 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{9}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{5}\right)$	$-p^2\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{9}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{5}\right)$	$222\cdot 401^2+\dots$
$H^{2,2}$	$y_1^3 y_2^9 y_3^1 y_4^5$	p^2	p^2	160801

Surface #56: $Q = (2, 4, 5, 9)$, $m = 20$

Equation: $x_0^{10} + x_1^5 + x_2^4 + x_0 x_3^2 = 0$

$p = 401$

Table 56: Trace contributions for $x_0^{10} + x_1^5 + x_2^4 + x_0 x_3^2 = 0$ at $p = 401$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{20}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-\Gamma_p\left(\frac{1}{20}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)$	$46+154\cdot 401+158\cdot 401^2+\dots$
$H^{1,1}$	$y_1^1 y_2^1 y_3^2 y_4^2$	p	p	401
	$y_1^2 y_2^2 y_3^2 y_4^2$	p	p	401
	$y_1^3 y_2^3 y_3^1 y_4^2$	p	p	401
	$y_1^3 y_2^3 y_3^3 y_4^1$	p	p	401
	$y_1^4 y_2^4 y_3^2 y_4^1$	p	p	401
	$y_1^6 y_2^1 y_3^2 y_4^2$	p	p	401
	$y_1^7 y_2^2 y_3^1 y_4^2$	p	p	401
	$y_1^1 y_2^2 y_3^3 y_4^1$	p	p	401
	$y_1^3 y_2^3 y_3^2 y_4^1$	p	p	401
	$y_1^9 y_2^4 y_3^1 y_4^1$	p	p	401
	$x_1^2 x_2^3 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{20}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{3}{20}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)$	$149\cdot 401+190\cdot 401^2+\dots$
	$x_1^3 x_2^4 y_3^1 e_1 e_2$	$-p\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{4}{5}\right)$	$-p\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{4}{5}\right)$	401
	$x_1^4 x_2^2 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{20}\right)\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{7}{20}\right)\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)$	$43\cdot 401+175\cdot 401^2+\dots$
	$x_1^4 x_2^3 y_3^1 e_1 e_2$	$-p\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{3}{5}\right)$	$-p\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{3}{5}\right)$	401
	$x_1^5 x_2^4 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{9}{20}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{9}{20}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)$	$191\cdot 401+387\cdot 401^2+\dots$
	$x_1^6 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{11}{20}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{11}{20}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)$	$21\cdot 401+350\cdot 401^2+\dots$
	$x_1^6 x_2^2 y_3^1 e_1 e_2$	$-p\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{2}{5}\right)$	$-p\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{2}{5}\right)$	401
	$x_1^7 x_2^3 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{13}{20}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{13}{20}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)$	$28\cdot 401+259\cdot 401^2+\dots$
	$x_1^8 x_2^1 y_3^1 e_1 e_2$	$-p\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{5}\right)$	$-p\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{5}\right)$	401
	$x_1^9 x_2^2 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{17}{20}\right)\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{17}{20}\right)\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)$	$218\cdot 401+144\cdot 401^2+\dots$
$H^{2,0}$	$x_1^{10} x_2^4 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{19}{20}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p^2\Gamma_p\left(\frac{19}{20}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)$	$340\cdot 401^2+\dots$
$H^{2,2}$	$y_1^9 y_2^4 y_3^2 y_4^2$	p^2	p^2	160801

Surface #57: $Q = (1, 3, 7, 10)$, $m = 21$

Equation: $x_0^{21} + x_1^7 + x_2^3 + x_0 x_3^2 = 0$

$p = 883$

Table 57: Trace contributions for $x_0^{21} + x_1^7 + x_2^3 + x_0 x_3^2 = 0$ at $p = 883$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{42}\right) \Gamma_p\left(\frac{1}{7}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$-\Gamma_p\left(\frac{1}{42}\right) \Gamma_p\left(\frac{1}{7}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$636+341\cdot 883+181\cdot 883^2+\dots$
$H^{1,1}$	$y_1^2 y_2^2 y_3^2 y_4^2$	p	p	883
	$y_1^4 y_2^4 y_3^4 y_4^4$	p	p	883
	$y_1^5 y_2^5 y_3^5 y_4^1$	p	p	883
	$y_1^8 y_2^1 y_3^2 y_4^2$	p	p	883
	$y_1^{10} y_2^3 y_3^3 y_4^2$	p	p	883
	$y_1^{11} y_2^4 y_3^2 y_4^1$	p	p	883
	$y_1^{13} y_2^6 y_3^1 y_4^1$	p	p	883
	$y_1^{16} y_2^2 y_3^1 y_4^2$	p	p	883
	$y_1^{17} y_2^3 y_3^2 y_4^1$	p	p	883
	$y_1^{19} y_2^5 y_3^1 y_4^1$	p	p	883
	$x_1^3 x_2^5 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{5}{42}\right) \Gamma_p\left(\frac{5}{7}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{5}{42}\right) \Gamma_p\left(\frac{5}{7}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$126\cdot 883+762\cdot 883^2+\dots$
	$x_1^6 x_2^4 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{11}{42}\right) \Gamma_p\left(\frac{4}{7}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{11}{42}\right) \Gamma_p\left(\frac{4}{7}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$175\cdot 883+70\cdot 883^2+\dots$
	$x_1^7 x_2^6 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{13}{42}\right) \Gamma_p\left(\frac{6}{7}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{13}{42}\right) \Gamma_p\left(\frac{6}{7}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$290\cdot 883+142\cdot 883^2+\dots$
	$x_1^9 x_2^3 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{17}{42}\right) \Gamma_p\left(\frac{3}{7}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{17}{42}\right) \Gamma_p\left(\frac{3}{7}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$443\cdot 883+378\cdot 883^2+\dots$
	$x_1^{10} x_2^5 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{19}{42}\right) \Gamma_p\left(\frac{5}{7}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{19}{42}\right) \Gamma_p\left(\frac{5}{7}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$572\cdot 883+479\cdot 883^2+\dots$
	$x_1^{12} x_2^2 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{23}{42}\right) \Gamma_p\left(\frac{2}{7}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{23}{42}\right) \Gamma_p\left(\frac{2}{7}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$724\cdot 883+262\cdot 883^2+\dots$
	$x_1^{13} x_2^1 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{25}{42}\right) \Gamma_p\left(\frac{4}{7}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{25}{42}\right) \Gamma_p\left(\frac{4}{7}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$295\cdot 883+322\cdot 883^2+\dots$
	$x_1^{15} x_2^1 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{29}{42}\right) \Gamma_p\left(\frac{1}{7}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{29}{42}\right) \Gamma_p\left(\frac{1}{7}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$679\cdot 883+23\cdot 883^2+\dots$
	$x_1^{16} x_2^3 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{31}{42}\right) \Gamma_p\left(\frac{3}{7}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{31}{42}\right) \Gamma_p\left(\frac{3}{7}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$772\cdot 883+427\cdot 883^2+\dots$
	$x_1^{19} x_2^2 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{37}{42}\right) \Gamma_p\left(\frac{2}{7}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{37}{42}\right) \Gamma_p\left(\frac{2}{7}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$876\cdot 883+623\cdot 883^2+\dots$
$H^{2,0}$	$x_1^{21} x_2^6 x_3^4 x_4^1 e_1 e_2 e_3 e_4$	$-p^2 \Gamma_p\left(\frac{41}{42}\right) \Gamma_p\left(\frac{6}{7}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$-p^2 \Gamma_p\left(\frac{41}{42}\right) \Gamma_p\left(\frac{6}{7}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right)$	$740\cdot 883^2+\dots$
$H^{2,2}$	$y_1^{20} y_2^6 y_3^2 y_4^2$	p^2	p^2	779689

Surface #58: $Q = (3, 8, 12, 1)$, $m = 24$

Equation: $x_0^8 + x_1^3 + x_2^2 + x_0 x_3^{21} = 0$

$p = 1009$

Table 58: Trace contributions for $x_0^8 + x_1^3 + x_2^2 + x_0 x_3^{21} = 0$ at $p = 1009$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{5}{42}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{21}\right)$	$-\Gamma_p\left(\frac{5}{42}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{21}\right)$	$191+719\cdot 1009+202\cdot 1009^2+\dots$
$H^{1,1}$	$y_1^1 y_2^2 y_3^1 y_4^{15}$	p	p	1009
	$y_1^1 y_2^1 y_3^1 y_4^{17}$	p	p	1009
	$y_1^1 y_2^2 y_3^1 y_4^{10}$	p	p	1009
	$y_1^1 y_2^1 y_3^1 y_4^{12}$	p	p	1009
	$y_1^1 y_2^2 y_3^1 y_4^5$	p	p	1009
	$y_1^1 y_2^1 y_3^1 y_4^7$	p	p	1009
	$x_1^1 x_2^2 x_3^1 x_4^{17} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{42}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{17}{21}\right)$	$p\Gamma_p\left(\frac{1}{42}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{17}{21}\right)$	$541\cdot 1009+471\cdot 1009^2+\dots$
	$x_1^2 x_2^2 x_3^1 x_4^{14} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$374\cdot 1009+734\cdot 1009^2+\dots$
	$x_1^3 x_2^2 x_3^1 x_4^{19} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{11}{42}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{19}{21}\right)$	$p\Gamma_p\left(\frac{11}{42}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{19}{21}\right)$	$840\cdot 1009+121\cdot 1009^2+\dots$
	$x_1^3 x_2^2 x_3^1 x_4^{11} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{13}{42}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{11}{21}\right)$	$p\Gamma_p\left(\frac{13}{42}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{11}{21}\right)$	$663\cdot 1009+879\cdot 1009^2+\dots$
	$x_1^4 x_3^1 y_2^1 y_4^{14} e_1 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	1009
	$x_1^4 x_3^1 y_2^1 y_4^7 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	1009
	$x_1^4 x_2^1 x_3^1 x_4^{16} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{17}{42}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{16}{21}\right)$	$p\Gamma_p\left(\frac{17}{42}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{16}{21}\right)$	$35\cdot 1009+610\cdot 1009^2+\dots$
	$x_1^4 x_2^2 x_3^1 x_4^8 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{19}{42}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{8}{21}\right)$	$p\Gamma_p\left(\frac{19}{42}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{8}{21}\right)$	$857\cdot 1009+669\cdot 1009^2+\dots$
	$x_1^5 x_2^1 x_3^1 x_4^{13} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{23}{42}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{13}{21}\right)$	$p\Gamma_p\left(\frac{23}{42}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{13}{21}\right)$	$697\cdot 1009+808\cdot 1009^2+\dots$
	$x_1^5 x_2^2 x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{25}{42}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{21}\right)$	$p\Gamma_p\left(\frac{25}{42}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{21}\right)$	$173\cdot 1009+127\cdot 1009^2+\dots$
	$x_1^6 x_2^1 x_3^1 x_4^{10} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{29}{42}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{10}{21}\right)$	$p\Gamma_p\left(\frac{29}{42}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{10}{21}\right)$	$487\cdot 1009+512\cdot 1009^2+\dots$
	$x_1^6 x_2^2 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{31}{42}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{21}\right)$	$p\Gamma_p\left(\frac{31}{42}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{21}\right)$	$806\cdot 1009+187\cdot 1009^2+\dots$
	$x_1^7 x_2^1 x_3^1 x_4^7 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$634\cdot 1009+274\cdot 1009^2+\dots$
	$x_1^8 x_2^1 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{41}{42}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{21}\right)$	$p\Gamma_p\left(\frac{41}{42}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{21}\right)$	$235\cdot 1009+696\cdot 1009^2+\dots$
$H^{2,0}$	$x_1^8 x_2^2 x_3^1 x_4^{20} e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{37}{42}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{20}{21}\right)$	$-p^2\Gamma_p\left(\frac{37}{42}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{20}{21}\right)$	$766\cdot 1009^2+\dots$
$H^{2,2}$	$y_1^7 y_2^2 y_3^1 y_4^{21}$	p^2	p^2	1018081

Surface #59: $Q = (8, 3, 12, 1)$, $m = 24$

Equation: $x_0^3 + x_1^8 + x_2^2 + x_0 x_3^{16} = 0$

$p = 769$

Table 59: Trace contributions for $x_0^3 + x_1^8 + x_2^2 + x_0 x_3^{16} = 0$ at $p = 769$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{5}{16}\right)\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{16}\right)$	$-\Gamma_p\left(\frac{5}{16}\right)\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{16}\right)$	$580+465\cdot 769+4\cdot 769^2+\dots$
$H^{1,1}$	$y_1^1 y_2^3 y_3^1 y_4^{13}$	p	p	769
	$y_1^1 y_2^5 y_3^1 y_4^9$	p	p	769
	$y_1^1 y_2^7 y_3^1 y_4^5$	p	p	769
	$y_1^2 y_2^1 y_3^1 y_4^{12}$	p	p	769
	$y_1^2 y_2^3 y_3^1 y_4^8$	p	p	769
	$y_1^2 y_2^5 y_3^1 y_4^4$	p	p	769
	$x_2^4 x_3^1 y_1^1 y_4^{11} e_2 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	769
	$x_2^4 x_3^1 y_1^2 y_4^6 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	769
	$x_1^1 x_2^4 x_3^1 x_4^{16} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	769
	$x_1^1 x_2^5 x_3^1 x_4^{13} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{16}\right)\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{13}{16}\right)$	$p\Gamma_p\left(\frac{1}{16}\right)\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{13}{16}\right)$	$184\cdot 769+488\cdot 769^2+\dots$
	$x_1^1 x_2^6 x_3^1 x_4^{10} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{8}\right)$	$p\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{8}\right)$	-769
	$x_1^1 x_2^7 x_3^1 x_4^7 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{16}\right)\Gamma_p\left(\frac{7}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{16}\right)$	$p\Gamma_p\left(\frac{3}{16}\right)\Gamma_p\left(\frac{7}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{16}\right)$	$406\cdot 769+396\cdot 769^2+\dots$
	$x_1^2 x_2^5 x_3^1 x_4^{14} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{8}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{8}\right)$	$p\Gamma_p\left(\frac{3}{8}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{8}\right)$	-769
	$x_1^2 x_2^3 x_3^1 x_4^{11} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{16}\right)\Gamma_p\left(\frac{3}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{11}{16}\right)$	$p\Gamma_p\left(\frac{7}{16}\right)\Gamma_p\left(\frac{3}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{11}{16}\right)$	$204\cdot 769+327\cdot 769^2+\dots$
	$x_1^2 x_2^4 x_3^1 x_4^8 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	769
	$x_1^2 x_2^5 x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{9}{16}\right)\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{16}\right)$	$p\Gamma_p\left(\frac{9}{16}\right)\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{16}\right)$	$720\cdot 769+153\cdot 769^2+\dots$
	$x_1^2 x_2^6 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{8}\right)$	$p\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{8}\right)$	-769
	$x_1^3 x_2^1 x_3^1 x_4^9 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{13}{16}\right)\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{9}{16}\right)$	$p\Gamma_p\left(\frac{13}{16}\right)\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{9}{16}\right)$	$161\cdot 769+53\cdot 769^2+\dots$
	$x_1^3 x_2^2 x_3^1 x_4^6 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{8}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{8}\right)$	$p\Gamma_p\left(\frac{7}{8}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{8}\right)$	-769
	$x_1^3 x_2^3 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{15}{16}\right)\Gamma_p\left(\frac{3}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{16}\right)$	$p\Gamma_p\left(\frac{15}{16}\right)\Gamma_p\left(\frac{3}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{16}\right)$	$163\cdot 769+232\cdot 769^2+\dots$
$H^{2,0}$	$x_1^3 x_2^7 x_3^1 x_4^{15} e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{11}{16}\right)\Gamma_p\left(\frac{7}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{15}{16}\right)$	$-p^2\Gamma_p\left(\frac{11}{16}\right)\Gamma_p\left(\frac{7}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{15}{16}\right)$	$651\cdot 769^2+\dots$
$H^{2,2}$	$y_1^2 y_2^7 y_3^1 y_4^{16}$	p^2	p^2	591361

Surface #60: $Q = (12, 1, 8, 3)$, $m = 24$

Equation: $x_0^2 + x_1^{24} + x_2^3 + x_0 x_3^4 = 0$

$p = 577$

Table 60: Trace contributions for $x_0^2 + x_1^{24} + x_2^3 + x_0 x_3^4 = 0$ at $p = 577$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{3}{8}\right)\Gamma_p\left(\frac{1}{24}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{4}\right)$	$-\Gamma_p\left(\frac{3}{8}\right)\Gamma_p\left(\frac{1}{24}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{4}\right)$	$543+23\cdot 577+401\cdot 577^2+\dots$
$H^{1,1}$	$y_1^1 y_2^5 y_3^2 y_4^3$	p	p	577
	$y_1^1 y_2^2 y_3^3 y_4^4$	p	p	577
	$y_1^1 y_2^{11} y_3^2 y_4^2$	p	p	577
	$y_1^1 y_2^{13} y_3^1 y_4^3$	p	p	577
	$y_1^1 y_2^{17} y_3^2 y_4^1$	p	p	577
	$y_1^1 y_2^{19} y_3^1 y_4^2$	p	p	577
	$x_1^1 x_4^4 y_2^8 y_3^2 e_1 e_4$	$-p\Gamma_p\left(\frac{1}{1}\right)$	$-p\Gamma_p\left(\frac{1}{1}\right)$	577
	$x_1^1 x_4^4 y_2^{16} y_3^1 e_1 e_4$	$-p\Gamma_p\left(\frac{1}{1}\right)$	$-p\Gamma_p\left(\frac{1}{1}\right)$	577
	$x_1^1 x_2^8 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{1}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{1}\right)$	577
	$x_1^1 x_2^{11} x_3^3 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{11}{24}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{3}{4}\right)$	$p\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{11}{24}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{3}{4}\right)$	$38\cdot 577+73\cdot 577^2+\dots$
	$x_1^1 x_2^{14} x_3^2 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^1 x_2^{16} x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{1}\right)$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{1}\right)$	577
	$x_1^1 x_2^{17} x_3^4 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{17}{24}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{4}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{17}{24}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{4}\right)$	$2\cdot 577+526\cdot 577^2+\dots$
	$x_1^1 x_2^{19} x_3^3 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{19}{24}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{3}{4}\right)$	$p\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{19}{24}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{3}{4}\right)$	$531\cdot 577+433\cdot 577^2+\dots$
	$x_1^1 x_2^{22} x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^2 x_2^2 x_3^2 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^2 x_2^5 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{5}{24}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{4}\right)$	$p\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{5}{24}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{4}\right)$	$439\cdot 577+215\cdot 577^2+\dots$
	$x_1^2 x_2^7 x_3^3 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{8}\right)\Gamma_p\left(\frac{7}{24}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{3}{4}\right)$	$p\Gamma_p\left(\frac{7}{8}\right)\Gamma_p\left(\frac{7}{24}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{3}{4}\right)$	$289\cdot 577+445\cdot 577^2+\dots$
	$x_1^2 x_2^{10} x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^2 x_2^{13} x_3^4 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{13}{24}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{4}\right)$	$p\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{13}{24}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{4}\right)$	$410\cdot 577+229\cdot 577^2+\dots$
$H^{2,0}$	$x_1^2 x_2^{23} x_3^3 x_4^1 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{23}{24}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{3}{4}\right)$	$-p^2\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{23}{24}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{3}{4}\right)$	$560\cdot 577^2+\dots$
$H^{2,2}$	$y_1^1 y_2^{23} y_3^2 y_4^4$	p^2	p^2	332929

Surface #61: $Q = (12, 3, 8, 1)$, $m = 24$

Equation: $x_0^2 + x_1^8 + x_2^3 + x_0 x_3^{12} = 0$

$p = 577$

Table 61: Trace contributions for $x_0^2 + x_1^8 + x_2^3 + x_0 x_3^{12} = 0$ at $p = 577$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{11}{24}\right)\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{12}\right)$	$-\Gamma_p\left(\frac{11}{24}\right)\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{12}\right)$	$259+342\cdot 577+361\cdot 577^2+\dots$
$H^{1,1}$	$y_1^1 y_2^1 y_3^2 y_4^9$	p	p	577
	$y_1^1 y_2^1 y_3^2 y_4^{10}$	p	p	577
	$y_1^1 y_2^1 y_3^2 y_4^6$	p	p	577
	$y_1^1 y_2^1 y_3^2 y_4^7$	p	p	577
	$y_1^1 y_2^1 y_3^2 y_4^3$	p	p	577
	$y_1^1 y_2^1 y_3^2 y_4^4$	p	p	577
	$x_1^1 x_2^3 x_3^4 x_4^{11} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{24}\right)\Gamma_p\left(\frac{3}{8}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{11}{12}\right)$	$p\Gamma_p\left(\frac{1}{24}\right)\Gamma_p\left(\frac{3}{8}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{11}{12}\right)$	$523\cdot 577+22\cdot 577^2+\dots$
	$x_1^1 x_2^4 y_3^8 y_4^1 e_1 e_2$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^1 x_2^4 y_3^2 y_4^1 e_1 e_2$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^1 x_2^4 x_3^8 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$363\cdot 577+487\cdot 577^2+\dots$
	$x_1^1 x_2^5 x_3^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{24}\right)\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{5}{12}\right)$	$p\Gamma_p\left(\frac{7}{24}\right)\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{5}{12}\right)$	$105\cdot 577+152\cdot 577^2+\dots$
	$x_1^1 x_2^6 x_3^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{5}{6}\right)$	$p\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{5}{6}\right)$	$213\cdot 577+89\cdot 577^2+\dots$
	$x_1^1 x_2^6 x_3^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{6}\right)$	$p\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{6}\right)$	$363\cdot 577+487\cdot 577^2+\dots$
	$x_1^1 x_2^7 x_3^7 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{24}\right)\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{7}{12}\right)$	$p\Gamma_p\left(\frac{5}{24}\right)\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{7}{12}\right)$	$395\cdot 577+19\cdot 577^2+\dots$
	$x_1^2 x_2^1 x_3^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{19}{24}\right)\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{5}{12}\right)$	$p\Gamma_p\left(\frac{19}{24}\right)\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{5}{12}\right)$	$149\cdot 577+349\cdot 577^2+\dots$
	$x_1^2 x_2^2 x_3^{10} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{5}{6}\right)$	$p\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{5}{6}\right)$	$213\cdot 577+89\cdot 577^2+\dots$
	$x_1^2 x_2^2 x_3^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{6}\right)$	$p\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{6}\right)$	$363\cdot 577+487\cdot 577^2+\dots$
	$x_1^2 x_2^3 x_3^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{17}{24}\right)\Gamma_p\left(\frac{3}{8}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{7}{12}\right)$	$p\Gamma_p\left(\frac{17}{24}\right)\Gamma_p\left(\frac{3}{8}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{7}{12}\right)$	$11\cdot 577+50\cdot 577^2+\dots$
	$x_1^2 x_2^4 x_3^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	$213\cdot 577+89\cdot 577^2+\dots$
	$x_1^2 x_2^3 x_3^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{23}{24}\right)\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{12}\right)$	$p\Gamma_p\left(\frac{23}{24}\right)\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{12}\right)$	$203\cdot 577+22\cdot 577^2+\dots$
$H^{2,0}$	$x_1^2 x_2^7 x_3^4 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{13}{24}\right)\Gamma_p\left(\frac{7}{8}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{11}{12}\right)$	$-p^2\Gamma_p\left(\frac{13}{24}\right)\Gamma_p\left(\frac{7}{8}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{11}{12}\right)$	$176\cdot 577^2+\dots$
$H^{2,2}$	$y_1^1 y_2^7 y_3^2 y_4^{12}$	p^2	p^2	332929

Surface #62: $Q = (6, 1, 8, 9)$, $m = 24$

Equation: $x_0^4 + x_1^{24} + x_2^3 + x_0 x_3^2 = 0$

$p = 577$

Table 62: Trace contributions for $x_0^4 + x_1^{24} + x_2^3 + x_0 x_3^2 = 0$ at $p = 577$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{1}{24}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{1}{24}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$543+23\cdot 577+401\cdot 577^2+\dots$
$H^{1,1}$	$y_1^1 y_2^5 y_3^2 y_4^2$	p	p	577
	$y_1^1 y_2^{13} y_3^1 y_4^2$	p	p	577
	$y_1^1 y_2^{17} y_3^2 y_4^1$	p	p	577
	$y_1^2 y_2^2 y_3^2 y_4^2$	p	p	577
	$y_1^2 y_2^{10} y_3^1 y_4^2$	p	p	577
	$y_1^2 y_2^{14} y_3^2 y_4^1$	p	p	577
	$y_1^2 y_2^{22} y_3^1 y_4^1$	p	p	577
	$y_1^3 y_2^7 y_3^1 y_4^2$	p	p	577
	$y_1^3 y_2^{11} y_3^2 y_4^1$	p	p	577
	$y_1^3 y_2^{19} y_3^1 y_4^1$	p	p	577
	$x_1^1 x_2^2 y_3^2 y_4^1 e_1 e_4$	$-p\Gamma_p\left(\frac{1}{1}\right)$	$-p\Gamma_p\left(\frac{1}{1}\right)$	577
	$x_1^1 x_2^4 y_2^{16} y_3^1 e_1 e_4$	$-p\Gamma_p\left(\frac{1}{1}\right)$	$-p\Gamma_p\left(\frac{1}{1}\right)$	577
	$x_1^1 x_2^8 x_3^2 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{1}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{1}\right)$	577
	$x_1^1 x_2^{16} x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{1}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{1}\right)$	577
	$x_1^1 x_2^{17} x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{17}{24}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{17}{24}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$2\cdot 577+526\cdot 577^2+\dots$
	$x_1^2 x_2^{11} x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{11}{24}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{11}{24}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$38\cdot 577+73\cdot 577^2+\dots$
	$x_1^2 x_2^{19} x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{19}{24}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{19}{24}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$531\cdot 577+433\cdot 577^2+\dots$
	$x_1^3 x_2^5 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{5}{24}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{5}{24}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$439\cdot 577+215\cdot 577^2+\dots$
	$x_1^3 x_2^{13} x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{13}{24}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{13}{24}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$410\cdot 577+229\cdot 577^2+\dots$
	$x_1^4 x_2^7 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{7}{24}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{7}{24}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$289\cdot 577+445\cdot 577^2+\dots$
$H^{2,0}$	$x_1^4 x_2^{23} x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{7}{8}\right)\Gamma_p\left(\frac{23}{24}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p^2\Gamma_p\left(\frac{7}{8}\right)\Gamma_p\left(\frac{23}{24}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$560\cdot 577^2+\dots$
$H^{2,2}$	$y_1^3 y_2^{23} y_3^2 y_4^2$	p^2	p^2	332929

Surface #63: $Q = (3, 2, 12, 7)$, $m = 24$

Equation: $x_0^8 + x_1^{12} + x_2^2 + x_0 x_3^3 = 0$

$p = 577$

Table 63: Trace contributions for $x_0^8 + x_1^{12} + x_2^2 + x_0 x_3^3 = 0$ at $p = 577$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$-\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$22+73\cdot 577+361\cdot 577^2+\dots$
$H^{1,1}$	$y_1^1 y_2^5 y_3^1 y_4^3$	p	p	577
	$y_1^1 y_2^9 y_3^1 y_4^2$	p	p	577
	$y_1^3 y_2^3 y_3^1 y_4^3$	p	p	577
	$y_1^3 y_2^7 y_3^1 y_4^2$	p	p	577
	$y_1^3 y_2^{11} y_3^1 y_4^1$	p	p	577
	$y_1^5 y_2^1 y_3^1 y_4^3$	p	p	577
	$y_1^5 y_2^5 y_3^1 y_4^2$	p	p	577
	$y_1^5 y_2^9 y_3^1 y_4^1$	p	p	577
	$y_1^7 y_2^3 y_3^1 y_4^2$	p	p	577
	$y_1^7 y_2^7 y_3^1 y_4^1$	p	p	577
	$x_2^6 x_3^1 y_1^4 y_4^2 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^1 x_2^6 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	577
	$x_1^2 x_2^8 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$363\cdot 577+487\cdot 577^2+\dots$
	$x_1^3 x_2^{10} x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$213\cdot 577+89\cdot 577^2+\dots$
	$x_1^4 x_3^1 y_2^4 y_4^2 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^4 x_3^1 y_2^8 y_4^1 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^4 x_2^5 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$210\cdot 577+142\cdot 577^2+\dots$
	$x_1^5 x_2^7 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$294\cdot 577+309\cdot 577^2+\dots$
	$x_1^6 x_2^2 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$363\cdot 577+487\cdot 577^2+\dots$
	$x_1^7 x_2^4 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$213\cdot 577+89\cdot 577^2+\dots$
$H^{2,0}$	$x_1^8 x_2^{11} x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$-p^2\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$341\cdot 577^2+\dots$
$H^{2,2}$	$y_1^7 y_2^{11} y_3^1 y_4^3$	p^2	p^2	332929

Surface #64: $Q = (2, 3, 8, 11)$, $m = 24$

Equation: $x_0^{12} + x_1^8 + x_2^3 + x_0 x_3^2 = 0$

$p = 577$

Table 64: Trace contributions for $x_0^{12} + x_1^8 + x_2^3 + x_0 x_3^2 = 0$ at $p = 577$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{24}\right)\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-\Gamma_p\left(\frac{1}{24}\right)\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$543+23\cdot 577+401\cdot 577^2+\dots$
$H^{1,1}$	$y_1^1 y_2^5 y_3^1 y_4^2$	p	p	577
	$y_1^2 y_2^5 y_3^2 y_4^2$	p	p	577
	$y_1^2 y_2^6 y_3^2 y_4^1$	p	p	577
	$y_1^4 y_2^4 y_3^1 y_4^2$	p	p	577
	$y_1^3 y_2^1 y_3^5 y_4^2$	p	p	577
	$y_1^5 y_2^5 y_3^2 y_4^1$	p	p	577
	$y_1^7 y_2^3 y_3^1 y_4^2$	p	p	577
	$y_1^1 y_2^7 y_3^1 y_4^1$	p	p	577
	$y_1^8 y_2^4 y_3^2 y_4^1$	p	p	577
	$y_1^{10} y_2^2 y_3^1 y_4^2$	p	p	577
	$y_1^{10} y_2^6 y_3^1 y_4^1$	p	p	577
	$y_1^{11} y_2^3 y_3^2 y_4^1$	p	p	577
	$x_1^3 x_2^5 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{24}\right)\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{5}{24}\right)\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$439\cdot 577+215\cdot 577^2+\dots$
	$x_1^4 x_2^2 y_3^4 y_4^1 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	577
	$x_1^4 x_2^7 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{24}\right)\Gamma_p\left(\frac{7}{8}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{7}{24}\right)\Gamma_p\left(\frac{7}{8}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$289\cdot 577+445\cdot 577^2+\dots$
	$x_1^6 x_2^3 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{11}{24}\right)\Gamma_p\left(\frac{11}{8}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{11}{24}\right)\Gamma_p\left(\frac{11}{8}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$38\cdot 577+73\cdot 577^2+\dots$
	$x_1^7 x_2^5 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{13}{24}\right)\Gamma_p\left(\frac{13}{8}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{13}{24}\right)\Gamma_p\left(\frac{13}{8}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$410\cdot 577+229\cdot 577^2+\dots$
	$x_1^8 x_3^1 y_2^4 y_4^1 e_1 e_3$	$-p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	$-p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	577
	$x_1^9 x_2^1 x_3^2 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{17}{24}\right)\Gamma_p\left(\frac{17}{8}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{17}{24}\right)\Gamma_p\left(\frac{17}{8}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$2\cdot 577+526\cdot 577^2+\dots$
	$x_1^{10} x_2^3 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{19}{24}\right)\Gamma_p\left(\frac{19}{8}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{19}{24}\right)\Gamma_p\left(\frac{19}{8}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$531\cdot 577+433\cdot 577^2+\dots$
$H^{2,0}$	$x_1^{12} x_2^7 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{23}{24}\right)\Gamma_p\left(\frac{7}{8}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p^2\Gamma_p\left(\frac{23}{24}\right)\Gamma_p\left(\frac{7}{8}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)$	$560\cdot 577^2+\dots$
$H^{2,2}$	$y_1^{11} y_2^7 y_3^2 y_4^2$	p^2	p^2	332929

Surface #65: $Q = (4, 3, 12, 5)$, $m = 24$

Equation: $x_0^6 + x_1^8 + x_2^2 + x_0x_3^4 = 0$

$p = 769$

Table 65: Trace contributions for $x_0^6 + x_1^8 + x_2^2 + x_0x_3^4 = 0$ at $p = 769$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$-\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$331+373\cdot 769+498\cdot 769^2+\dots$
$H^{1,1}$	$y_1^1 y_2^3 y_3^1 y_4^4$	p	p	769
	$y_1^1 y_2^3 y_3^1 y_4^4$	p	p	769
	$y_1^1 y_2^3 y_3^1 y_4^4$	p	p	769
	$y_1^3 y_2^1 y_3^1 y_4^4$	p	p	769
	$y_1^3 y_2^3 y_3^1 y_4^4$	p	p	769
	$y_1^3 y_2^3 y_3^1 y_4^4$	p	p	769
	$y_1^3 y_2^3 y_3^1 y_4^4$	p	p	769
	$y_1^3 y_2^3 y_3^1 y_4^4$	p	p	769
	$y_1^5 y_2^1 y_3^1 y_4^4$	p	p	769
	$y_1^5 y_2^1 y_3^1 y_4^4$	p	p	769
	$y_1^5 y_2^1 y_3^1 y_4^4$	p	p	769
	$x_2^4 x_3^1 y_1^2 y_4^3 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	769
	$x_2^4 x_3^1 y_1^2 y_4^3 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	769
	$x_1^1 x_2^4 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	769
	$x_1^2 x_2^6 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	769
	$x_1^3 x_2^1 y_2^2 y_4^3 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	769
	$x_1^3 x_2^1 y_2^2 y_4^3 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	769
	$x_1^3 x_2^1 y_2^2 y_4^3 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	769
	$x_1^3 x_2^1 y_2^2 y_4^3 e_1 e_3$	$p\Gamma_p\left(\frac{3}{8}\right)\Gamma_p\left(\frac{3}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$p\Gamma_p\left(\frac{3}{8}\right)\Gamma_p\left(\frac{3}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$185\cdot 769+702\cdot 769^2+\dots$
	$x_1^4 x_2^5 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$p\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{4}\right)$	$212\cdot 769+567\cdot 769^2+\dots$
	$x_1^5 x_2^2 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	769
$H^{2,0}$	$x_1^6 x_2^7 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{7}{8}\right)\Gamma_p\left(\frac{7}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$-p^2\Gamma_p\left(\frac{7}{8}\right)\Gamma_p\left(\frac{7}{8}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{4}\right)$	$539\cdot 769^2+\dots$
$H^{2,2}$	$y_1^5 y_2^7 y_3^1 y_4^4$	p^2	p^2	591361

Surface #66: $Q = (3, 6, 8, 7)$, $m = 24$

Equation: $x_0^8 + x_1^4 + x_2^3 + x_0 x_3^3 = 0$

$p = 577$

Table 66: Trace contributions for $x_0^8 + x_1^4 + x_2^3 + x_0 x_3^3 = 0$ at $p = 577$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{12}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	$-\Gamma_p\left(\frac{1}{12}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	$22+73\cdot 577+361\cdot 577^2+\dots$
$H^{1,1}$	$y_1^1 y_2^1 y_3^2 y_4^3$	p	p	577
	$y_1^2 y_2^1 y_3^1 y_4^3$	p	p	577
	$y_1^2 y_2^2 y_3^2 y_4^2$	p	p	577
	$y_1^3 y_2^3 y_3^1 y_4^2$	p	p	577
	$y_1^3 y_2^3 y_3^2 y_4^1$	p	p	577
	$y_1^5 y_2^1 y_3^1 y_4^3$	p	p	577
	$y_1^5 y_2^1 y_3^2 y_4^2$	p	p	577
	$y_1^6 y_2^2 y_3^1 y_4^2$	p	p	577
	$y_1^6 y_2^2 y_3^2 y_4^1$	p	p	577
	$y_1^7 y_2^3 y_3^1 y_4^1$	p	p	577
	$x_1^2 x_2^2 x_3^2 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	$p \Gamma_p\left(\frac{1}{6}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	$363\cdot 577+487\cdot 577^2+\dots$
	$x_1^2 x_2^3 y_3^1 y_4^2 e_1 e_2$	$-p \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{3}{4}\right)$	$-p \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{3}{4}\right)$	577
	$x_1^2 x_2^3 y_3^2 y_4^1 e_1 e_2$	$-p \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{3}{4}\right)$	$-p \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{3}{4}\right)$	577
	$x_1^4 x_2^1 x_3^2 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{5}{12}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	$p \Gamma_p\left(\frac{5}{12}\right) \Gamma_p\left(\frac{1}{4}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	$210\cdot 577+142\cdot 577^2+\dots$
	$x_1^4 x_2^2 y_3^1 y_4^2 e_1 e_2$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^4 x_2^2 y_3^2 y_4^1 e_1 e_2$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	577
	$x_1^5 x_2^3 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{7}{12}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	$p \Gamma_p\left(\frac{7}{12}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	$294\cdot 577+309\cdot 577^2+\dots$
	$x_1^6 x_2^1 y_3^1 y_4^2 e_1 e_2$	$-p \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{4}\right)$	$-p \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{4}\right)$	577
	$x_1^6 x_2^1 y_3^2 y_4^1 e_1 e_2$	$-p \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{4}\right)$	$-p \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{1}{4}\right)$	577
	$x_1^7 x_2^2 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	$p \Gamma_p\left(\frac{5}{6}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{3}\right)$	$213\cdot 577+89\cdot 577^2+\dots$
$H^{2,0}$	$x_1^8 x_2^3 x_3^2 x_4^2 e_1 e_2 e_3 e_4$	$-p^2 \Gamma_p\left(\frac{11}{12}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	$-p^2 \Gamma_p\left(\frac{11}{12}\right) \Gamma_p\left(\frac{3}{4}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{2}{3}\right)$	$341\cdot 577^2+\dots$
$H^{2,2}$	$y_1^7 y_2^3 y_3^2 y_4^3$	p^2	p^2	332929

Surface #67: $Q = (1, 4, 14, 9)$, $m = 28$

Equation: $x_0^{28} + x_1^7 + x_2^2 + x_0 x_3^3 = 0$

$p = 3529$

Table 67: Trace contributions for $x_0^{28} + x_1^7 + x_2^2 + x_0 x_3^3 = 0$ at $p = 3529$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{42}\right)\Gamma_p\left(\frac{1}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$-\Gamma_p\left(\frac{1}{42}\right)\Gamma_p\left(\frac{1}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$1146+1267\cdot 3529+2875\cdot 3529^2+\dots$
$H^{1,1}$	$y_1^3 y_2^3 y_3^1 y_4^3$	p	p	3529
	$y_1^5 y_2^5 y_3^1 y_4^2$	p	p	3529
	$y_1^9 y_2^2 y_3^1 y_4^3$	p	p	3529
	$y_1^{11} y_2^4 y_3^1 y_4^2$	p	p	3529
	$y_1^{13} y_2^5 y_3^1 y_4^1$	p	p	3529
	$y_1^{15} y_2^1 y_3^1 y_4^3$	p	p	3529
	$y_1^{17} y_2^3 y_3^1 y_4^2$	p	p	3529
	$y_1^{19} y_2^5 y_3^1 y_4^1$	p	p	3529
	$y_1^{23} y_2^2 y_3^1 y_4^2$	p	p	3529
	$y_1^{25} y_2^4 y_3^1 y_4^1$	p	p	3529
	$x_1^4 x_2^5 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{42}\right)\Gamma_p\left(\frac{5}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{5}{42}\right)\Gamma_p\left(\frac{5}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$3521\cdot 3529+531\cdot 3529^2+\dots$
	$x_1^8 x_2^3 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{11}{42}\right)\Gamma_p\left(\frac{4}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{11}{42}\right)\Gamma_p\left(\frac{4}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$2987\cdot 3529+2101\cdot 3529^2+\dots$
	$x_1^9 x_2^6 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{13}{42}\right)\Gamma_p\left(\frac{6}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{13}{42}\right)\Gamma_p\left(\frac{6}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$1761\cdot 3529+2946\cdot 3529^2+\dots$
	$x_1^{12} x_2^3 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{17}{42}\right)\Gamma_p\left(\frac{3}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{17}{42}\right)\Gamma_p\left(\frac{3}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$1678\cdot 3529+663\cdot 3529^2+\dots$
	$x_1^{13} x_2^5 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{19}{42}\right)\Gamma_p\left(\frac{5}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{19}{42}\right)\Gamma_p\left(\frac{5}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$276\cdot 3529+2589\cdot 3529^2+\dots$
	$x_1^{16} x_2^2 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{23}{42}\right)\Gamma_p\left(\frac{2}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{23}{42}\right)\Gamma_p\left(\frac{2}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$1675\cdot 3529+1651\cdot 3529^2+\dots$
	$x_1^{17} x_2^4 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{25}{42}\right)\Gamma_p\left(\frac{4}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{25}{42}\right)\Gamma_p\left(\frac{4}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$3325\cdot 3529+3228\cdot 3529^2+\dots$
	$x_1^{20} x_2^1 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{29}{42}\right)\Gamma_p\left(\frac{1}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{29}{42}\right)\Gamma_p\left(\frac{1}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$1008\cdot 3529+2640\cdot 3529^2+\dots$
	$x_1^{21} x_2^3 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{31}{42}\right)\Gamma_p\left(\frac{3}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{31}{42}\right)\Gamma_p\left(\frac{3}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$3236\cdot 3529+3414\cdot 3529^2+\dots$
	$x_1^{25} x_2^2 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{37}{42}\right)\Gamma_p\left(\frac{2}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{37}{42}\right)\Gamma_p\left(\frac{2}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$441\cdot 3529+3300\cdot 3529^2+\dots$
$H^{2,0}$	$x_1^{28} x_2^6 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{41}{42}\right)\Gamma_p\left(\frac{6}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$-p^2\Gamma_p\left(\frac{41}{42}\right)\Gamma_p\left(\frac{6}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$2288\cdot 3529^2+\dots$
$H^{2,2}$	$y_1^{27} y_2^6 y_3^1 y_4^3$	p^2	p^2	12453841

Surface #68: $Q = (4, 7, 14, 3)$, $m = 28$

Equation: $x_0^7 + x_1^4 + x_2^2 + x_0x_3^8 = 0$

$p = 449$

Table 68: Trace contributions for $x_0^7 + x_1^4 + x_2^2 + x_0x_3^8 = 0$ at $p = 449$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{8}\right)$	$-\Gamma_p\left(\frac{1}{8}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{8}\right)$	$139+297\cdot 449+407\cdot 449^2+\dots$
$H^{1,1}$	$y_1^1 y_2^3 y_3^1 y_4^5$	p	p	449
	$y_1^2 y_2^3 y_3^1 y_4^4$	p	p	449
	$y_1^2 y_2^3 y_3^1 y_4^4$	p	p	449
	$y_1^3 y_2^1 y_3^1 y_4^7$	p	p	449
	$y_1^3 y_2^3 y_3^1 y_4^1$	p	p	449
	$y_1^4 y_2^1 y_3^1 y_4^6$	p	p	449
	$y_1^4 y_2^3 y_3^1 y_4^2$	p	p	449
	$y_1^5 y_2^1 y_3^1 y_4^5$	p	p	449
	$y_1^5 y_2^3 y_3^1 y_4^1$	p	p	449
	$y_1^6 y_2^1 y_3^1 y_4^4$	p	p	449
	$x_2^2 x_3^1 y_1^1 y_4^7 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	449
	$x_2^2 x_3^1 y_1^2 y_4^6 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	449
	$x_2^2 x_3^1 y_1^3 y_4^5 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	449
	$x_2^2 x_3^1 y_1^4 y_4^4 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	449
	$x_2^2 x_3^1 y_1^5 y_4^3 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	449
	$x_2^2 x_3^1 y_1^6 y_4^2 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	449
	$x_1^1 x_2^2 x_3^1 x_4^8 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	449
	$x_1^3 x_2^3 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{8}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{8}\right)$	$p\Gamma_p\left(\frac{3}{8}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{8}\right)$	$3\cdot 449+336\cdot 449^2+\dots$
	$x_1^4 x_2^2 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	449
	$x_1^5 x_2^2 x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{8}\right)$	$p\Gamma_p\left(\frac{5}{8}\right)\Gamma_p\left(\frac{1}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{8}\right)$	$150\cdot 449+112\cdot 449^2+\dots$
$H^{2,0}$	$x_1^7 x_2^3 x_3^1 x_4^7 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{7}{8}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{8}\right)$	$-p^2\Gamma_p\left(\frac{7}{8}\right)\Gamma_p\left(\frac{3}{4}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{8}\right)$	$42\cdot 449^2+\dots$
$H^{2,2}$	$y_1^6 y_2^3 y_3^1 y_4^8$	p^2	p^2	201601

Surface #69: $Q = (7, 4, 14, 3)$, $m = 28$

Equation: $x_0^4 + x_1^7 + x_2^2 + x_0 x_3^7 = 0$

$p = 3137$

Table 69: Trace contributions for $x_0^4 + x_1^7 + x_2^2 + x_0 x_3^7 = 0$ at $p = 3137$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{3}{14}\right) \Gamma_p\left(\frac{1}{7}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{7}\right)$	$-\Gamma_p\left(\frac{3}{14}\right) \Gamma_p\left(\frac{1}{7}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{7}\right)$	$3024+1234 \cdot 3137+2028 \cdot 3137^2+\dots$
$H^{1,1}$	$y_1^1 y_2^2 y_3^1 y_4^7$	p	p	3137
	$y_1^1 y_2^2 y_3^2 y_4^4$	p	p	3137
	$y_1^1 y_2^4 y_3^1 y_4^5$	p	p	3137
	$y_1^1 y_2^5 y_3^1 y_4^4$	p	p	3137
	$y_1^1 y_2^6 y_3^1 y_4^3$	p	p	3137
	$y_1^3 y_2^1 y_3^1 y_4^5$	p	p	3137
	$y_1^3 y_2^2 y_3^1 y_4^4$	p	p	3137
	$y_1^3 y_2^3 y_3^1 y_4^3$	p	p	3137
	$y_1^3 y_2^4 y_3^1 y_4^2$	p	p	3137
	$y_1^3 y_2^5 y_3^1 y_4^1$	p	p	3137
	$x_1^1 x_2^5 x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{14}\right) \Gamma_p\left(\frac{5}{7}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{5}{7}\right)$	$p \Gamma_p\left(\frac{1}{14}\right) \Gamma_p\left(\frac{5}{7}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{5}{7}\right)$	$727 \cdot 3137+1093 \cdot 3137^2+\dots$
	$x_1^2 x_3^1 y_2^1 y_4^6 e_1 e_3$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	3137
	$x_1^2 x_3^1 y_2^2 y_4^5 e_1 e_3$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	3137
	$x_1^2 x_3^1 y_2^3 y_4^4 e_1 e_3$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	3137
	$x_1^2 x_3^1 y_2^4 y_4^3 e_1 e_3$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	3137
	$x_1^2 x_3^1 y_2^5 y_4^2 e_1 e_3$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	3137
	$x_1^2 x_3^1 y_2^6 y_4^1 e_1 e_3$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	3137
	$x_1^2 x_2^4 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{5}{14}\right) \Gamma_p\left(\frac{4}{7}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{4}{7}\right)$	$p \Gamma_p\left(\frac{5}{14}\right) \Gamma_p\left(\frac{4}{7}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{4}{7}\right)$	$2216 \cdot 3137+2139 \cdot 3137^2+\dots$
	$x_1^3 x_2^3 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{9}{14}\right) \Gamma_p\left(\frac{3}{7}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{3}{7}\right)$	$p \Gamma_p\left(\frac{9}{14}\right) \Gamma_p\left(\frac{3}{7}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{3}{7}\right)$	$1233 \cdot 3137+2776 \cdot 3137^2+\dots$
	$x_1^4 x_2^2 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{13}{14}\right) \Gamma_p\left(\frac{2}{7}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{7}\right)$	$p \Gamma_p\left(\frac{13}{14}\right) \Gamma_p\left(\frac{2}{7}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{7}\right)$	$863 \cdot 3137+96 \cdot 3137^2+\dots$
$H^{2,0}$	$x_1^4 x_2^6 x_3^1 x_4^6 e_1 e_2 e_3 e_4$	$-p^2 \Gamma_p\left(\frac{11}{14}\right) \Gamma_p\left(\frac{6}{7}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{6}{7}\right)$	$-p^2 \Gamma_p\left(\frac{11}{14}\right) \Gamma_p\left(\frac{6}{7}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{6}{7}\right)$	$1277 \cdot 3137^2+\dots$
$H^{2,2}$	$y_1^3 y_2^6 y_3^1 y_4^7$	p^2	p^2	9840769

Surface #70: $Q = (10, 1, 15, 4)$, $m = 30$

Equation: $x_0^3 + x_1^{30} + x_2^2 + x_0 x_3^5 = 0$

$p = 1801$

Table 70: Trace contributions for $x_0^3 + x_1^{30} + x_2^2 + x_0 x_3^5 = 0$ at $p = 1801$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{4}{15}\right)\Gamma_p\left(\frac{1}{30}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{5}\right)$	$-\Gamma_p\left(\frac{4}{15}\right)\Gamma_p\left(\frac{1}{30}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{5}\right)$	$75+724\cdot 1801+1243\cdot 1801^2+\dots$
$H^{1,1}$	$y_1^1 y_2^7 y_3^1 y_4^5$	p	p	1801
	$y_1^1 y_2^{13} y_3^1 y_4^4$	p	p	1801
	$y_1^1 y_2^{19} y_3^1 y_4^3$	p	p	1801
	$y_1^1 y_2^{25} y_3^1 y_4^2$	p	p	1801
	$y_1^2 y_2^5 y_3^1 y_4^4$	p	p	1801
	$y_1^2 y_2^{11} y_3^1 y_4^3$	p	p	1801
	$y_1^2 y_2^{17} y_3^1 y_4^2$	p	p	1801
	$y_1^2 y_2^{23} y_3^1 y_4^1$	p	p	1801
	$x_1^1 x_2^5 y_3^5 y_4^5 e_1 e_4$	$-p\Gamma_p\left(\frac{1}{1}\right)$	$-p\Gamma_p\left(\frac{1}{1}\right)$	1801
	$x_1^1 x_2^{15} x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{1}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{1}\right)$	1801
	$x_1^1 x_2^{19} x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{15}\right)\Gamma_p\left(\frac{19}{30}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{5}\right)$	$p\Gamma_p\left(\frac{1}{15}\right)\Gamma_p\left(\frac{19}{30}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{5}\right)$	$351\cdot 1801+499\cdot 1801^2+\dots$
	$x_1^1 x_2^{23} x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{15}\right)\Gamma_p\left(\frac{23}{30}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{5}\right)$	$p\Gamma_p\left(\frac{1}{15}\right)\Gamma_p\left(\frac{23}{30}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{5}\right)$	$1083\cdot 1801+617\cdot 1801^2+\dots$
	$x_1^1 x_2^{27} x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{15}\right)\Gamma_p\left(\frac{9}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{5}\right)$	$p\Gamma_p\left(\frac{1}{15}\right)\Gamma_p\left(\frac{9}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{5}\right)$	$350\cdot 1801+1081\cdot 1801^2+\dots$
	$x_1^2 x_2^9 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{9}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{5}\right)$	$p\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{9}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{5}\right)$	$32\cdot 1801+348\cdot 1801^2+\dots$
	$x_1^2 x_2^{13} x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{15}\right)\Gamma_p\left(\frac{13}{30}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{5}\right)$	$p\Gamma_p\left(\frac{7}{15}\right)\Gamma_p\left(\frac{13}{30}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{5}\right)$	$1475\cdot 1801+587\cdot 1801^2+\dots$
	$x_1^2 x_2^{17} x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{8}{15}\right)\Gamma_p\left(\frac{17}{30}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{5}\right)$	$p\Gamma_p\left(\frac{8}{15}\right)\Gamma_p\left(\frac{17}{30}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{5}\right)$	$1464\cdot 1801+1452\cdot 1801^2+\dots$
	$x_1^2 x_2^{21} x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{7}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{5}\right)$	$p\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{7}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{5}\right)$	$394\cdot 1801+1512\cdot 1801^2+\dots$
	$x_1^3 x_2^3 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{5}\right)$	$p\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{10}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{5}\right)$	$1024\cdot 1801+660\cdot 1801^2+\dots$
	$x_1^3 x_2^7 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{13}{15}\right)\Gamma_p\left(\frac{7}{30}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{5}\right)$	$p\Gamma_p\left(\frac{13}{15}\right)\Gamma_p\left(\frac{7}{30}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{5}\right)$	$1051\cdot 1801+1343\cdot 1801^2+\dots$
	$x_1^3 x_2^{11} x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{14}{15}\right)\Gamma_p\left(\frac{11}{30}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{5}\right)$	$p\Gamma_p\left(\frac{14}{15}\right)\Gamma_p\left(\frac{11}{30}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{5}\right)$	$1057\cdot 1801+1483\cdot 1801^2+\dots$
$H^{2,0}$	$x_1^3 x_2^{29} x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{11}{15}\right)\Gamma_p\left(\frac{29}{30}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{5}\right)$	$-p^2\Gamma_p\left(\frac{11}{15}\right)\Gamma_p\left(\frac{29}{30}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{5}\right)$	$1777\cdot 1801^2+\dots$
$H^{2,2}$	$y_1^2 y_2^{29} y_3^1 y_4^5$	p^2	p^2	3243601

Surface #71: $Q = (6, 1, 15, 8)$, $m = 30$

Equation: $x_0^5 + x_1^{30} + x_2^2 + x_0 x_3^3 = 0$

$p = 1801$

Table 71: Trace contributions for $x_0^5 + x_1^{30} + x_2^2 + x_0 x_3^3 = 0$ at $p = 1801$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{2}{15}\right)\Gamma_p\left(\frac{1}{30}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$-\Gamma_p\left(\frac{2}{15}\right)\Gamma_p\left(\frac{1}{30}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$1353+564\cdot 1801+428\cdot 1801^2+\dots$
$H^{1,1}$	$y_1^1 y_2^{11} y_3^1 y_4^3$	p	p	1801
	$y_1^1 y_2^{21} y_3^1 y_4^2$	p	p	1801
	$y_1^2 y_2^7 y_3^1 y_4^3$	p	p	1801
	$y_1^2 y_2^{17} y_3^1 y_4^2$	p	p	1801
	$y_1^2 y_2^{27} y_3^1 y_4^1$	p	p	1801
	$y_1^3 y_2^3 y_3^1 y_4^3$	p	p	1801
	$y_1^3 y_2^{13} y_3^1 y_4^2$	p	p	1801
	$y_1^3 y_2^{23} y_3^1 y_4^1$	p	p	1801
	$y_1^4 y_2^9 y_3^1 y_4^2$	p	p	1801
	$y_1^4 y_2^{19} y_3^1 y_4^1$	p	p	1801
	$x_1^1 x_4^3 y_2^{15} y_3^1 e_1 e_4$	$-p\Gamma_p\left(\frac{1}{1}\right)$	$-p\Gamma_p\left(\frac{1}{1}\right)$	1801
	$x_1^1 x_2^{15} x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{1}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{1}\right)$	1801
	$x_1^1 x_2^{23} x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{15}\right)\Gamma_p\left(\frac{23}{30}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{1}{15}\right)\Gamma_p\left(\frac{23}{30}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$759\cdot 1801+569\cdot 1801^2+\dots$
	$x_1^2 x_2^{17} x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{4}{15}\right)\Gamma_p\left(\frac{17}{30}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{4}{15}\right)\Gamma_p\left(\frac{17}{30}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$654\cdot 1801+1472\cdot 1801^2+\dots$
	$x_1^2 x_2^{25} x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	1801
	$x_1^3 x_2^{11} x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{15}\right)\Gamma_p\left(\frac{11}{30}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{7}{15}\right)\Gamma_p\left(\frac{11}{30}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$820\cdot 1801+74\cdot 1801^2+\dots$
	$x_1^3 x_2^{19} x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{8}{15}\right)\Gamma_p\left(\frac{19}{30}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{8}{15}\right)\Gamma_p\left(\frac{19}{30}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$481\cdot 1801+512\cdot 1801^2+\dots$
	$x_1^4 x_2^5 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	1801
	$x_1^4 x_2^{13} x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{11}{15}\right)\Gamma_p\left(\frac{13}{30}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{11}{15}\right)\Gamma_p\left(\frac{13}{30}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$179\cdot 1801+1208\cdot 1801^2+\dots$
	$x_1^5 x_2^7 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{14}{15}\right)\Gamma_p\left(\frac{7}{30}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{14}{15}\right)\Gamma_p\left(\frac{7}{30}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$140\cdot 1801+137\cdot 1801^2+\dots$
$H^{2,0}$	$x_1^5 x_2^{29} x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{13}{15}\right)\Gamma_p\left(\frac{29}{30}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$-p^2\Gamma_p\left(\frac{13}{15}\right)\Gamma_p\left(\frac{29}{30}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$1001\cdot 1801^2+\dots$
$H^{2,2}$	$y_1^4 y_2^{29} y_3^1 y_4^3$	p^2	p^2	3243601

Surface #72: $Q = (10, 3, 15, 2)$, $m = 30$

Equation: $x_0^3 + x_1^{10} + x_2^2 + x_0 x_3^{10} = 0$

$p = 601$

Table 72: Trace contributions for $x_0^3 + x_1^{10} + x_2^2 + x_0 x_3^{10} = 0$ at $p = 601$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{3}{10}\right) \Gamma_p\left(\frac{1}{10}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{10}\right)$	$-\Gamma_p\left(\frac{3}{10}\right) \Gamma_p\left(\frac{1}{10}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{10}\right)$	$327+86\cdot 601+474\cdot 601^2+\dots$
$H^{1,1}$	$y_1^1 y_2^3 y_3^1 y_4^9$	p	p	601
	$y_1^1 y_2^5 y_3^1 y_4^7$	p	p	601
	$y_1^1 y_2^7 y_3^1 y_4^5$	p	p	601
	$y_1^1 y_2^9 y_3^1 y_4^3$	p	p	601
	$y_1^2 y_2^1 y_3^3 y_4^8$	p	p	601
	$y_1^2 y_2^3 y_3^1 y_4^6$	p	p	601
	$y_1^2 y_2^5 y_3^1 y_4^4$	p	p	601
	$y_1^2 y_2^7 y_3^1 y_4^2$	p	p	601
	$x_2^3 x_3^1 y_1^1 y_4^1 e_2 e_3$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	601
	$x_2^5 x_3^1 y_1^1 y_4^1 e_2 e_3$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	601
	$x_1^1 x_4^{10} y_2^5 y_3^1 e_1 e_4$	$-p \Gamma_p\left(\frac{1}{1}\right)$	$-p \Gamma_p\left(\frac{1}{1}\right)$	601
	$x_1^1 x_2^5 x_3^1 x_4^{10} e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{1}\right)$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{1}\right)$	601
	$x_1^1 x_2^5 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{10}\right) \Gamma_p\left(\frac{7}{10}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{7}{10}\right)$	$p \Gamma_p\left(\frac{1}{10}\right) \Gamma_p\left(\frac{7}{10}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{7}{10}\right)$	$248\cdot 601+308\cdot 601^2+\dots$
	$x_1^1 x_2^9 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{5}\right) \Gamma_p\left(\frac{9}{10}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{5}\right)$	$p \Gamma_p\left(\frac{1}{5}\right) \Gamma_p\left(\frac{9}{10}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{5}\right)$	$32\cdot 601+517\cdot 601^2+\dots$
	$x_1^2 x_4^5 y_2^5 y_3^1 e_1 e_4$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	601
	$x_1^2 x_3^3 x_2^1 x_4^8 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{2}{5}\right) \Gamma_p\left(\frac{3}{10}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{4}{5}\right)$	$p \Gamma_p\left(\frac{2}{5}\right) \Gamma_p\left(\frac{3}{10}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{4}{5}\right)$	$423\cdot 601+34\cdot 601^2+\dots$
	$x_1^2 x_3^5 x_2^1 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	601
	$x_1^2 x_2^3 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{3}{5}\right) \Gamma_p\left(\frac{7}{10}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{5}\right)$	$p \Gamma_p\left(\frac{3}{5}\right) \Gamma_p\left(\frac{7}{10}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{5}\right)$	$314\cdot 601+436\cdot 601^2+\dots$
	$x_1^3 x_2^1 x_3^1 x_4^6 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{4}{5}\right) \Gamma_p\left(\frac{1}{10}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{3}{5}\right)$	$p \Gamma_p\left(\frac{4}{5}\right) \Gamma_p\left(\frac{1}{10}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{3}{5}\right)$	$432\cdot 601+213\cdot 601^2+\dots$
	$x_1^3 x_2^3 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{9}{10}\right) \Gamma_p\left(\frac{3}{10}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{3}{10}\right)$	$p \Gamma_p\left(\frac{9}{10}\right) \Gamma_p\left(\frac{3}{10}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{3}{10}\right)$	$269\cdot 601+487\cdot 601^2+\dots$
$H^{2,0}$	$x_1^3 x_2^9 x_3^1 x_4^9 e_1 e_2 e_3 e_4$	$-p^2 \Gamma_p\left(\frac{7}{10}\right) \Gamma_p\left(\frac{9}{10}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{9}{10}\right)$	$-p^2 \Gamma_p\left(\frac{7}{10}\right) \Gamma_p\left(\frac{9}{10}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{9}{10}\right)$	$533\cdot 601^2+\dots$
$H^{2,2}$	$y_1^2 y_2^9 y_3^1 y_4^{10}$	p^2	p^2	361201

Surface #73: $Q = (15, 2, 10, 3)$, $m = 30$

Equation: $x_0^2 + x_1^{15} + x_2^3 + x_0 x_3^5 = 0$

$p = 1801$

Table 73: Trace contributions for $x_0^2 + x_1^{15} + x_2^3 + x_0 x_3^5 = 0$ at $p = 1801$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{15}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{5}\right)$	$-\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{15}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{5}\right)$	$1787+788\cdot 1801+1301\cdot 1801^2+\dots$
$H^{1,1}$	$y_1^1 y_2^2 y_3^2 y_4^4$	p	p	1801
	$y_1^4 y_2^2 y_3^2 y_4^5$	p	p	1801
	$y_1^1 y_2^5 y_3^2 y_4^3$	p	p	1801
	$y_1^1 y_2^7 y_3^1 y_4^4$	p	p	1801
	$y_1^1 y_2^8 y_3^2 y_4^2$	p	p	1801
	$y_1^1 y_2^{10} y_3^1 y_4^3$	p	p	1801
	$y_1^1 y_2^{11} y_3^2 y_4^1$	p	p	1801
	$y_1^1 y_2^{13} y_3^1 y_4^2$	p	p	1801
	$x_2^5 x_3^2 y_1^1 y_4^3 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{5}\right)$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{5}\right)$	1801
	$x_2^{10} x_3^1 y_1^1 y_4^3 e_2 e_3$	$-p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{5}\right)$	$-p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{5}\right)$	1801
	$x_1^1 x_4^5 y_2^2 y_3^1 e_1 e_4$	$-p\Gamma_p\left(\frac{1}{1}\right)$	$-p\Gamma_p\left(\frac{1}{1}\right)$	1801
	$x_1^1 x_4^5 y_2^{10} y_3^1 e_1 e_4$	$-p\Gamma_p\left(\frac{1}{1}\right)$	$-p\Gamma_p\left(\frac{1}{1}\right)$	1801
	$x_1^1 x_2^5 x_3^2 y_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{1}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{1}\right)$	1801
	$x_1^1 x_2^8 x_3^2 y_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{8}{15}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{3}{5}\right)$	$p\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{8}{15}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{3}{5}\right)$	$985\cdot 1801+1626\cdot 1801^2+\dots$
	$x_1^1 x_2^{10} x_3^1 y_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{1}\right)$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{1}\right)$	1801
	$x_1^1 x_2^{11} x_3^2 y_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{11}{15}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{1}\right)$	$p\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{11}{15}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{1}\right)$	$701\cdot 1801+1271\cdot 1801^2+\dots$
	$x_1^1 x_2^{13} x_3^1 y_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{13}{15}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{1}\right)$	$p\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{13}{15}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{1}\right)$	$1395\cdot 1801+881\cdot 1801^2+\dots$
	$x_2^2 x_3^2 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{2}{15}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{4}{5}\right)$	$p\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{2}{15}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{4}{5}\right)$	$173\cdot 1801+1223\cdot 1801^2+\dots$
	$x_1^2 x_2^4 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{4}{15}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{4}{5}\right)$	$p\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{4}{15}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{4}{5}\right)$	$984\cdot 1801+78\cdot 1801^2+\dots$
	$x_1^2 x_2^7 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{7}{15}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{5}\right)$	$p\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{7}{15}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{5}\right)$	$373\cdot 1801+1207\cdot 1801^2+\dots$
$H^{2,0}$	$x_1^2 x_2^{14} x_3^4 x_4^1 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{14}{15}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{4}{5}\right)$	$-p^2\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{14}{15}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{4}{5}\right)$	$1415\cdot 1801^2+\dots$
$H^{2,2}$	$y_1^1 y_2^{14} y_3^2 y_4^5$	p^2	p^2	3243601

Surface #74: $Q = (2, 6, 15, 7)$, $m = 30$

Equation: $x_0^{15} + x_1^5 + x_2^2 + x_0 x_3^4 = 0$

$p = 601$

Table 74: Trace contributions for $x_0^{15} + x_1^5 + x_2^2 + x_0 x_3^4 = 0$ at $p = 601$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{20}\right) \Gamma_p\left(\frac{1}{5}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{4}\right)$	$-\Gamma_p\left(\frac{1}{20}\right) \Gamma_p\left(\frac{1}{5}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{4}\right)$	$241+157\cdot 601+221\cdot 601^2+\dots$
$H^{1,1}$	$y_1^2 y_2^2 y_3^1 y_4^1$	p	p	601
	$y_1^3 y_2^3 y_3^1 y_4^1$	p	p	601
	$y_1^4 y_2^4 y_3^1 y_4^1$	p	p	601
	$y_1^6 y_2^1 y_3^1 y_4^1$	p	p	601
	$y_1^7 y_2^2 y_3^1 y_4^1$	p	p	601
	$y_1^8 y_2^3 y_3^1 y_4^1$	p	p	601
	$y_1^9 y_2^4 y_3^1 y_4^1$	p	p	601
	$y_1^{11} y_2^1 y_3^1 y_4^3$	p	p	601
	$y_1^{12} y_2^2 y_3^1 y_4^2$	p	p	601
	$y_1^{13} y_2^3 y_3^1 y_4^1$	p	p	601
	$x_1^3 x_2^3 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{3}{20}\right) \Gamma_p\left(\frac{3}{5}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{3}{4}\right)$	$p \Gamma_p\left(\frac{3}{20}\right) \Gamma_p\left(\frac{3}{5}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{3}{4}\right)$	$435\cdot 601+323\cdot 601^2+\dots$
	$x_1^3 x_2^4 y_3^1 y_4^2 e_1 e_2$	$-p \Gamma_p\left(\frac{1}{5}\right) \Gamma_p\left(\frac{4}{5}\right)$	$-p \Gamma_p\left(\frac{1}{5}\right) \Gamma_p\left(\frac{4}{5}\right)$	601
	$x_1^6 x_2^2 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{7}{20}\right) \Gamma_p\left(\frac{2}{5}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{3}{4}\right)$	$p \Gamma_p\left(\frac{7}{20}\right) \Gamma_p\left(\frac{2}{5}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{3}{4}\right)$	$512\cdot 601+273\cdot 601^2+\dots$
	$x_1^6 x_2^3 y_3^1 y_4^2 e_1 e_2$	$-p \Gamma_p\left(\frac{2}{5}\right) \Gamma_p\left(\frac{3}{5}\right)$	$-p \Gamma_p\left(\frac{2}{5}\right) \Gamma_p\left(\frac{3}{5}\right)$	601
	$x_1^7 x_2^4 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{9}{20}\right) \Gamma_p\left(\frac{4}{5}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{4}\right)$	$p \Gamma_p\left(\frac{9}{20}\right) \Gamma_p\left(\frac{4}{5}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{4}\right)$	$164\cdot 601+180\cdot 601^2+\dots$
	$x_1^9 x_2^1 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{11}{20}\right) \Gamma_p\left(\frac{1}{5}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{3}{4}\right)$	$p \Gamma_p\left(\frac{11}{20}\right) \Gamma_p\left(\frac{1}{5}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{3}{4}\right)$	$11\cdot 601+424\cdot 601^2+\dots$
	$x_1^9 x_2^2 y_3^1 y_4^2 e_1 e_2$	$-p \Gamma_p\left(\frac{3}{5}\right) \Gamma_p\left(\frac{2}{5}\right)$	$-p \Gamma_p\left(\frac{3}{5}\right) \Gamma_p\left(\frac{2}{5}\right)$	601
	$x_1^{10} x_2^3 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{13}{20}\right) \Gamma_p\left(\frac{3}{5}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{4}\right)$	$p \Gamma_p\left(\frac{13}{20}\right) \Gamma_p\left(\frac{3}{5}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{4}\right)$	$27\cdot 601+495\cdot 601^2+\dots$
	$x_1^{12} x_2^1 y_3^1 y_4^2 e_1 e_2$	$-p \Gamma_p\left(\frac{4}{5}\right) \Gamma_p\left(\frac{1}{5}\right)$	$-p \Gamma_p\left(\frac{4}{5}\right) \Gamma_p\left(\frac{1}{5}\right)$	601
	$x_1^{13} x_2^2 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{17}{20}\right) \Gamma_p\left(\frac{2}{5}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{4}\right)$	$p \Gamma_p\left(\frac{17}{20}\right) \Gamma_p\left(\frac{2}{5}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{4}\right)$	$496\cdot 601+283\cdot 601^2+\dots$
$H^{2,0}$	$x_1^{15} x_2^4 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$-p^2 \Gamma_p\left(\frac{19}{20}\right) \Gamma_p\left(\frac{4}{5}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{3}{4}\right)$	$-p^2 \Gamma_p\left(\frac{19}{20}\right) \Gamma_p\left(\frac{4}{5}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{3}{4}\right)$	$202\cdot 601^2+\dots$
$H^{2,2}$	$y_1^{14} y_2^4 y_3^1 y_4^1$	p^2	p^2	361201

Surface #75: $Q = (6, 5, 15, 4)$, $m = 30$

Equation: $x_0^5 + x_1^6 + x_2^2 + x_0 x_3^6 = 0$

$p = 1801$

Table 75: Trace contributions for $x_0^5 + x_1^6 + x_2^2 + x_0 x_3^6 = 0$ at $p = 1801$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{6}\right)$	$-\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{6}\right)$	$73+1801+74\cdot 1801^2+\dots$
$H^{1,1}$	$y_1^1 y_2^3 y_3^1 y_4^5$	p	p	1801
	$y_1^1 y_2^3 y_3^1 y_4^3$	p	p	1801
	$y_1^2 y_2^1 y_3^1 y_4^6$	p	p	1801
	$y_1^2 y_2^3 y_3^1 y_4^4$	p	p	1801
	$y_1^2 y_2^3 y_3^1 y_4^4$	p	p	1801
	$y_1^2 y_2^5 y_3^1 y_4^2$	p	p	1801
	$y_1^3 y_2^1 y_3^1 y_4^5$	p	p	1801
	$y_1^3 y_2^3 y_3^1 y_4^3$	p	p	1801
	$y_1^3 y_2^5 y_3^1 y_4^1$	p	p	1801
	$y_1^4 y_2^1 y_3^1 y_4^4$	p	p	1801
	$y_1^4 y_2^3 y_3^1 y_4^2$	p	p	1801
	$x_2^3 x_3^1 y_1^1 y_4^5 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	1801
	$x_2^3 x_3^1 y_1^2 y_4^4 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	1801
	$x_2^3 x_3^1 y_1^3 y_4^3 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	1801
	$x_2^3 x_3^1 y_1^4 y_4^2 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	1801
	$x_1^1 x_4^6 y_2^3 y_3^1 e_1 e_4$	$-p\Gamma_p\left(\frac{1}{1}\right)$	$-p\Gamma_p\left(\frac{1}{1}\right)$	1801
	$x_1^1 x_2^3 x_3^6 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	1801
	$x_1^2 x_2^5 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	1801
	$x_1^3 x_4^3 y_2^3 y_3^1 e_1 e_4$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	1801
	$x_1^3 x_2^3 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	1801
	$x_1^4 x_2^1 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	1801
$H^{2,0}$	$x_1^5 x_2^1 x_3^5 x_4^1 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{6}\right)$	$-p^2\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{5}{6}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{6}\right)$	$1727\cdot 1801^2+\dots$
$H^{2,2}$	$y_1^4 y_2^5 y_3^1 y_4^6$	p^2	p^2	3243601

Surface #76: $Q = (1, 12, 18, 5)$, $m = 36$

Equation: $x_0^{36} + x_1^3 + x_2^2 + x_0 x_3^7 = 0$

$p = 7561$

Table 76: Trace contributions for $x_0^{36} + x_1^3 + x_2^2 + x_0 x_3^7 = 0$ at $p = 7561$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{42}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{7}\right)$	$-\Gamma_p\left(\frac{1}{42}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{7}\right)$	$3600+5964\cdot 7561+2430\cdot 7561^2+\dots$
$H^{1,1}$	$y_1^5 y_2^2 y_3^1 y_4^5$	p	p	7561
	$y_1^4 y_2^2 y_3^1 y_4^4$	p	p	7561
	$y_1^{11} y_2^2 y_3^1 y_4^4$	p	p	7561
	$y_1^{13} y_2^2 y_3^1 y_4^6$	p	p	7561
	$y_1^{17} y_2^2 y_3^1 y_4^3$	p	p	7561
	$y_1^{19} y_2^2 y_3^1 y_4^5$	p	p	7561
	$y_1^{23} y_2^2 y_3^1 y_4^2$	p	p	7561
	$y_1^{25} y_2^2 y_3^1 y_4^4$	p	p	7561
	$y_1^{29} y_2^2 y_3^1 y_4^1$	p	p	7561
	$y_1^{31} y_2^2 y_3^1 y_4^3$	p	p	7561
	$x_1^5 x_2^2 x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{42}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{7}\right)$	$p\Gamma_p\left(\frac{5}{42}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{7}\right)$	$1720\cdot 7561+6274\cdot 7561^2+\dots$
	$x_1^{10} x_2^2 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{11}{42}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{7}\right)$	$p\Gamma_p\left(\frac{11}{42}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{7}\right)$	$3092\cdot 7561+3870\cdot 7561^2+\dots$
	$x_1^{12} x_2^2 x_3^1 x_4^6 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{13}{42}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{6}{7}\right)$	$p\Gamma_p\left(\frac{13}{42}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{6}{7}\right)$	$3900\cdot 7561+2898\cdot 7561^2+\dots$
	$x_1^{15} x_2^2 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{17}{42}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{7}\right)$	$p\Gamma_p\left(\frac{17}{42}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{7}\right)$	$5183\cdot 7561+834\cdot 7561^2+\dots$
	$x_1^{17} x_2^2 x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{19}{42}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{7}\right)$	$p\Gamma_p\left(\frac{19}{42}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{7}\right)$	$5013\cdot 7561+1734\cdot 7561^2+\dots$
	$x_1^{20} x_2^2 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{23}{42}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{7}\right)$	$p\Gamma_p\left(\frac{23}{42}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{7}\right)$	$6650\cdot 7561+3824\cdot 7561^2+\dots$
	$x_1^{22} x_2^2 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{25}{42}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{7}\right)$	$p\Gamma_p\left(\frac{25}{42}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{7}\right)$	$2178\cdot 7561+7343\cdot 7561^2+\dots$
	$x_1^{25} x_2^2 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{29}{42}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{7}\right)$	$p\Gamma_p\left(\frac{29}{42}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{7}\right)$	$696\cdot 7561+5390\cdot 7561^2+\dots$
	$x_1^{27} x_2^2 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{31}{42}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{7}\right)$	$p\Gamma_p\left(\frac{31}{42}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{7}\right)$	$604\cdot 7561+6959\cdot 7561^2+\dots$
	$x_1^{32} x_2^2 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{37}{42}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{7}\right)$	$p\Gamma_p\left(\frac{37}{42}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{7}\right)$	$2809\cdot 7561+3051\cdot 7561^2+\dots$
$H^{2,0}$	$x_1^{36} x_2^2 x_3^1 x_4^6 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{41}{42}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{6}{7}\right)$	$-p^2\Gamma_p\left(\frac{41}{42}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{6}{7}\right)$	$754\cdot 7561^2+\dots$
$H^{2,2}$	$y_1^{35} y_2^2 y_3^1 y_4^7$	p^2	p^2	57168721

Surface #77: $Q = (3, 4, 18, 11)$, $m = 36$

Equation: $x_0^{12} + x_1^9 + x_2^2 + x_0 x_3^3 = 0$

$p = 1297$

Table 77: Trace contributions for $x_0^{12} + x_1^9 + x_2^2 + x_0 x_3^3 = 0$ at $p = 1297$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{18}\right)\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$-\Gamma_p\left(\frac{1}{18}\right)\Gamma_p\left(\frac{1}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$1065+544\cdot 1297+1271\cdot 1297^2+\dots$
$H^{1,1}$	$y_1^1 y_2^4 y_3^1 y_4^3$	p	p	1297
	$y_1^1 y_2^7 y_3^1 y_4^2$	p	p	1297
	$y_1^3 y_2^3 y_3^1 y_4^3$	p	p	1297
	$y_1^3 y_2^6 y_3^1 y_4^2$	p	p	1297
	$y_1^5 y_2^2 y_3^1 y_4^3$	p	p	1297
	$y_1^5 y_2^5 y_3^1 y_4^2$	p	p	1297
	$y_1^5 y_2^8 y_3^1 y_4^1$	p	p	1297
	$y_1^7 y_2^1 y_3^1 y_4^3$	p	p	1297
	$y_1^7 y_2^4 y_3^1 y_4^2$	p	p	1297
	$y_1^7 y_2^7 y_3^1 y_4^1$	p	p	1297
	$y_1^9 y_2^3 y_3^1 y_4^2$	p	p	1297
	$y_1^9 y_2^6 y_3^1 y_4^1$	p	p	1297
	$y_1^{11} y_2^2 y_3^1 y_4^2$	p	p	1297
	$y_1^{11} y_2^5 y_3^1 y_4^1$	p	p	1297
	$x_1^4 x_2^5 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{18}\right)\Gamma_p\left(\frac{5}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{5}{18}\right)\Gamma_p\left(\frac{5}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$156\cdot 1297+1237\cdot 1297^2+\dots$
	$x_1^5 x_2^7 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{18}\right)\Gamma_p\left(\frac{7}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{7}{18}\right)\Gamma_p\left(\frac{7}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$479\cdot 1297+957\cdot 1297^2+\dots$
	$x_1^6 x_3^1 y_2^3 y_4^2 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	1297
	$x_1^6 x_3^1 y_2^6 y_4^1 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	1297
	$x_1^8 x_2^2 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{11}{18}\right)\Gamma_p\left(\frac{2}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$p\Gamma_p\left(\frac{11}{18}\right)\Gamma_p\left(\frac{2}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$1121\cdot 1297+432\cdot 1297^2+\dots$
	$x_1^9 x_2^4 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{13}{18}\right)\Gamma_p\left(\frac{4}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$p\Gamma_p\left(\frac{13}{18}\right)\Gamma_p\left(\frac{4}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{3}\right)$	$291\cdot 1297+702\cdot 1297^2+\dots$
$H^{2,0}$	$x_1^{12} x_2^8 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{17}{18}\right)\Gamma_p\left(\frac{8}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$-p^2\Gamma_p\left(\frac{17}{18}\right)\Gamma_p\left(\frac{8}{9}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{3}\right)$	$587\cdot 1297^2+\dots$
$H^{2,2}$	$y_1^{11} y_2^8 y_3^1 y_4^3$	p^2	p^2	1682209

Surface #78: $Q = (5, 8, 20, 7)$, $m = 40$

Equation: $x_0^8 + x_1^5 + x_2^2 + x_0x_3^5 = 0$

$p = 401$

Table 78: Trace contributions for $x_0^8 + x_1^5 + x_2^2 + x_0x_3^5 = 0$ at $p = 401$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{10}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{5}\right)$	$-\Gamma_p\left(\frac{1}{10}\right)\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{5}\right)$	$56+94\cdot 401+205\cdot 401^2+\dots$
$H^{1,1}$	$y_1^1 y_2^2 y_3^1 y_4^5$	p	p	401
	$y_1^1 y_2^3 y_3^1 y_4^4$	p	p	401
	$y_1^1 y_2^4 y_3^1 y_4^3$	p	p	401
	$y_1^3 y_2^1 y_3^1 y_4^5$	p	p	401
	$y_1^3 y_2^2 y_3^1 y_4^4$	p	p	401
	$y_1^3 y_2^3 y_3^1 y_4^3$	p	p	401
	$y_1^3 y_2^4 y_3^1 y_4^2$	p	p	401
	$y_1^5 y_2^1 y_3^1 y_4^4$	p	p	401
	$y_1^5 y_2^2 y_3^1 y_4^3$	p	p	401
	$y_1^5 y_2^3 y_3^1 y_4^2$	p	p	401
	$y_1^5 y_2^4 y_3^1 y_4^1$	p	p	401
	$y_1^7 y_2^1 y_3^1 y_4^3$	p	p	401
	$y_1^7 y_2^2 y_3^1 y_4^2$	p	p	401
	$y_1^7 y_2^3 y_3^1 y_4^1$	p	p	401
	$x_1^3 x_2^3 x_3^3 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{10}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{5}\right)$	$p\Gamma_p\left(\frac{3}{10}\right)\Gamma_p\left(\frac{3}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{5}\right)$	$35\cdot 401+69\cdot 401^2+\dots$
	$x_1^4 x_3^1 y_2^1 y_4^4 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)$	401
	$x_1^4 x_3^1 y_2^2 y_4^3 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)$	401
	$x_1^4 x_3^1 y_2^3 y_4^2 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)$	401
	$x_1^4 x_3^1 y_2^4 y_4^1 e_1 e_3$	$-p\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{5}\right)\Gamma_p\left(\frac{1}{2}\right)$	401
	$x_1^6 x_2^2 x_3^2 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{10}\right)\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{5}\right)$	$p\Gamma_p\left(\frac{7}{10}\right)\Gamma_p\left(\frac{2}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{5}\right)$	$275\cdot 401+305\cdot 401^2+\dots$
$H^{2,0}$	$x_1^8 x_2^4 x_3^4 x_4^4 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{9}{10}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{5}\right)$	$-p^2\Gamma_p\left(\frac{9}{10}\right)\Gamma_p\left(\frac{4}{5}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{5}\right)$	$222\cdot 401^2+\dots$
$H^{2,2}$	$y_1^7 y_2^4 y_3^1 y_4^5$	p^2	p^2	160801

Surface #79: $Q = (6, 14, 21, 1)$, $m = 42$

Equation: $x_0^7 + x_1^3 + x_2^2 + x_0 x_3^{36} = 0$

$p = 7561$

Table 79: Trace contributions for $x_0^7 + x_1^3 + x_2^2 + x_0 x_3^{36} = 0$ at $p = 7561$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{5}{36}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{36}\right)$	$-\Gamma_p\left(\frac{5}{36}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{36}\right)$	$2459+1660\cdot 7561+5631\cdot 7561^2+\dots$
$H^{1,1}$	$y_1^1 y_2^2 y_3^1 y_4^{25}$	p	p	7561
	$y_1^2 y_2^1 y_3^1 y_4^{32}$	p	p	7561
	$y_1^2 y_2^2 y_3^1 y_4^{20}$	p	p	7561
	$y_1^3 y_2^1 y_3^1 y_4^{27}$	p	p	7561
	$y_1^3 y_2^2 y_3^1 y_4^{15}$	p	p	7561
	$y_1^4 y_2^1 y_3^1 y_4^{22}$	p	p	7561
	$y_1^4 y_2^2 y_3^1 y_4^{10}$	p	p	7561
	$y_1^5 y_2^1 y_3^1 y_4^{17}$	p	p	7561
	$y_1^5 y_2^2 y_3^1 y_4^5$	p	p	7561
	$y_1^6 y_2^1 y_3^1 y_4^{12}$	p	p	7561
	$x_1^1 x_2^2 x_3^1 x_4^{29} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{36}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{29}{36}\right)$	$p\Gamma_p\left(\frac{1}{36}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{29}{36}\right)$	$7034\cdot 7561+3407\cdot 7561^2+\dots$
	$x_1^2 x_2^1 x_3^1 x_4^{23} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{36}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{23}{36}\right)$	$p\Gamma_p\left(\frac{7}{36}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{23}{36}\right)$	$5789\cdot 7561+634\cdot 7561^2+\dots$
	$x_1^3 x_2^1 x_3^1 x_4^{31} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{11}{36}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{31}{36}\right)$	$p\Gamma_p\left(\frac{11}{36}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{31}{36}\right)$	$6774\cdot 7561+908\cdot 7561^2+\dots$
	$x_1^3 x_2^2 x_3^1 x_4^{17} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{13}{36}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{17}{36}\right)$	$p\Gamma_p\left(\frac{13}{36}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{17}{36}\right)$	$2283\cdot 7561+2201\cdot 7561^2+\dots$
	$x_1^4 x_2^1 x_3^1 x_4^{25} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{17}{36}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{25}{36}\right)$	$p\Gamma_p\left(\frac{17}{36}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{25}{36}\right)$	$2400\cdot 7561+7388\cdot 7561^2+\dots$
	$x_1^4 x_2^2 x_3^1 x_4^{11} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{19}{36}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{11}{36}\right)$	$p\Gamma_p\left(\frac{19}{36}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{11}{36}\right)$	$1131\cdot 7561+1770\cdot 7561^2+\dots$
	$x_1^5 x_2^1 x_3^1 x_4^{19} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{23}{36}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{19}{36}\right)$	$p\Gamma_p\left(\frac{23}{36}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{19}{36}\right)$	$1646\cdot 7561+3490\cdot 7561^2+\dots$
	$x_1^5 x_2^2 x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{25}{36}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{36}\right)$	$p\Gamma_p\left(\frac{25}{36}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{36}\right)$	$1566\cdot 7561+1088\cdot 7561^2+\dots$
	$x_1^6 x_2^1 x_3^1 x_4^{13} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{29}{36}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{13}{36}\right)$	$p\Gamma_p\left(\frac{29}{36}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{13}{36}\right)$	$6490\cdot 7561+6963\cdot 7561^2+\dots$
	$x_1^7 x_2^1 x_3^1 x_4^7 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{35}{36}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{36}\right)$	$p\Gamma_p\left(\frac{35}{36}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{36}\right)$	$1033\cdot 7561+3868\cdot 7561^2+\dots$
$H^{2,0}$	$x_1^7 x_2^2 x_3^1 x_4^{35} e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{31}{36}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{35}{36}\right)$	$-p^2\Gamma_p\left(\frac{31}{36}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{35}{36}\right)$	$452\cdot 7561^2+\dots$
$H^{2,2}$	$y_1^6 y_2^2 y_3^1 y_4^{36}$	p^2	p^2	57168721

Surface #80: $Q = (14, 6, 21, 1)$, $m = 42$

Equation: $x_0^3 + x_1^7 + x_2^2 + x_0 x_3^{28} = 0$

$p = 3529$

Table 80: Trace contributions for $x_0^3 + x_1^7 + x_2^2 + x_0 x_3^{28} = 0$ at $p = 3529$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{9}{28}\right)\Gamma_p\left(\frac{1}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{28}\right)$	$-\Gamma_p\left(\frac{9}{28}\right)\Gamma_p\left(\frac{1}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{28}\right)$	$287+1756\cdot 3529+3308\cdot 3529^2+\dots$
$H^{1,1}$	$y_1^1 y_2^2 y_3^1 y_4^{25}$	p	p	3529
	$y_1^1 y_2^2 y_3^1 y_4^{21}$	p	p	3529
	$y_1^1 y_2^4 y_3^1 y_4^{17}$	p	p	3529
	$y_1^1 y_2^5 y_3^1 y_4^{13}$	p	p	3529
	$y_1^1 y_2^6 y_3^1 y_4^9$	p	p	3529
	$y_1^2 y_2^1 y_3^1 y_4^{20}$	p	p	3529
	$y_1^2 y_2^2 y_3^1 y_4^{16}$	p	p	3529
	$y_1^2 y_2^3 y_3^1 y_4^{12}$	p	p	3529
	$y_1^2 y_2^4 y_3^1 y_4^8$	p	p	3529
	$y_1^2 y_2^5 y_3^1 y_4^4$	p	p	3529
	$x_1^1 x_2^4 x_3^1 x_4^{25} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{28}\right)\Gamma_p\left(\frac{4}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{25}{28}\right)$	$p\Gamma_p\left(\frac{1}{28}\right)\Gamma_p\left(\frac{4}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{25}{28}\right)$	$3101\cdot 3529+2896\cdot 3529^2+\dots$
	$x_1^1 x_2^5 x_3^1 x_4^{19} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{28}\right)\Gamma_p\left(\frac{5}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{19}{28}\right)$	$p\Gamma_p\left(\frac{3}{28}\right)\Gamma_p\left(\frac{5}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{19}{28}\right)$	$2660\cdot 3529+589\cdot 3529^2+\dots$
	$x_1^1 x_2^6 x_3^1 x_4^{13} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{28}\right)\Gamma_p\left(\frac{6}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{13}{28}\right)$	$p\Gamma_p\left(\frac{5}{28}\right)\Gamma_p\left(\frac{6}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{13}{28}\right)$	$3143\cdot 3529+1843\cdot 3529^2+\dots$
	$x_1^2 x_2^2 x_3^1 x_4^{23} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{11}{28}\right)\Gamma_p\left(\frac{2}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{23}{28}\right)$	$p\Gamma_p\left(\frac{11}{28}\right)\Gamma_p\left(\frac{2}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{23}{28}\right)$	$1210\cdot 3529+2972\cdot 3529^2+\dots$
	$x_1^2 x_2^3 x_3^1 x_4^{17} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{13}{28}\right)\Gamma_p\left(\frac{3}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{17}{28}\right)$	$p\Gamma_p\left(\frac{13}{28}\right)\Gamma_p\left(\frac{3}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{17}{28}\right)$	$1433\cdot 3529+1192\cdot 3529^2+\dots$
	$x_1^2 x_2^4 x_3^1 x_4^{11} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{15}{28}\right)\Gamma_p\left(\frac{4}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{11}{28}\right)$	$p\Gamma_p\left(\frac{15}{28}\right)\Gamma_p\left(\frac{4}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{11}{28}\right)$	$1847\cdot 3529+1768\cdot 3529^2+\dots$
	$x_1^2 x_2^5 x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{17}{28}\right)\Gamma_p\left(\frac{5}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{28}\right)$	$p\Gamma_p\left(\frac{17}{28}\right)\Gamma_p\left(\frac{5}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{28}\right)$	$1782\cdot 3529+1966\cdot 3529^2+\dots$
	$x_1^3 x_2^1 x_3^1 x_4^{15} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{23}{28}\right)\Gamma_p\left(\frac{1}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{15}{28}\right)$	$p\Gamma_p\left(\frac{23}{28}\right)\Gamma_p\left(\frac{1}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{15}{28}\right)$	$3465\cdot 3529+3012\cdot 3529^2+\dots$
	$x_1^3 x_2^2 x_3^1 x_4^9 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{25}{28}\right)\Gamma_p\left(\frac{2}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{9}{28}\right)$	$p\Gamma_p\left(\frac{25}{28}\right)\Gamma_p\left(\frac{2}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{9}{28}\right)$	$333\cdot 3529+2224\cdot 3529^2+\dots$
	$x_1^3 x_2^3 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{27}{28}\right)\Gamma_p\left(\frac{3}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{28}\right)$	$p\Gamma_p\left(\frac{27}{28}\right)\Gamma_p\left(\frac{3}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{28}\right)$	$437\cdot 3529+2595\cdot 3529^2+\dots$
$H^{2,0}$	$x_1^3 x_2^6 x_3^1 x_4^{27} e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{19}{28}\right)\Gamma_p\left(\frac{6}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{27}{28}\right)$	$-p^2\Gamma_p\left(\frac{19}{28}\right)\Gamma_p\left(\frac{6}{7}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{27}{28}\right)$	$332\cdot 3529^2+\dots$
$H^{2,2}$	$y_1^2 y_2^6 y_3^1 y_4^{28}$	p^2	p^2	12453841

Surface #81: $Q = (21, 6, 14, 1)$, $m = 42$

Equation: $x_0^2 + x_1^7 + x_2^3 + x_0 x_3^{21} = 0$

$p = 883$

Table 81: Trace contributions for $x_0^2 + x_1^7 + x_2^3 + x_0 x_3^{21} = 0$ at $p = 883$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{10}{21}\right)\Gamma_p\left(\frac{1}{7}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{21}\right)$	$-\Gamma_p\left(\frac{10}{21}\right)\Gamma_p\left(\frac{1}{7}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{21}\right)$	$50+77\cdot 883+704\cdot 883^2+\dots$
$H^{1,1}$	$y_1^1 y_2^1 y_3^2 y_4^{15}$	p	p	883
	$y_1^1 y_2^2 y_3^1 y_4^{19}$	p	p	883
	$y_1^1 y_2^2 y_3^2 y_4^{12}$	p	p	883
	$y_1^1 y_2^3 y_3^1 y_4^{16}$	p	p	883
	$y_1^1 y_2^3 y_3^2 y_4^9$	p	p	883
	$y_1^1 y_2^4 y_3^1 y_4^{13}$	p	p	883
	$y_1^1 y_2^4 y_3^2 y_4^6$	p	p	883
	$y_1^1 y_2^5 y_3^1 y_4^{10}$	p	p	883
	$y_1^1 y_2^5 y_3^2 y_4^3$	p	p	883
	$y_1^1 y_2^6 y_3^1 y_4^7$	p	p	883
	$x_1^1 x_2^3 x_3^2 x_4^{17} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{2}{21}\right)\Gamma_p\left(\frac{3}{7}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{17}{21}\right)$	$p\Gamma_p\left(\frac{2}{21}\right)\Gamma_p\left(\frac{3}{7}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{17}{21}\right)$	$845\cdot 883+356\cdot 883^2+\dots$
	$x_1^1 x_2^4 x_3^2 x_4^{11} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{21}\right)\Gamma_p\left(\frac{4}{7}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{11}{21}\right)$	$p\Gamma_p\left(\frac{5}{21}\right)\Gamma_p\left(\frac{4}{7}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{11}{21}\right)$	$170\cdot 883+875\cdot 883^2+\dots$
	$x_1^1 x_2^5 x_3^1 x_4^{19} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{21}\right)\Gamma_p\left(\frac{5}{7}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{19}{21}\right)$	$p\Gamma_p\left(\frac{1}{21}\right)\Gamma_p\left(\frac{5}{7}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{19}{21}\right)$	$860\cdot 883+642\cdot 883^2+\dots$
	$x_1^1 x_2^5 x_3^2 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{8}{21}\right)\Gamma_p\left(\frac{5}{7}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{5}{21}\right)$	$p\Gamma_p\left(\frac{8}{21}\right)\Gamma_p\left(\frac{5}{7}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{5}{21}\right)$	$37\cdot 883+685\cdot 883^2+\dots$
	$x_1^1 x_2^6 x_3^1 x_4^{13} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{4}{21}\right)\Gamma_p\left(\frac{6}{7}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{13}{21}\right)$	$p\Gamma_p\left(\frac{4}{21}\right)\Gamma_p\left(\frac{6}{7}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{13}{21}\right)$	$688\cdot 883+41\cdot 883^2+\dots$
	$x_1^2 x_2^1 x_3^2 x_4^8 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{17}{21}\right)\Gamma_p\left(\frac{1}{7}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{8}{21}\right)$	$p\Gamma_p\left(\frac{17}{21}\right)\Gamma_p\left(\frac{1}{7}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{8}{21}\right)$	$643\cdot 883+577\cdot 883^2+\dots$
	$x_1^2 x_2^2 x_3^1 x_4^{16} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{13}{21}\right)\Gamma_p\left(\frac{2}{7}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{16}{21}\right)$	$p\Gamma_p\left(\frac{13}{21}\right)\Gamma_p\left(\frac{2}{7}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{16}{21}\right)$	$358\cdot 883+746\cdot 883^2+\dots$
	$x_1^2 x_2^2 x_3^2 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{20}{21}\right)\Gamma_p\left(\frac{2}{7}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{2}{21}\right)$	$p\Gamma_p\left(\frac{20}{21}\right)\Gamma_p\left(\frac{2}{7}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{2}{21}\right)$	$691\cdot 883+659\cdot 883^2+\dots$
	$x_1^2 x_2^3 x_3^1 x_4^{10} e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{16}{21}\right)\Gamma_p\left(\frac{3}{7}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{10}{21}\right)$	$p\Gamma_p\left(\frac{16}{21}\right)\Gamma_p\left(\frac{3}{7}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{10}{21}\right)$	$535\cdot 883+705\cdot 883^2+\dots$
	$x_1^2 x_2^4 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{19}{21}\right)\Gamma_p\left(\frac{4}{7}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{4}{21}\right)$	$p\Gamma_p\left(\frac{19}{21}\right)\Gamma_p\left(\frac{4}{7}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{4}{21}\right)$	$395\cdot 883+132\cdot 883^2+\dots$
$H^{2,0}$	$x_1^2 x_2^6 x_3^2 x_4^{20} e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{11}{21}\right)\Gamma_p\left(\frac{6}{7}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{20}{21}\right)$	$-p^2\Gamma_p\left(\frac{11}{21}\right)\Gamma_p\left(\frac{6}{7}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{20}{21}\right)$	$53\cdot 883^2+\dots$
$H^{2,2}$	$y_1^1 y_2^6 y_3^2 y_4^{21}$	p^2	p^2	779689

Surface #82: $Q = (2, 14, 21, 5)$, $m = 42$

Equation: $x_0^{21} + x_1^3 + x_2^2 + x_0 x_3^8 = 0$

$p = 1009$

Table 82: Trace contributions for $x_0^{21} + x_1^3 + x_2^2 + x_0 x_3^8 = 0$ at $p = 1009$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{24}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{8}\right)$	$-\Gamma_p\left(\frac{1}{24}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{8}\right)$	$490+188\cdot 1009+739\cdot 1009^2+\dots$
$H^{1,1}$	$y_1^2 y_2^2 y_3^1 y_4^6$	p	p	1009
	$y_1^4 y_2^1 y_3^1 y_4^8$	p	p	1009
	$y_1^5 y_2^2 y_3^1 y_4^5$	p	p	1009
	$y_1^7 y_2^1 y_3^1 y_4^7$	p	p	1009
	$y_1^8 y_2^2 y_3^1 y_4^4$	p	p	1009
	$y_1^{10} y_2^1 y_3^1 y_4^6$	p	p	1009
	$y_1^{11} y_2^2 y_3^1 y_4^3$	p	p	1009
	$y_1^{13} y_2^1 y_3^1 y_4^5$	p	p	1009
	$y_1^{14} y_2^2 y_3^1 y_4^2$	p	p	1009
	$y_1^{16} y_2^1 y_3^1 y_4^4$	p	p	1009
	$y_1^{17} y_2^2 y_3^1 y_4^1$	p	p	1009
	$y_1^{19} y_2^1 y_3^1 y_4^3$	p	p	1009
	$x_1^5 x_2^2 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{24}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{8}\right)$	$p\Gamma_p\left(\frac{5}{24}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{8}\right)$	$294\cdot 1009+720\cdot 1009^2+\dots$
	$x_1^7 x_2^1 x_3^1 x_4^7 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{24}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{8}\right)$	$p\Gamma_p\left(\frac{7}{24}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{8}\right)$	$830\cdot 1009+857\cdot 1009^2+\dots$
	$x_1^9 x_2^2 y_3^4 e_1 e_2$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	$-p\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{2}{3}\right)$	1009
	$x_1^{10} x_2^1 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{11}{24}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{8}\right)$	$p\Gamma_p\left(\frac{11}{24}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{8}\right)$	$704\cdot 1009+989\cdot 1009^2+\dots$
	$x_1^{12} x_2^1 x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{13}{24}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{8}\right)$	$p\Gamma_p\left(\frac{13}{24}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{8}\right)$	$526\cdot 1009+850\cdot 1009^2+\dots$
	$x_1^{14} x_2^1 y_3^4 e_1 e_2$	$-p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	$-p\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{3}\right)$	1009
	$x_1^{15} x_2^2 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{17}{24}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{8}\right)$	$p\Gamma_p\left(\frac{17}{24}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{8}\right)$	$62\cdot 1009+951\cdot 1009^2+\dots$
	$x_1^{17} x_2^1 x_3^1 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{19}{24}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{8}\right)$	$p\Gamma_p\left(\frac{19}{24}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{8}\right)$	$429\cdot 1009+284\cdot 1009^2+\dots$
$H^{2,0}$	$x_1^{21} x_2^1 x_3^1 x_4^7 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{23}{24}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{8}\right)$	$-p^2\Gamma_p\left(\frac{23}{24}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{8}\right)$	$661\cdot 1009^2+\dots$
$H^{2,2}$	$y_1^{20} y_2^2 y_3^1 y_4^8$	p^2	p^2	1018081

Surface #83: $Q = (14, 3, 21, 4)$, $m = 42$

Equation: $x_0^3 + x_1^{14} + x_2^2 + x_0 x_3^7 = 0$

$p = 3529$

Table 83: Trace contributions for $x_0^3 + x_1^{14} + x_2^2 + x_0 x_3^7 = 0$ at $p = 3529$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{2}{7}\right)\Gamma_p\left(\frac{1}{14}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{7}\right)$	$-\Gamma_p\left(\frac{2}{7}\right)\Gamma_p\left(\frac{1}{14}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{7}\right)$	$2806+664\cdot 3529+1683\cdot 3529^2+\dots$
$H^{1,1}$	$y_1^1 y_2^3 y_3^1 y_4^7$	p	p	3529
	$y_1^1 y_2^5 y_3^1 y_4^6$	p	p	3529
	$y_1^1 y_2^7 y_3^1 y_4^5$	p	p	3529
	$y_1^1 y_2^9 y_3^1 y_4^4$	p	p	3529
	$y_1^1 y_2^{11} y_3^1 y_4^3$	p	p	3529
	$y_1^1 y_2^{13} y_3^1 y_4^2$	p	p	3529
	$y_1^2 y_2^1 y_3^1 y_4^6$	p	p	3529
	$y_1^2 y_2^3 y_3^1 y_4^5$	p	p	3529
	$y_1^2 y_2^5 y_3^1 y_4^4$	p	p	3529
	$y_1^2 y_2^7 y_3^1 y_4^3$	p	p	3529
	$y_1^2 y_2^9 y_3^1 y_4^2$	p	p	3529
	$y_1^2 y_2^{11} y_3^1 y_4^1$	p	p	3529
	$x_2^7 x_3^1 y_1^1 y_4^5 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	3529
	$x_2^7 x_3^1 y_1^2 y_4^3 e_2 e_3$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	$-p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)$	3529
	$x_1^7 x_4^7 y_2^1 y_3^1 e_1 e_4$	$-p\Gamma_p\left(\frac{1}{1}\right)$	$-p\Gamma_p\left(\frac{1}{1}\right)$	3529
	$x_1^1 x_2^7 x_3^1 x_4^7 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{1}\right)$	$p\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{1}\right)$	3529
	$x_1^1 x_2^{11} x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{1}{7}\right)\Gamma_p\left(\frac{11}{14}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{7}\right)$	$p\Gamma_p\left(\frac{1}{7}\right)\Gamma_p\left(\frac{11}{14}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{4}{7}\right)$	$3509\cdot 3529+2601\cdot 3529^2+\dots$
	$x_2^2 x_3^5 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{3}{7}\right)\Gamma_p\left(\frac{5}{14}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{7}\right)$	$p\Gamma_p\left(\frac{3}{7}\right)\Gamma_p\left(\frac{5}{14}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{7}\right)$	$673\cdot 3529+2410\cdot 3529^2+\dots$
	$x_1^2 x_2^9 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{4}{7}\right)\Gamma_p\left(\frac{9}{14}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{7}\right)$	$p\Gamma_p\left(\frac{4}{7}\right)\Gamma_p\left(\frac{9}{14}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{2}{7}\right)$	$624\cdot 3529+83\cdot 3529^2+\dots$
	$x_1^3 x_2^3 x_3^3 x_4^3 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{6}{7}\right)\Gamma_p\left(\frac{3}{14}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{7}\right)$	$p\Gamma_p\left(\frac{6}{7}\right)\Gamma_p\left(\frac{3}{14}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{3}{7}\right)$	$1588\cdot 3529+1387\cdot 3529^2+\dots$
$H^{2,0}$	$x_1^3 x_2^{13} x_3^1 x_4^6 e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{5}{7}\right)\Gamma_p\left(\frac{13}{14}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{6}{7}\right)$	$-p^2\Gamma_p\left(\frac{5}{7}\right)\Gamma_p\left(\frac{13}{14}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{6}{7}\right)$	$2421\cdot 3529^2+\dots$
$H^{2,2}$	$y_1^2 y_2^{13} y_3^1 y_4^7$	p^2	p^2	12453841

Surface #84: $Q = (3, 16, 24, 5)$, $m = 48$

Equation: $x_0^{16} + x_1^3 + x_2^2 + x_0 x_3^9 = 0$

$p = 2593$

Table 84: Trace contributions for $x_0^{16} + x_1^3 + x_2^2 + x_0 x_3^9 = 0$ at $p = 2593$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{18}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{9}\right)$	$-\Gamma_p\left(\frac{1}{18}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{9}\right)$	$56+1272\cdot 2593+2209\cdot 2593^2+\dots$
$H^{1,1}$	$y_1^1 y_2^2 y_3^1 y_4^7$	p	p	2593
	$y_1^3 y_2^1 y_3^1 y_4^9$	p	p	2593
	$y_1^3 y_2^2 y_3^1 y_4^6$	p	p	2593
	$y_1^5 y_2^1 y_3^1 y_4^8$	p	p	2593
	$y_1^5 y_2^2 y_3^1 y_4^5$	p	p	2593
	$y_1^7 y_2^1 y_3^1 y_4^7$	p	p	2593
	$y_1^7 y_2^2 y_3^1 y_4^4$	p	p	2593
	$y_1^9 y_2^1 y_3^1 y_4^6$	p	p	2593
	$y_1^9 y_2^2 y_3^1 y_4^3$	p	p	2593
	$y_1^{11} y_2^1 y_3^1 y_4^5$	p	p	2593
	$y_1^{11} y_2^2 y_3^1 y_4^2$	p	p	2593
	$y_1^{13} y_2^1 y_3^1 y_4^4$	p	p	2593
	$y_1^{13} y_2^2 y_3^1 y_4^1$	p	p	2593
	$y_1^{15} y_2^1 y_3^1 y_4^3$	p	p	2593
	$x_1^5 x_2^2 x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{5}{18}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{5}{9}\right)$	$p \Gamma_p\left(\frac{5}{18}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{5}{9}\right)$	$975\cdot 2593+1021\cdot 2593^2+\dots$
	$x_1^7 x_2^1 x_3^1 x_4^7 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{7}{18}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{7}{9}\right)$	$p \Gamma_p\left(\frac{7}{18}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{7}{9}\right)$	$2073\cdot 2593+1639\cdot 2593^2+\dots$
	$x_1^8 x_3^1 y_2^1 y_4^6 e_1 e_3$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	2593
	$x_1^8 x_3^1 y_2^2 y_4^3 e_1 e_3$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	$-p \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{1}{2}\right)$	2593
	$x_1^{10} x_2^2 x_3^1 x_4^2 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{11}{18}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{9}\right)$	$p \Gamma_p\left(\frac{11}{18}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{2}{9}\right)$	$1481\cdot 2593+327\cdot 2593^2+\dots$
	$x_1^{12} x_2^1 x_3^1 x_4^4 e_1 e_2 e_3 e_4$	$p \Gamma_p\left(\frac{13}{18}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{4}{9}\right)$	$p \Gamma_p\left(\frac{13}{18}\right) \Gamma_p\left(\frac{1}{3}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{4}{9}\right)$	$1976\cdot 2593+1515\cdot 2593^2+\dots$
$H^{2,0}$	$x_1^{16} x_2^2 x_3^1 x_4^8 e_1 e_2 e_3 e_4$	$-p^2 \Gamma_p\left(\frac{17}{18}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{8}{9}\right)$	$-p^2 \Gamma_p\left(\frac{17}{18}\right) \Gamma_p\left(\frac{2}{3}\right) \Gamma_p\left(\frac{1}{2}\right) \Gamma_p\left(\frac{8}{9}\right)$	$1065\cdot 2593^2+\dots$
$H^{2,2}$	$y_1^{15} y_2^2 y_3^1 y_4^9$	p^2	p^2	6723649

Surface #85: $Q = (6, 22, 33, 5)$, $m = 66$

Equation: $x_0^{11} + x_1^3 + x_2^2 + x_0 x_3^{12} = 0$

$p = 2377$

Table 85: Trace contributions for $x_0^{11} + x_1^3 + x_2^2 + x_0 x_3^{12} = 0$ at $p = 2377$.

$H^{s,r}$	Element	tc_i	α_i	α_i expanded
$H^{0,0}$	$y_1^1 y_2^1 y_3^1 y_4^1$	1	1	1
$H^{0,2}$	$x_1^1 x_2^1 x_3^1 x_4^1 e_1 e_2 e_3 e_4$	$-\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{12}\right)$	$-\Gamma_p\left(\frac{1}{12}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{1}{12}\right)$	$941+1594\cdot 2377+1436\cdot 2377^2+\dots$
$H^{1,1}$	$y_1^1 y_2^2 y_3^1 y_4^9$	p	p	2377
	$y_1^2 y_2^1 y_3^1 y_4^{12}$	p	p	2377
	$y_1^2 y_2^2 y_3^1 y_4^8$	p	p	2377
	$y_1^3 y_2^1 y_3^1 y_4^{11}$	p	p	2377
	$y_1^3 y_2^2 y_3^1 y_4^7$	p	p	2377
	$y_1^4 y_2^1 y_3^1 y_4^{10}$	p	p	2377
	$y_1^4 y_2^2 y_3^1 y_4^6$	p	p	2377
	$y_1^5 y_2^1 y_3^1 y_4^9$	p	p	2377
	$y_1^5 y_2^2 y_3^1 y_4^5$	p	p	2377
	$y_1^6 y_2^1 y_3^1 y_4^8$	p	p	2377
	$y_1^6 y_2^2 y_3^1 y_4^4$	p	p	2377
	$y_1^7 y_2^1 y_3^1 y_4^7$	p	p	2377
	$y_1^7 y_2^2 y_3^1 y_4^3$	p	p	2377
	$y_1^8 y_2^1 y_3^1 y_4^6$	p	p	2377
	$y_1^8 y_2^2 y_3^1 y_4^2$	p	p	2377
	$y_1^9 y_2^1 y_3^1 y_4^5$	p	p	2377
	$y_1^9 y_2^2 y_3^1 y_4^1$	p	p	2377
	$y_1^{10} y_2^1 y_3^1 y_4^4$	p	p	2377
	$x_1^5 x_2^2 x_3^1 x_4^5 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{12}\right)$	$p\Gamma_p\left(\frac{5}{12}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{5}{12}\right)$	$2197\cdot 2377+160\cdot 2377^2+\dots$
	$x_1^7 x_2^1 x_3^1 x_4^7 e_1 e_2 e_3 e_4$	$p\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{12}\right)$	$p\Gamma_p\left(\frac{7}{12}\right)\Gamma_p\left(\frac{1}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{7}{12}\right)$	$964\cdot 2377+294\cdot 2377^2+\dots$
$H^{2,0}$	$x_1^{11} x_2^2 x_3^1 x_4^{11} e_1 e_2 e_3 e_4$	$-p^2\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{11}{12}\right)$	$-p^2\Gamma_p\left(\frac{11}{12}\right)\Gamma_p\left(\frac{2}{3}\right)\Gamma_p\left(\frac{1}{2}\right)\Gamma_p\left(\frac{11}{12}\right)$	$485\cdot 2377^2+\dots$
$H^{2,2}$	$y_1^{10} y_2^2 y_3^1 y_4^{12}$	p^2	p^2	5650129